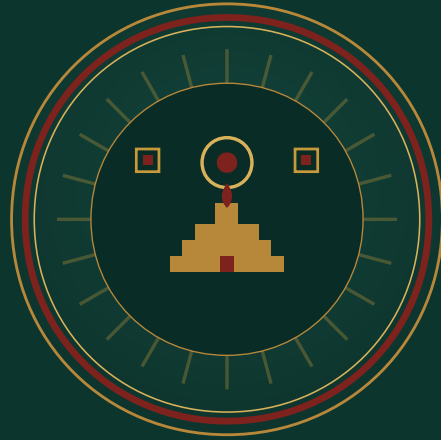




THE VAAL TONGUE



A RECONSTRUCTION

*A reverse-engineering of multiple constructed-language ("Vaal") texts from
Path of Exile / Path of Exile 2, read as a Mesoamerican-sourced constructed
language built on attested Maya and Nahuatl roots.*

PATH OF EXILE | MESOAMERICAN RECONSTRUCTION

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1. OVERVIEW

This document consolidates the full reconstruction of multiple written Vaal texts: the **Atziri Chant** (sung in Utzaal on the eve of the Cataclysm; §4), the **Cuachic Vault Litany** (the call-and-response of Blood Priestess Zelina and Blood Priest Zolin in PoE2 0.5.0; §5), the **Commander's Temple barks** (Elite Commander, in Atziri's Temple at Lira Vaal; §6), **The Drill Sergeant & the Vaal Regiment** (Drill Sergeant, in Atziri's Temple at Lira Vaal; §7), and **Stray captures** (Kamasan Smith, in Atziri's Temple at Lira Vaal; §8). The linguistic framework and a Vaal-internal analysis of the language's own syntax and grammar come first (§2, §3). Each text is then given as a Vaal original with an English translation, followed by a single combined lexicon of every token (§9), an appendix of recorded alternates and open questions (§10), the supporting lore and figures (§11, §12), the working method (§13), status (§14), a complete morpheme index (§15), contributor acknowledgments (§16), the statistical analysis (§17), and citations (§18). The syntax is derived in isolation and blind-confirmed (§3, §17.8); the phonological correspondence rules are collected in §2.6 and hardened orthographically in §17.9 (a phonetic phonology from audio is out of scope, §2.6).

The first three are **offerings flowing upward to Atziri**, the worshippers give themselves (blood, flesh, spirit, heart, their dead, their "new children") and the queen receives and consumes. Directionality is uniform across all of them.

Corpus basis. Everything here rests on the Vaal material available in *Path of Exile 2* as of patch 0.5.0 and the *Path of Exile 1* Vaal content (the Temple of Atzoatl and related lore, datamined through patch 3.27). The readings, rules, and confidence figures describe that corpus only; text or audio added by later patches has not been incorporated, and the reconstruction is versioned so it can be re-tested against them as they arrive (§14).

Statistical standing (full detail in §17). The reconstruction has been stress-tested against a null model of phonotactically Vaal-like nonsense, and two results frame everything that follows. First, Vaal is **not** a random jumble of Maya- and Nahuatl-sounding syllables: 26 of the 65 committed tokens (the enlarged base, §17.7; 23 of 62 in the original run) reconstruct under strict rules where pure noise would yield about 3 (binomial $p \approx 3 \times 10^{-18}$, using the strict-and-online noise floor from Battery D, the resources the real decode uses), and the effect holds when morphologically related tokens are collapsed to distinct roots (§17.7), so the random-match null, that every committed reading is a chance dictionary hit, is strongly disfavored under the tested model. Second, it is not wholly solid either: the robustly verified (hardened) core is **38.5% of the committed tokens (95% CI 27.6–50.6%)**, rising to **44.6% (CI 32.4–57.6%)** over the 56 distinct roots (§17.7); and per reading the confidence that a committed gloss is a genuine root rather than a chance dictionary hit spans, depending on how much of the language one assumes was built from real roots, **from about 45% (pessimistic) to 90% (optimistic)**, with the empirical base rate near 58% (from the name roster, §17.6) placing hardened readings around 75–81%. These headline figures are the enlarged 65-token base: three names added after the original run (*Quecholli*, *Panquetzaliztli*, *Ixchel*) were themselves blind-certified as plants (Battery E, §17.7), so the enlarged base is null-tested in its own right, and the original 62-token run (35.5% / 41.5%) is retained beside it as the prior epoch (§17.7). In short, part of the corpus shows structure the tested random model does not produce, while much of it stays statistically indistinguishable from that model; the honest unit of confidence is the individual token, and whole sentences are correspondingly less certain (§17.5).

What this reconstruction is, and is not. This is a fan reconstruction, rigorous in method but still speculative, not an authoritative or official account of the Vaal language. The statistics establish one thing only: that the corpus is very unlikely to be pure noise, that it carries structure the tested random model does not produce. They do not establish that the developers intended any particular meaning, that a given gloss is the correct translation, or that the recovered grammar reflects a designed system rather than a pattern the method imposed. No living speakers exist and no design documents are public, so idiom, connotation, wordplay, and intended register are largely unrecoverable and are likely to remain so. Every reading here is offered as the best-supported hypothesis under a stated method, and confidence is given as a range rather than a single number precisely because certainty is not available. The committed lexicon is probable; the sentences are less certain than their words; the whole is a serious but provisional effort, not a settled decipherment.





2. LINGUISTIC FRAMEWORK

2.1 THE STANDARD

Every reading rests on an **attested dictionary root** with workable phonology. No invented grammar; no "squinting until it fits." Where a root cannot be verified, the token is flagged and its candidates are preserved as alternates rather than promoted to a committed reading.

2.2 THE SOURCE PALETTE: THREE SOURCE LANGUAGES PLUS LOANS

The conlang is not built from one language but three, plus Romance (Spanish) loans:

LAYER	ROLE	EVIDENCE
Yucatec Maya	Core, most content roots, the article <i>le</i> , presentatives, blood/spirit/flesh vocabulary	<i>ik'</i> , <i>ba'</i> , <i>le</i> , <i>kuxtal</i> , <i>k'imil</i> , <i>iitz</i> , <i>il</i>
Classical Nahuatl	Layer, divinity, place-names, negation, several adjectives	<i>teōtl</i> , <i>-tlān</i> , <i>aocmo</i> , <i>cualli</i> , <i>nochi</i> , <i>yancuic</i> , <i>yocoya</i>
Highland Maya (K'iche')	Thin seam, surfaces only where the other two cannot account for a sound	<i>jare</i> (the /r/)
Romance (Spanish)	Loans, abstract/martial concepts	<i>ascensión</i> + <i>-ada</i> ; <i>jefe</i> → <i>xefe</i> ; <i>otra</i> → <i>u'tra</i> ; <i>échelo</i> ; <i>daca</i> ; <i>fulgor</i> → <i>fukuur</i> (soft)

The **"alien" letters were the clues**. Three sounds do not belong to the Yucatec-plus-Nahuatl core, and each turned out to mark a different source:

- **/g/** in *Gyan'uks* → Nahuatl *yancuic* "new." Nahuatl has no native /g/; the written *g* is silent (Text 2 audio), an orthographic flag for the Nahuatl source, not a sound.[15]
- **/r/** in *jare* → Highland Maya. Yucatec shifted proto-Mayan *r to /j/ and Nahuatl never had /r/, but **K'iche' keeps it**, its core definite articles are *we* / *le* / *ri* ("this / that / the"). An /r/ in this text points squarely at K'iche'an.[17]
- **/f/** in *fukuur* → Spanish. No Mayan or Nahuatl word carries /f/ (in Yucatec, *d f g j r v* occur only in loans), so the /f/ marks a Romance loan, most likely *fulgor* "blaze, lightning-flash" (see Text 3, §6, and §10.4).[9,15]
- **/d/** in *daka* → Spanish, by the same logic, and checked against all three source languages, not just two. A voiced stop /d/ is loan-only in Yucatec Maya, Classical Nahuatl, **and K'iche'**; none has it natively. K'iche' is the test that matters, since it was K'iche' that supplied the /r/ in *jare*, but even there the only non-plain stop is the implosive /b/ (written *b'*, a *b*-sound, not *d*), and the one genuinely odd consonant, a dialectal [ð], is an intervocalic allophone of /l/, not a word-initial /d/ stop. So a /d/-initial root cannot be native to the palette; in a line already carrying Spanish loans (*xefe*, *u'tra*), it points to *daca* "give it here" (see §10.4).[9,15,17,18]

2.3 THE PHONEME INVENTORIES OF THE SOURCE LANGUAGES

The diagnostic key in §2.4 rests on the actual phoneme inventories of the three source languages, set out here from primary descriptive and peer-reviewed sources rather than assumed[50]. (These tables were added, and the key

below corrected, after reader AdministratorQotra flagged that the working framework had misstated the Nahuatl consonant system; see §16.)

Classical Nahuatl. A small inventory: no voiced obstruents, no ejectives, no native /r/, no uvulars.

	BILABIAL	ALVEOLAR	POSTALVEOLAR	LATERAL	VELAR	LABIOVELAR	GLOTTAL
Stop	p	t			k	k ^w (<i>cu</i>)	ʔ (<i>saltillo</i>)
Affricate		ts (<i>tz</i>)	tʃ (<i>ch</i>)	tɬ (<i>tl</i>)			
Fricative		s (<i>z</i> , <i>c</i>)	ʃ (<i>x</i>)				(h)
Nasal	m	n					
Sonorant		l	j (<i>γ</i>)			w	

Vowels: four qualities /i e a o/, each short or long. There is **no phonemic /u/** (the *o* vowel spans [o]–[u]) and **no tone**. The saltillo /ʔ ~ h/ is one plain glottal phoneme, postvocalic only, not part of any ejective series. /k^w/ and /tɬ/ are the strongest Nahuatl tells.

Yucatec Maya. The core layer: a full glottalized series and an implosive, plus tone, but no uvulars and no native /r/.

	BILABIAL	ALVEOLAR	POSTALVEOLAR	VELAR	GLOTTAL
Plain stop	p	t		k	ʔ
Ejective	p'	t'		k'	
Implosive	ɓ (<i>b</i>)				
Plain affricate		ts (<i>tz</i>)	tʃ (<i>ch</i>)		
Ejective affricate		ts' (<i>tz'</i>)	tʃ' (<i>ch'</i>)		
Fricative		s	ʃ (<i>x</i>)		h (<i>j</i>)
Nasal	m	n			
Sonorant		l, (<i>r loan</i>)	j (<i>γ</i>)	w	

Vowels: five qualities /a e i o u/, each in four shapes, short, long low-tone, long high-tone, and glottalized (rearticulated). **Tone sits only on long vowels** (the acute marks in *kíimil*, *náach*). No uvulars; /r/ is loan-only (Proto-Mayan *r became /j/).

K'iche' (Highland Maya). The seam: shares the glottalized series and implosive, but adds uvular /q q'/ and a native tap /r/, lacks tone, and has no ejective /p'/.

	BILABIAL	ALVEOLAR	POSTALVEOLAR	VELAR	UVULAR	GLOTTAL
Plain stop	p	t		k	q	ʔ
Glottalized	ɓ (<i>b'</i>)	t'		k'	q'	
Plain affricate		ts (<i>tz</i>)	tʃ (<i>ch</i>)			
Ejective affricate		ts' (<i>tz'</i>)	tʃ' (<i>ch'</i>)			
Fricative		s	ʃ (<i>x</i>)	x ~ χ (<i>j</i>)		

	BILABIAL	ALVEOLAR	POSTALVEOLAR	VELAR	UVULAR	GLOTTAL
Nasal	m	n				
Sonorant		l, r (<i>tap</i>)	j (<i>γ</i>)	w		

Vowels: five qualities /a e i o u/ with a length contrast (historically ten vowels; six in some dialects, the reduced short set written *ä* /ə/). **No tone.** Here *j* is the velar/uvular fricative /x ~ χ/, not the Yucatec /h/.

What each layer's sounds mark. The sorting the reconstruction depends on:

FEATURE	YUCATEC	NAHUATL	K'ICHE'	TELLS
Ejective /p'/	yes	no	no	/p'/ = Yucatec, not K'iche'
Ejective /t' k' ts' tʃ'/	yes	no	yes	Maya, not Nahuatl
Implosive /b/	yes (<i>b</i>)	no	yes (<i>b'</i>)	Maya, not Nahuatl
Saltillo /ʔ ~ h/	yes	yes	yes	shared, not diagnostic
Uvular /q q'/	no	no	yes	K'iche'
Native /r/	no	no	yes	K'iche'
/kʷ/, /tʃ/	no	yes	no	Nahuatl
Lexical tone	yes	no	no	Yucatec
Phonemic /u/	yes	no	yes	Nahuatl lacks /u/

Two consequences for parsing follow directly. Because Nahuatl has no /u/, a corpus /u/ inside a Nahuatl-sourced root is a stylization of /o/ (or of the /kʷ/ in *cu-*), as in *yuquia* ← *γocoya* "to create" (confirmed in the Text 2 audio, where the *u* is heard as /o/). And because the plain glottal stop is shared by all three, only the ejectives, the implosive, the uvulars, the tap /r/, /kʷ/, /tʃ/, and tone actually carry layer information.

2.4 A DIAGNOSTIC KEY FOR UNFAMILIAR SOUNDS

The lesson of /r/ is that an "alien" sound is not automatically a loan: /r/ looked foreign but was K'iche', inside the palette, just a different layer. So a not-yet-listed sound splits into two predictive classes. These are heuristics for parsing future tokens, to be tested against new text rather than treated as law (the spelling is the game's stylization, not IPA).

Two kinds of "alien" letter. One distinction governs every row below. A letter can be alien in two different ways. It can be a **true loan phoneme**: a sound actually pronounced that none of the three palette languages has (/d/, /f/, /v/, /z/, and the like). Such a sound marks a genuine Spanish loanword and points *outside* the palette. Or it can be an **orthographic flag**: a written letter that is not pronounced as it looks (*g*, and to a degree a plain *b*). This is the game's stylized spelling of a native root and points *back into* the palette, not out of it. The clearest case is the *g* of *Gyan'uks*, written but silent in the audio, spelling the Nahuatl *γa-* of *γancuic*. The rule of thumb: a spoken loan sound is evidence of a loan; a written flag is not.

Points to a specific layer *inside* the palette:

SOUND (SPELLING)	POINTS TO	WHY
ejective /t' k' ts' ch'/ and implosive /b/ (<i>b</i> , <i>b'</i>)	Maya (Yucatec or K'iche')	Nahuatl has no ejective series; its one glottal segment, the saltillo /ʔ ~ h/, is a plain glottal stop, not a glottalized consonant, so a bare glottal stop (<i>a'tul</i> , <i>U'te</i>) is not itself a Maya tell, only the ejectives and the implosive are (correction flagged by reader AdministratorQotra; see §16)
bilabial ejective /p'/	Yucatec , not K'iche'	Yucatec has the bilabial ejective /p'/; K'iche' does not, its only bilabial glottalic is the implosive /b/
uvular /q ~ q'/	K'iche' , not Yucatec	the uvular series is K'iche'-only; Yucatec and Nahuatl have no uvulars
/tʰ/ (<i>tʰ</i>), /kʷ/ (<i>cu</i> , <i>uc</i>)	Nahuatl	neither Maya language has them, cf. <i>tlayeb</i> , <i>atla</i> , the <i>cua-</i> of <i>Cuachic</i>
/r/ (tap), uvular /q/ ~ /q'/	K'iche' / Highland Maya	Yucatec lost *r→j and has no uvulars; Nahuatl has neither. /r/ cracked <i>jare</i> ; a true <i>q</i> (≠ <i>k</i>) would be the next such tell, as yet unseen
lexical tone / acute-marked vowels (<i>í</i> , <i>â</i>)	Yucatec (soft)	tone is Yucatec, not Nahuatl or (mostly) K'iche', cf. <i>kíimil</i> , <i>náach</i>

Points *outside* the palette (loan, in practice Spanish):

SOUND (SPELLING)	CALL	CAVEAT
/d/, /f/	Spanish loan	established (<i>daka</i> , <i>fukuur</i>); /f/ (and /v/) <i>could</i> be a Nahuatl /w/→[v~f] allophone, so rule out a <i>w</i> -word first
/v/, /z/, /ʒ/, /dʒ/, /θ/, /ɲ/ (<i>ñ</i>), trilled /rr/	loan, almost certainly Spanish	none is native to any of the three
/g/	ambiguous → often Nahuatl	not native anywhere, but recurrently a stylized spelling of Nahuatl. In <i>Gyan'uks</i> it spells <i>yancuic</i> and is silent in the audio (the onset being /j/); and, the cleaner, reusable rule, word-initial Gua- / Gue- = Nahuatl <i>cua-</i> / <i>cui-</i> / kʷ / (the Spanish treatment of /kʷ/, as in <i>Cuauhtémoc</i> → <i>Guatemoc</i>), so a <i>G-</i> before a back vowel flags a hidden Nahuatl /kʷ/ root: <i>Guatelitzí</i> ← <i>cuauh-</i> "tree" (§10.8), exactly as <i>Guatemala</i> ← <i>Cuauhtēmallān</i> . Read the rest of the word before deciding
plain voiced [b] (non-implosive)	weak flag	⟨b⟩ normally spells the native implosive /b/; only a clearly plain, voiced [b] in loan context points out

So the reliable "look outside immediately" set is narrow: /d/, /f/, /v/, /z/, /ʒ/, /dʒ/, /θ/, /ɲ/, and a trilled /rr/. Everything else either belongs to one of the three layers or (like /g/ and ⟨b⟩) is ambiguous enough to check first.



2.5 SEMANTIC FIELDS THE SEARCH TARGETS (LORE ANCHORS, IN BRIEF)

The corpus is Vaal liturgy and military command centred on sacrifice offered up to the queen **Atziri**, so the dictionary passes deliberately probe the semantic fields the lore foregrounds rather than searching blind: **blood**, **heart**, and **essence**; **flesh**; **spirit and breath**; **fire and burning**; **death and the dead**; **offering and sacrifice**; **water and purity**; and **immortality / the undying**. These are not imposed on the tokens; they bound where a plausible reading is looked for once a root's phonology fits, which is why so many committed glosses cluster in

those fields. Proper names skew Nahuatl (Atziri, Cuachic, Atzoatl, Ketzuli). The full narrative basis, Atziri and the Beast *pochiti*, Utzaal and Lira Vaal, the Cuachic blood-priests, the tomes *Eztli Pilli* and *Guatelitzi*, is in §11–§12.

2.6 STYLIZATION AND CORRESPONDENCE RULES (WORKING RULES; HARDENED ORTHOGRAPHICALLY IN §17.9)

Where §2.4 identifies which source a sound belongs to, this subsection collects the other half: the recurring ways a source form is reshaped into its Vaal surface form. These rules were read off the token analyses (§9, §10), not derived from audio, so they are catalogued here as working correspondences, not as a hardened phonology. Their reliability is already bounded from one side: the strict-versus-loose latitude test in §17 (and `TIGHTENED_LATITUDE.md`) measures how much the committed readings depend on the more permissive of these operations, and most committed readings do depend on them, since the strict-latitude pass more than halves the reconstructable set. These are the initial working hypotheses, not left unexamined: §17.9 hardens them at the orthographic level, sorting the strict-safe correspondences from the permissive ones, which is the one dimension the corpus can test without audio.

Two rules are corroborated by in-game audio and are the most secure:

- **Written *u* is realized /o/.** Nahuatl-sourced tokens show it (*yuquia* left from *yocoya*, the *u* of *U'Te* heard as /o/, *ukto* from *ocotl*), consistent with Nahuatl having no /u/. Audio-confirmed on Text 2.
- **Word-initial *g* is silent.** In *Gyan'uks* the *g* is not pronounced; the onset is /j/, the *ya-* of *yancuic*. Audio-confirmed.

The remainder are orthographic or morphological stylizations inferred from the written forms:

RULE	EXAMPLE	NOTE
Nahuatl absolutive <i>-tl</i> reduces and reorders to <i>-to</i>	<i>ukto</i> from <i>ocotl</i> ; <i>tlapec</i> from <i>tlapechtli</i>	cluster simplification; forbidden under the strict latitude (§17)
Verb root takes a terminal <i>-a</i>	<i>Ela</i> from <i>el</i> "burn"; <i>Xatlene</i> from <i>xotla</i>	regular on the burn-root
Reverential <i>-tzin</i> reduces to <i>-tzi</i> / <i>-tli</i>	<i>Atziri</i> from <i>atl</i> + <i>-tzin</i> ; <i>Guatelitzi</i>	name-forming
Word-initial <i>Gua-</i> / <i>Gue-</i> spells Nahuatl <i>cua-</i> / <i>cui-</i> / <i>kʷ-</i>	<i>Guatelitzi</i> from <i>cuauh-</i>	Spanish colonial orthography (as <i>Cuauhtemoc</i> to <i>Guatemoc</i>); also §2.4
Terminal-vowel padding on closed roots	<i>Ma'oxe</i> , <i>puxe</i> , <i>buxa</i> from <i>bíuk</i>	adds an open final syllable
Non-morphemic coda <i>-uks</i> / <i>-ks</i> / <i>-s</i>	<i>Otsuks</i> , <i>Gyan'uks</i> , <i>Donuks</i>	a closing romanization tail, carries no meaning (§10.9)
Soft non-morphemic tail <i>-che</i> / <i>-zeh</i>	<i>upulche</i> , <i>quxzeh</i>	appended to a committed root, left unparsed
Ejective written plain, and <i>q</i> for ejective / <i>k'</i>	<i>k'áak'</i> to <i>kahk</i> ; the <i>q</i> of <i>quxzeh</i> = / <i>k'</i>	the corpus is inconsistent in marking ejectives
Loan-only /d f r g/ mark a non-core source	<i>daka</i> , <i>fukuur</i> (Spanish); <i>jare</i> , <i>koriek</i> (K'iche' /r/)	the source-identification side is in §2.4

Every rule above is a candidate for the treatment the lexicon and syntax received: state it, then test it. That test is the open phonology work, and it needs systematic audio, which the present corpus and sources do not provide, so these rules are the honest current state, collected in one place, and explicitly not yet carrying confidence intervals

of their own. **Scope note.** A phonetic phonology, the actual sounds behind the spellings, requires trained transcription of the in-game audio, which is a specialist skill outside the maintainer's expertise, so it is not being pursued. Anyone who wishes to take it up is welcome to. If a specific audio detail is caught by ear it will be added to the corpus (as the /o/ and silent-/g/ points already were); otherwise the line is drawn here, and the orthographic hardening in §17.9 is the ceiling this project sets for phonology.



3. VAAL-INTERNAL SYNTAX AND GRAMMAR

This section analyses Vaal as a language in its own right: its clause structure, word order, and morphology as they actually appear in the corpus, independent of the source palette. It is the syntactic counterpart to the lexical work in the sections that follow, and it is kept deliberately separate from them in method (see §3.1).

The findings here are the analyst's first reading. They are then tested by a blind multi-agent confirmation pass (documented in `SYNTAX_CONFIRMATION_PROTOCOL.md` and `SYNTAX_EXPERIMENT_LOG.md`) and quantified with confidence intervals in §17. Where the corpus does not settle a question, it is left open rather than forced (§3.7).

3.1 METHOD: ANALYSIS IN ISOLATION

Every grammatical claim in this section is derived from Vaal's own internal distribution: which element appears in which position, adjacent to what, and how consistently across the corpus. The rule that governs the lexical work has a direct analogue here.

- **Distribution is the only evidence.** A word-order or morpheme-placement claim must rest on the positions the Vaal strings actually show, counted across the corpus, not on the grammar of any palette language. "Yucatec is verb-initial, therefore Vaal is" is forbidden in exactly the way gloss-led root search is forbidden (§2, rule 2a): it reverse-engineers the answer from the source rather than reading it off the data.
- **Glosses label, they do not license.** A token's committed meaning (from §9) may be used to say what a clause is doing (a command, a question, a presentation), because that is needed to group like with like. It may never be used to import the source language's rule for how that clause is built.
- **Palette comparison is a closing step, never a premise.** Once a Vaal-internal pattern is established, it may be noted in passing whether it happens to match or diverge from Maya or Nahuatl. That observation is commentary; it is never the source of a grammatical conclusion, and no conclusion in this section depends on it.

3.2 THE SYNTACTIC CORPUS

The analysable material is the set of connected Vaal utterances in Texts 1 through 4 (§4 to §7) and the one full stray line in Text 5 (§8). The boss ability-cries in Text 5 (Ahuatotli, Mektul) are one- and two-word skill-names rather than clauses and carry little syntactic information; they are noted but not weighted. In total roughly two dozen clauses are available. This is a small corpus, and the confidence statement in §17 is scaled to it; the value here is that the corpus is closed and every clause is examined.

3.3 CLAUSE TYPES ARE SET BY A CLAUSE-INITIAL ELEMENT

The most robust pattern in the corpus is that the first element of a clause fixes its type. Five openers recur, each with a consistent following shape.

- **Presentative: A'te / U'Te / Yatle (and the reduced u'te) + a noun.** This is the single most frequent clause type: *A'te 'Ibil*, *A'te ik'el*, *A'te liimek*, *A'te yuquia*, *A'te fukuur*, *U'Te kuxkal*, *u'te ik'el*, *u'te puyao*. The opener is

clause-initial and is followed by a noun or noun phrase; the clause carries no verb. It presents or points to something.

- **Interrogative: Máax + the rest.** *Máax tlayeb mucane?*, *Máax ka ti a'tul?*. The question word is clause-initial in every attested question.
- **Imperative: Xi + a verb.** *Xi daka puxe*, *Xi u'te cha'tsoke*. *Xi* sits immediately before the verb it commands and is clause-initial.
- **Privative: 'Ayok / Aiokmo + the rest.** *'Ayok ta' en*, *'Ayok kifba Atziri...*, *Aiokmo til xu'te*, *Aiokmo tul jare elba*. The negator opens the clause and scopes over what follows.
- **Vocative + predicate: (Atziri, Atziri,) + a nominal predicate.** The Text 1 chant frame is a doubled vocative followed by a verbless predicate: *ek te mucane Vaal*, *ma kilya Zerphi*, *ikba'yucane Vaal*, *ascenada akal*. The predicate is nominal or adjectival and there is no copula between subject and predicate (see §3.4).

That the clause type is announced first, before the content, is itself a Vaal-internal structural fact and one of the clearest signals in the corpus.

3.4 WORD ORDER WITHIN THE PHRASE

- **Dependents precede their noun.** Attributive modifiers, articles, and possessive markers all stand before the noun, with no counterexample in the corpus. Modifier before noun: *tlayeb kutsen*, *tlayeb kifba*, *tlayeb ukto*, *tlayeb mucane*, and *niách i'chian* ("far" before "house"). Article before noun: *le mucane*, *le'itzil*, *le le Vaal*. Possessive proclitic before noun: *u mujuk'*, *Ik'eche*. The Vaal noun phrase is consistently head-final for its dependents. One caveat, surfaced by the blind confirmation (§17.8): *tlayeb* "dark" may in some lines be a fronted locative ("in the dark") rather than a pure attributive. Either way it stands before the noun, so the prenominal generalisation holds; only the adjective-versus-adverbial status of *tlayeb* is left open (§3.7).
- **Possessed precedes possessor.** In *u mujuk' le mucane* ("the strength of the mighty") the possessed noun, marked with the proclitic *u*, comes first and the possessor phrase *le mucane* follows. The genitive is head-initial.
- **Verb precedes object.** In the one unambiguous marker-verb-object string, *Xi daka puxe*, the verb *daka* precedes its object *puxe*. Predicate-first order also appears in *Ela tlayeb ukto* (the burning predicate before its noun phrase).
- **Equational clauses have no copula.** Subject and nominal predicate are simply juxtaposed: *Atziri le'itzil* ("Atziri, the essence"), *Atziri le'itzil* and the whole Text 1 frame *Atziri ... ascenada akal*. Nothing links them; the juxtaposition is the predication.

3.5 MORPHOLOGICAL SHAPE

The corpus shows a consistent split by position.

- **Grammatical operators are proclitic (they attach or stand before their host):** the presentatives *A'te / U'Te*, the imperative *Xi*, the modal or optative *ma* (*ma kilya*), the privative *'Ayok*, the article *le*, and the possessives *u-* and *a-*. Clause-type and determiner marking is front-loaded.
- **Lexical morphology is suffixing (built on the right edge of the root):** the honorific or plural *-ane* (*ikba'yucane*, *mucane*), the reflexive *-ba'* (*ik'bala*, *elba*, *kifba*), the abstractive *-el* (*ik'el*), the second-person *-ech* (*Ik'eche*). Nominal derivation stacks to the right of the root.

The picture is a language that marks clause type and reference at the front of the clause and the noun phrase, and derives its nouns by suffixation.

3.6 A FIRST GRAMMATICAL SKETCH

Read only from the corpus, Vaal presents as follows: a clause opens with a function word that names its type (presentative, interrogative, imperative, privative) or, in the chant register, a vocative; nominal predication needs no copula; noun phrases place every dependent (article, possessive, modifier) before the head; genitives place the possessed before the possessor; and where a verb governs an object, the verb comes first. Morphologically, operators cliticise to the front and lexical affixes suffix to the rear. These are the templates the confirmation pass (§17) is asked to reproduce blind.

3.7 OPEN QUESTIONS

Held open, because the corpus does not yet settle them:

- The two particles *ka* and *ti* in *Máax ka ti a'tul*, standing between the question word and the verb; their function (aspect, relational, interrogative reinforcement) is undetermined.
- The doubling in *le le Vaal*: a reduplicated article, two stacked determiners, or a scribal repetition.
- The suffix string in *jare'yantul* (*jare'* + *yan* + *tul*): the segmentation is proposed but the role of the medial *-yan-* is open.
- The particle *ta'* in *'Ayok ta' en*, between the privative and the pronoun.
- Whether the presentative alternation *A'te* / *U'Te* / *Yatle* is purely phonological variation or encodes a deictic contrast (near versus far); the distribution does not yet disambiguate.
- Whether *tlayeb* "dark" is a true attributive adjective or a fronted locative phrase in lines like *tlayeb kutsen* and *Ela tlayeb ukto*; both readings keep it before the noun, so word order is unaffected, but its category is unsettled (raised in the confirmation pass, §17.8).

3.8 COMPARISON WITH THE SOURCE PALETTE (CLOSING NOTE)

Everything above was derived without reference to any source language, as §3.1 requires. Only now, with the grammar fixed from Vaal's own distribution, is it compared with the palette. This comparison is commentary: it feeds none of the confidence figures in §17.8 and is not evidence for any rule in §3. Typological features like prenominal articles or a missing copula are common across the world's languages, so a match is suggestive at most, never diagnostic.

Read against the palette, the independently derived grammar lines up with the same layering the lexicon shows (§9, §17): a Yucatec–Mayan core, a lighter Nahuatl presence, and Spanish only as loanwords.

The Mayan match is the closest. The clause-initial marking of type, the prenominal article and possessive, the possessed-before-possessor genitive built on a possessive proclitic, the verbless equational clause, and the split between front-standing operators and suffixing derivation are all characteristic of Yucatec Maya and K'iche'. The interrogative standing first (Vaal *Máax*) and attributive modifiers standing before their noun also fit the Mayan

profile. This is the typological company the Yucatec-core lexicon would predict, reached here from the syntax alone.

Nahuatl matches on some points (a prenominal possessive, no copula) but diverges on the one that matters most: Nahuatl is heavily prefixing and polysynthetic, packing subject and object agreement onto the verb, and Vaal shows none of that. Vaal's morphology is light, with proclitic operators and suffixing derivation, a Mayan shape rather than a Nahuatl one. The Nahuatl layer the lexicon places in divinity and place-names does not, on this corpus, extend to the grammar.

Spanish diverges plainly: it places attributive adjectives after the noun and requires a copula (*ser / estar*), both the opposite of what Vaal does. Consistent with the lexical finding, Spanish contributes words (*xefe*, *u'tra*, *ascenada*), not structure.

Two of the open questions in §3.7 gain candidate leads from the palette, offered as leads and not resolutions. The doubled article in *le le Vaal* resembles the Yucatec discontinuous article *le ... o'* that brackets its noun; a bracketing determiner is one thing the doubling could reflect. The particles *ka* and *ti* have common Yucatec functions (a subordinator *ka'*, a relational *ti'*) that would suit their slot between the question word and the verb. Both stay open in §3.7; the palette only narrows where to look next, which is the role §3.1 allows it.

The caveat stands: this is a post-hoc match on a small corpus, and typological convergence is weak evidence on its own. Its value is as a consistency check, that the grammar derived blind points to the same source stratification the lexicon does, not as independent proof of either.





4. TEXT 1: THE ATZIRI CHANT[4]

Sung by Atziri's followers in Utzaal on the eve of the Cataclysm. An exhortation and offering TO Atziri: the blood flows up to her; she is the recipient.

THE CHANT

#	VAAL	ENGLISH
1	Atziri, Atziri, ek te mucane Vaal!	Atziri, Atziri, star of the mighty Vaal!
2	Teoyuxtlane ascensionada! Teoyuxtlane Yutsal!	Divine, ascended; divine Utzaal!
3	Atziri, Atziri, ma kilya Zerphi!	Atziri, Atziri, undying as Zerphi!
4	Ti itsok anab nochira! Kextal xi kujkuali ik'bala!	To you, the blood of all your servants; transform us, make our spirit worthy!
5	Atziri, Atziri, ikba'yucane Vaal!	Atziri, Atziri, undying breath of the Vaal!
6	A'te 'Ibil tlayeb kutsen! A'te ik'el tlayeb kifba!	Here, our flesh laid bare in the dark, the offering; here, our hidden self in the dark, the heart!
7	Atziri, Atziri, ascenada akal!	Atziri, Atziri, risen into the eternal waters!
8	Aiokmo til xu'te! Aiokmo tul jare elba!	No more burning to ash, no ending; no more dwindling to embers, no going out!

LINE NOTES

- **Lines 3 & 5** both render *undying* (*kilya*, then *ikba'yucane*). The repetition is deliberate: an immortality motif, fitting a queen who sought eternal life.
- **Dramatic irony runs throughout.** "Eternal waters" (*akal*) is the still pond of the nightmare realm she ends up trapped in. "No more burning... no going out" is sung on the very night the Cataclysm burns Utzaal to nothing. "Undying as Zerphi," though Zerphi did, in the end, die; he is undying only in legend.
- **"Utzaal" kept here (*Yutsal*).** *Yutsal* is the Yucatec common noun *y-* (3rd-person possessive) + *uts* "good" + *-al*, "its goodness, the good place" (attested in Cordemex as *yutsal*, "todo es bueno") [51]. Capitalized in the chant, it is the proper-name use, the city *Utzaal* (itself *utz* "good" + *-aal*), so the translation keeps *Utzaal*. Contrast the lowercase *yutsal* in §5, which is translated as the common noun.
- **Source-validated against the game files.** The chant exists verbatim in the dataminced `NPCTextAudio` table as `VaalSermon_01-08`, in order, with zero drift. The internal name *VaalSermon* confirms the register, and the data fixes the canonical spelling *A'te 'Ibil* (an earlier draft wrote *Íbil*).





5. TEXT 2: THE CUACHIC VAULT LITANY[2,3]

Call by Blood Priestess Zelina; response by Blood Priest Zolin. The sealed-away Vaal survivors offering *THEMSELVES* up to Atziri, the same upward direction as the chant above, preserved as liturgy from the height of her power. In the game files each boss carries eight chant lines tagged ChantA–ChantH; the two columns below are those two pools, aligned by their shared A–H index.

THE LITANY

PAIR	SPEAKER	VAAL	ENGLISH
1	Zelina (call)	Kuxte' kíimil'!	Life and death!
1	Zolin (response)	Tlaxye' le Vaal!	The people of the Vaal!
2	Zelina (call)	Ma'oxe ik'el'!	The countless spirits!
2	Zolin (response)	Atziri le'itzil'!	Atziri, the essence!
3	Zelina (call)	Xatlene kujkuali'!	Kindle the worthy!
3	Zolin (response)	Gyan'uks ko'janti'!	Your new children, come, devour!
4	Zelina (call)	A'te 'Ibil'!	Behold the flesh!
4	Zolin (response)	A'te ik'el'!	Behold the spirit!
5	Zelina (call)	Yaxe chikula'!	The first sign!
5	Zolin (response)	Tzokan'te ik'el'!	The last spirit!
6	Zelina (call)	A'te líimek'!	Behold those in the earth!
6	Zolin (response)	Yatle yutsal'!	There stands the good place!
7	Zelina (call)	Tlayeb kutsen'!	In the dark, the offering!
7	Zolin (response)	Tlayeb kifba'!	In the dark, the heart!
8	Zelina (call)	U'Te kuxkal'!	Behold the living!
8	Zolin (response)	A'te yuquia'!	Behold the made!

LINE NOTES

- **Pair 2** is the multitude and the one: *Ma'oxe ik'el* "the countless spirits" (numberless faithful offered up) answered by *Atziri le'itzil* "Atziri, the essence": the queen who is the essence they all feed.
- **Pair 3** *Gyan'uks ko'janti* "your new children, come, devour" is metaphorical: the latter-day vault Vaal as Atziri's "new children," offered for her to consume. Not literal children.
- **Pair 6** *A'te líimek* / *Yatle yutsal* sets the buried dead against what still stands: those in the earth, and the good place still rising. Here *yutsal* is **lowercase**, so the translation uses the common noun, Yucatec *y-uts-al* "the good place, its goodness" (root *uts* "good," §9)[51], rather than the proper name. The good place that stands is almost certainly **Utzaal**, the capitalized *Yutsal* of the chant (§4) being that same *uts* root lexicalized as the city's name, but the lowercase form keeps the reading descriptive: "there stands the good place."

- **Source-validated against the game files.** All sixteen lines match poe2db's extracted audio subtitles exactly, with zero transcription drift. Zelina's call pool is tagged *BloodPriestess_ChantA-H* (*Kuxte' kíimil' ... U'Te kuxkal*); Zolin's response pool *BloodPriest_ChantA-H* (*Tlaxye' le Vaal ... A'te yuquia*). This independently confirms the *Ma'oxe* reading: it appears verbatim as *Ma'oxe ik'el*. Both bosses sit at *Metadata/Monsters/VaalMonsters/Living/BloodPriests/*: "Living" Vaal, i.e. present-day descendants, which fits their being recruited as hideout vendors once subdued.





6. TEXT 3: QUEMALANI, THE ELITE COMMANDER[1]

The Vaal voice-lines of the Commander boss in the Commander's Chamber of Atziri's Temple at Lira Vaal (PoE2's Fate of the Vaal). The encounter shows up under variant names, Quemalani, the Elite Commander (live chat feed) and Xolotl, the Royal Commander (the poe2db datamine page; internal asset BlackjawPastLiving), but it is one and the same boss and voice pool, and the particular label is not significant. The Commander is fire-based: it breathes fire, which squares with its blaze imagery: the Vaal line A'te fukuur! "Behold the blaze!" and the English bark "Flaming Glory!" (This temple is not the Temple of Atzoatl; that was the PoE1 Incursion temple, raided ~20 years earlier by the Exile/"Godslayer"; the PoE2 temple reuses its room-names and grid, and the Vaal mistake the player for that earlier "demon of Atzoatl.")

These are separate barks, not one speech. The lines below are independent voice-line phrases the Commander fires at different points in the fight: in the chat feed they appear interleaved with one another and with English barks ("Flaming Glory!", "No further!"), not as a continuous oration. Each is rendered on its own; the token readings stand, but there is no connective narrative tying them into a single address.

Orthography note. Two spelling conventions are clear from the in-game audio: **j** = /h/ (so *ja* = *ha* "water"; *mujuk'* = /muhuk'/) and **x** = /ʃ/ ("sh"). This also explains *xefe*: it is the **archaic Spanish spelling of jefe** (old Spanish *x* = /ʃ/). The "alien" letters stay source-clues: /g/ → Nahuatl (*yancuic*), /r/ → K'iche' (Highland Maya), and /f/ & /d/ → Spanish (both are loan-only across Maya, Nahuatl, and K'iche'); see *fukuur* and *daka* in §10.4. Two further pronunciations are now confirmed from the Text 2 audio, and both match the inventories of the source languages (§2.3). The **/g/ in Gyan'uks** is silent: the onset is heard as /j/, the *ya-* of Nahuatl *yancuic* "new." That fits Nahuatl exactly, since its inventory has /j/ but no voiced stops and so cannot carry a true /g/, the written *g** being an empty orthographic flag rather than a sound. And the **u** in U'Te and *yuquia* is pronounced /o/: for Nahuatl *yuquia* ← *yocoya* this is the expected reflex, since Nahuatl has no /u/ and its *o* spans [o]–[u] (§2.3); for the Maya presentative U'Te the /o/ is the vowel itself, Yucatec having both /o/ and /u/, so here the romanized *u** simply stands for an actual /o/.

THE BARKS (INDEPENDENT VOICE-LINES)

GROUP	VAAL	ENGLISH
Short cry	Eche lu* nochbe*!	Pour it out, all that we are!
Short cry	Kí' inib*!	Sweet, my flesh!
Bark 1	'Ayok ta' en, u mujuk' le mucane, níach i'chian.	Nothing more for me now; the strength of the mighty is in me, deep within her house.
Bark 1	A'te waja yatle u'tra buxa*!	Behold the offering; there stands another!
Bark 1	A'te fukuur*!	Behold the blaze!
Bark 2	Xi daka* puxe*...	Bring the draught...
Bark 2	'Ayok kifba Atziri kilya sakilja.	The heart beats no more; for Atziri, the undying, the white drink.
Bark 2	Ik'eche sakilja atla donuks*.	You are the breath within it, the sacred water, our draught.
Bark 2	Donuks*... ko'soxsal*.	Drink... raw and fresh!

GROUP	VAAL	ENGLISH
Bark 2	Donuks*... ko'mujuk!	Drink... and take strength!
Bark 3	Xi u'te cha'tsoke, a'te itsok xefe yotlapek le le Vaal te'moxti* qexcan...	See, the end comes round again; behold the blood. Commander of the Vaal altar, where all are remade...
Bark 3	Na' puyao*.	the reddened offering.
Bark 3	Ich tlapec u'te puyao*, a'te itsok pu uch' ta'nuk!	On the altar, watch it redden; behold the blood; drink deep, great one!

*Tokens marked * are soft or open parses, not admitted to the committed lexicon (§9); each is analysed in §10.*

Register note: the speaker is a military high leader, not a priest. The lines read as a commander's orders and declarations; he leads the sacrifice rather than presiding over it. "We drink" is a war-band's blood-draught before battle (he drinks first, then commands his men); "my flesh, and gladly" is a leader offering his own body first.

LINE NOTES

- **xefe** = Spanish *jefe* "**chief, commander**": a Romance loan beside *ascensión*, and a self-naming one: the line names the speaker's own office. The Romance seam is a confirmed pattern, not a one-off.
- **The blood-draught is sourced ritual.** *uch'* is Yucatec *uk'* "to drink" (*ba'ax taak a uk'ul?* "what do you want to drink?"). *sakilja* is *sak ha'* "white water," an attested Maya **ceremonial** drink, and inscriptions record *sak juy ch'ich'* "white stirred blood," a beverage built on the blood metaphor. So *sakilja* + *itsok* + *uch'* is one coherent, sourced image: the sacred white draught that is blood, drunk at the altar.

(Text 3's tokens are folded into the combined lexicon, §9; its soft parses and open items into the appendix, §10.4 and §10.7.)





7. TEXT 4: THE DRILL SERGEANT & THE VAAL REGIMENT[1]

Captured live from in-game global chat (not yet in the datamined pool). A call-and-response drill between a Drill Sergeant and the Vaal Regiment, the military cousin of the Cuachic Vault litany (§5): a fixed call met by a fixed war-cry. The sergeant's name randomizes across instances (Tizoc, Cotan, Axilo), the same variant-name device seen with the Commander (Quemalani, Xolotl); all are Nahuatl-style officer names. Only the green NPC lines are Vaal; the player-character's English banter and "Error:" UI lines are excluded. The fuller catechism below, with its second response Atziri!, was filled in from additional information provided by Substantial_Bat_8440; an earlier capture had only the opening and the variant calls.

THE DRILL

A call-and-response catechism. The sergeant calls; the regiment answers with one of two shouts, the war-cry U'te mucane! "Behold, the mighty!" or the queen's name, Atziri! The sergeant's name varies per instance (Tizoc, Cotan, Axilo); the lines do not.

SPEAKER	VAAL	ENGLISH
Sergeant	Otsuks*! Tzokan'te u'te ik'el...	Company! Behold the last spirit...
Regiment	Atziri!	Atziri!
Sergeant	A'te Kuxkal tlayeb kutsen. Ela tlayeb ukto*! Máax ka ti a'tul*?	Behold the living, the offering in the dark. The dark torch burns! Who goes there?
Regiment	Atziri!	Atziri!
Sergeant	Otsuks*! Máax ka ti a'tul*?	Company! Who goes there?
Regiment	U'te mucane!	Behold, the mighty!
Sergeant	Máax ka ti cheyel*?	Who are you to stand watch here?
Regiment	U'te mucane!	Behold, the mighty!
Sergeant	Máax tlayeb mucane?	Who is the mighty of the dark?
Regiment	Atziri!	Atziri!

Other one-off sergeant calls, captured outside the catechism sequence:

VAAL	ENGLISH
Otsuks*, quxzeh!	Company, bite! Be fierce!
Axba*!?! Kíibsa* ta' en!	What!? Kill, before me!

*Tokens marked * are soft or open parses, not admitted to the committed lexicon (§9); each is analysed in §10.*

LINE NOTES

- **A call-and-response catechism.** The sergeant calls; the regiment answers with either the war-cry *U'te mucane!* "Behold, the mighty!" or the queen's name *Atziri!* It is the military cousin of the Cuachic Vault litany (§5), a fixed liturgy in a drill register.

- **Otsuks** "Company! Fall in!": Nahuatl presentative *o* "behold!"[34] + Yucatec *tsuk* "cluster, company"[31][36]; the fixed opener, with only the *-s* coda unresolved.
- **The catechism runs on committed vocabulary.** *Tzokan'te u'te ik'el* "behold the last spirit" (cf. Text 2's *Tzokan'te ik'el* "the last spirit"); *A'te Kuxkal tlayeb kutsen* "behold the living, the offering in the dark" (the same frame as Text 3's *A'te 'Ibil tlayeb kutsen*); *Máax tlayeb mucane?* "who is the mighty of the dark?", answered *Atziri!* All of *Máax, ti, U'te / A'te, ik'el, Kuxkal, tlayeb, kutsen, mucane, Tzokan'te* are in §9.
- **Ela** = Maya *el / el* "to burn, blaze" (the root of the committed *elba* "burn out"), with the regular *el* → *ela* stylization; now in §9. **ukto** reads as Nahuatl *ocotl / ocutl* "pine torch, resinous firebrand" (root *oco-/ocu-* "pine"); the torch that burns in the dark, which fits *Ela* "burns" far better than the earlier "drink" guesses. The colonial sources already spell the variant *ocutl* with /u/, and the Florentine Codex pairs it directly with the verb "to burn"[52]. The final *-tl* cluster reducing to *-to* is a stylization, so it stays soft (§10.7).
- The soft tokens (*a'tul* = Set A *a-* "you" + *taal* "come"; *cheyel* = Nah. *chiya* "to watch/wait" + *-el*; *Axba* = *ba'ax* "what" by metathesis; *quxzeh* = *k'ux* "bite/be fierce"; *Kiibsa'* = *kíim* "die" + causative; plus *ka* and *ta'*) are analysed token-by-token in §10.9; none was promoted to §9 until the closure pass (§10.11), which moved *quxzeh*'s root *k'ux* across.
- One further line, cut off at a screenshot edge, ends "...*ta' nuk!*"; *ta'nuk* "great one" is in §9, but the full line isn't legible and is left unrecorded.





8. TEXT 5: STRAY CAPTURES

A log of single Vaal lines caught in footage that do not yet form a text of their own. Each is parsed against the committed lexicon; any new root it establishes is folded into §9.

LINE 1: THE KAMASAN SMITH (XOMATL, ZANTICO)[37]

A fire-and-forge boss in Atziri's Temple at Lira Vaal, encountered at the height of Vaal power, before Atziri's communion with the Beast and with no knowledge of the coming Cataclysm. He works at the forge (molten metal, heat, hammer-strikes) until the player enters the room, then speaks this line and attacks. The boss appears under randomized names, captured so far as Xomatl and Zantico, the same variant-name device seen with the Drill Sergeant (Tizoc / Cotan / Axilo) and the Commander (Quemalani / Xolotl); the two captures below are the same speaker.

SPEAKER	VAAL	ENGLISH
Xomatl	Ti ek tala jare'yantul!	Into the dark you come; and so, your waning!
Zantico	Ti' ek'le upulché*	Into the dark, cast down!

*Tokens marked * are soft or open parses, not admitted to the committed lexicon (§9); each is analysed in §10.*

Line notes

- Every token is committed vocabulary: *Ti* "to / for," *ek* = *ek'* "dark, black" (the death-bat Smith takes the shadow sense of the root; the same root also means "star," Atziri as hailed in Text 1, kept here as the alternate), *tala* = *taal* "to come," *jare'* "and so," *yan* "there is / here stands," *tul* "decline, wane" (§9, the same Text 1 sense). Both Kamasan Smith lines open *Ti ek'* "into the dark," now glossed identically.
- It is a threat, not Vaal foreboding. The smith is not lamenting the empire's fall, he has no knowledge of the Cataclysm and stands at the height of Vaal power; he is promising the *intruder's* end. The player has come into the star's domain (Atziri's temple), and the smith answers, *jare' yan tul* "and so, your waning," turning from the forge to fight. The line reads literally "and so there is the waning"; the "your" is supplied from context, the doom is aimed at the one who has just come.
- *jare'* here is the **apostrophe form** the lexicon had flagged as undocumented (§10.2); this capture attests it in-game.
- The one soft point is syntax, not vocabulary: *jare'yantul* is written solid but reads as *jare' + yan + tul*. The words are committed; the clause-internal order is provisional.
- **Line 2 (Zantico).** *Ti' ek'le upulché* repeats the opener *Ti ek'* "into the dark" (*ek'* + the Yucatec determiner *le* "the"), then *upulché* = *u-* (3rd person) + *pul* "to throw, cast, hurl" + the *-ché* tail. The working gloss is "into the dark, cast down" (the wordlist gives *pul kaabal* literally for "cast down"). Because *u-* is third person the literal is "it casts / its casting," the object unexpressed, with the intruder the contextual target. The *-ché* tail is the soft point, readable as *che'* "wood, stave" but possibly lexicalized, so *upulché* is logged soft (§10.7); the new root *pul* "to throw, cast" is committed (§9).
- **A second variant name.** *Zantico* is the same boss as *Xomatl* under a randomized name, confirming the variant-name device once more; the two lines share the *Ti ek'* opener and the casting register. The "casting

out" theme also links to the Commander's *Eche lu nochbe* (§10.4), but realized there with Spanish *echar* rather than Yucatec *pul*, the layered-loan behavior seen throughout.



LINE 2: AHUATOTLI, THE BLIND[44]

A Vaal boss (the Delve boss "Ahuatotli, the Blind"). The shouts are Nahuatl-flavoured ability-cries more than sentences, so most tokens stay soft, but each is given a candidate root below rather than left unparsed.

WHEN	VAAL	TOKEN-BY-TOKEN (CANDIDATE ROOTS, SOFT)
start	tatlat Atziri	<i>tatlat</i> : Nah. <i>tlatla</i> "to burn" (redup./stylised) or <i>tlahtoa</i> "to speak, proclaim"[10]; <i>Atziri</i> (committed)
start	Vaal a xomaplat	<i>Vaal</i> (committed); <i>a</i> : poss. Maya Set A <i>a</i> - "your"[7]; <i>xoma</i> :- cf. the name <i>Xomatl</i> (Text 5); <i>-plat</i> : poss. Nah. <i>tlapal</i> - "colour, strength"[10]; open
start	xictep cutlotl	<i>xi</i> :- Nah. imperative "do!"[10]; <i>-tep</i> : poss. Nah. <i>tepē</i> - "hill" / locative <i>-tepec</i> [10]; <i>cutlotl</i> : a Nah. <i>-tl</i> noun, poss. <i>cuitlatl</i> "residue"[10]; open
during	azcado	poss. Nah. <i>āzcatl</i> "ant"[10]; the <i>-do</i> /d/ would mark a Spanish overlay; open
during	huatat	poss. Nah. <i>cuauh</i> - "tree, eagle" (<i>hua</i> - ← <i>cua</i> -)[10] + redup. <i>-tat</i> ; open
during	quiquate	poss. Nah. <i>qui</i> - (3sg object) + <i>cua</i> "to eat" (redup.) → "devours it"[10]; open
during	zahua moti	<i>zahua</i> : Nah. <i>zāhua</i> "to fast" or <i>zāhuatl</i> "pox, scab"[10]; <i>moti</i> : <i>mo</i> - reflexive + open

LINE 3: MEKTUL[44]

A Vaal boss. Unlike Ahuatotli's, several of Mektul's shouts carry recognisable Yucatec roots, fire, wind, gold, and a request-verb, though the full phrases stay soft.

WHEN	VAAL	TOKEN-BY-TOKEN (CANDIDATE ROOTS)
during	táak'iin	<i>táak'in</i> "gold, money" (Yucatec)[36], attested
during	kahk-tche	<i>kahk</i> = <i>k'áak'</i> "fire" + <i>tche</i> = <i>che'</i> "tree, wood"[7][36], both now committed in §9 → "fire-wood, firebrand"; closed
during	ka'tse-ik	<i>ka'</i> "two, again"[36] + <i>tse</i> (open) + <i>ik</i> = <i>ik'</i> "wind, spirit"[7]; soft
during	ipkaat	<i>-kaat</i> = <i>k'áat</i> "to ask, want"[36]; <i>ip</i> - open
during	koriek	the <i>r</i> marks the K'iche' seam (cf. <i>jare</i> , §10.2); re-cut past <i>kor</i> :- leading candidate <i>ko</i> - "strong, hard" + inchoative <i>-(i)rik</i> (cf. <i>kowirik</i> "to become strong")[47], a "harden / grow strong" cry; soft, see §10.11
during	xecwa	<i>-wa</i> : poss. <i>waaj</i> "bread, offering"[7] or <i>wáa</i> "or, if"[36]; <i>xec</i> - open

Line notes (2 and 3). These are boss combat-cries, mostly skill-names, so every token above is given a candidate root, but most stay soft rather than committed. The split runs clean along the corpus's seams: Ahuatotli's tokens are Nahuatl (the *xi*- imperative, *-tl* nouns, candidate roots *tlatla* "burn," *cua* "eat," *āzcatl* "ant"), while Mektul's are core Yucatec (*k'áak'* "fire," *che'* "wood," *ik'* "wind/spirit," *k'áat* "ask," and the attested *táak'in* "gold"), with one *r* (*koriek*) flagging the K'iche' seam. None is promoted to §9. Both speakers are gathered with the other figures in §12.





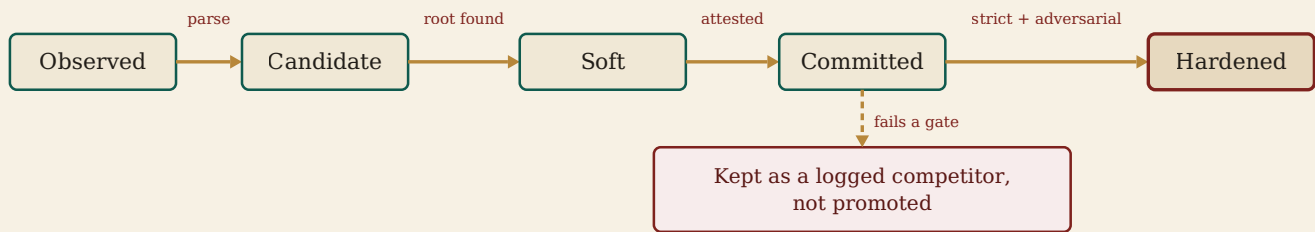
9. COMBINED LEXICON

Confidence key (revised after the statistical hardening pass, §17). The tier now reflects how a token behaves under adversarial and strict-latitude testing (the *lexicon hardening protocol*, `HARDENING_PROTOCOL.md`), not merely whether a root exists:

- **H** hardened (top tier): the root reconstructs the surface under the *strict* latitude (no cross-grafts, no free vowel shifts, no metathesis, no ad hoc coda drops) AND survives a blind adversarial pass (no equal-or-stronger competitor in a different language). These are the readings that clear the noise floor established in §17.
- **C** committed but latitude-dependent: an attested root with sound phonology under the standard (looser) latitude, but it either fails the strict test or has not yet been run through both gates ("hardening pending"). Real, but standing on permissive rules.
- **L** lore-anchored (fits the narrative); an orthogonal tag combined as **H+L**, **C+L**.
- A star (*) on a C entry flags a logged cross-language competitor from the adversarial pass (see the Notes column and §10).

Tokens with a plausible-to-verified root but one unresolved derivational step are held in §10.4 as **soft candidates** (S), which the testing showed are statistically close to noise and are therefore not evidence on their own. Standalone confirmed Vaal names (*pochiti*, *Eztli Pilli*, *Guatelitzi*) share this lexicon. Superscripts in the **Source** column link to §18; the hardening methodology and results are in §17.

The five confidence stages, and how a token moves between them, are shown below; promotion is always earned by passing a test, never granted by default.



Promotion is earned by test, not granted by default. Tiers in section 9, gates in section 17.3.

TOKEN	LAYER	SOURCE	GLOSS	CONF.	NOTES
Aiokmo	Nahuatl	Nah. <i>aocmo</i> [10]	no more, no longer	C	Text 1 form; cf. Text 3 ' <i>Ayok</i> .
akal	Maya	Maya <i>akal</i> "pond"[7]	still / eternal waters	H	Shadow sense: the nightmare realm.
anab	Maya	Classic Maya <i>anaab</i> [7]	court-servant, attendant	C	Attested courtier title; replaced an earlier Taíno guess.
ascensionada / ascenada	Spanish	Rom. <i>ascensión</i> + <i>-ada</i> [21]	ascended, risen	H	First-identified Romance loan (others: <i>xeŋe</i> , <i>u'tra</i> , <i>échelo</i> , <i>fukuur</i>).

TOKEN	LAYER	SOURCE	GLOSS	CONF.	NOTES
A'te / U'Te / Yatle	Maya	Maya <i>at</i> / <i>yan</i> presentative[7]	behold / here is / there stands	C+L*	Latitude-dependent; adversarial competitor: Nah. <i>ahtle</i> "nothing" (form-close, opposite sense; context favors the presentative). The offering gesture. The <i>u</i> of <i>U'Te</i> is pronounced /o/ (Text 2 audio).
atla	Nahuatl	Nah. <i>atl</i> "water" (+ <i>-tlan</i> "in/at")[10]	the water / upon the water	C	
Atziri	Nahuatl	Nah. <i>ātl</i> "water" + rev. <i>-tzin</i> [10,13]	the queen (name)	C+L	Water / reflection / vanity motif.
'Ayok	Nahuatl	Nah. <i>aocmo</i> / <i>ayoc</i> [10]	no more, no longer	C	Same root as <i>Aiokmo</i> .
cha'tsoke	Maya	Maya <i>ka'a</i> "again" + <i>ts'ook</i> "end"[7]	the ending once more	C	
che'	Maya	Maya <i>che'</i> "tree, wood"[7][36]	tree, wood	H	In <i>kahk-tche</i> (Text 5, Mektul) = <i>k'áak' che'</i> "fire-wood, firebrand."
chikula'	Maya	Maya <i>chiikul</i> [7]	sign, omen	L	
ek	Maya	Maya <i>ek'</i> [7]	star; also black / dark	H	Hardened (§17): reconstructs under strict latitude (<i>ek'</i> , glottal-optional) and survives the adversarial pass. The Kamasan Smith (Text 5) takes the "dark" sense (<i>Ti ek'</i> "into the dark"); Text 1's hail of Atziri takes "star."
Ela	Maya	Maya <i>el</i> / <i>elel</i> "to burn"[7][36]	burns, it burns	C	Text 4 catechism; same root as <i>elba</i> ; regular <i>el</i> → <i>ela</i> stylization.
elba	Maya	Maya <i>el</i> + <i>ba'</i> "self"[7]	burn out, extinguish	C	
-en	Maya	Maya <i>-en</i> / <i>teen</i> [7]	me, I	H	In <i>ta'en</i> (Text 3, speech 1).
Eztli Pilli	Nahuatl	Nah. <i>eztli</i> "blood" + <i>pilli</i> "noble, prince"[10]	"the Blood Prince" (a forbidden Vaal tome)	H+L	Hardened (§17): <i>eztli</i> reconstructs strictly (5/5 blind strict decodes) and survives the adversarial pass. Standalone tome name[6]; full roots, no stylization; attested collocation (<i>cualli eztli</i> "good blood"). See §10.6.
Guatelitzi	onomastic	(name)[38]	the Architect of Flesh (a Vaal architect of the Temple of Atzoatl)	C+L	Standalone name, from Zelina's " <i>Tome of Guatelitzi</i> "; the Vaal flesh-and-immortality architect whose rooms culminate in the <i>Sanctum of Immortality</i> (" <i>Flesh rending for life unending</i> "). Nahuatl-flavoured; <i>-tzi</i> from reverential <i>-tzin</i> . See §10.8.
Gyan'uks	Nahuatl	Nah. <i>yancuic</i> "new"[10]	the new ones → "(your) new children"	L	The written <i>g</i> is silent (Text 2 audio); the onset is /j/, the <i>ya-</i> of <i>yancuic</i> , which fits Nahuatl (it has /j/, no /g/). See the orthography note in §5.
'Ibil	Maya	Maya <i>il</i> "see" + <i>-bil</i> [7]	the seen → the flesh (laid bare)	C+L	Canonical game form ' <i>Ibil</i> (per <i>VaalSermon_06</i>); an earlier draft wrote <i>Íbil</i> .
ich	Maya	Maya <i>ich(il)</i> [7]	in, within	H	"within the altar."

TOKEN	LAYER	SOURCE	GLOSS	CONF.	NOTES
i'chian	Nahuatl	Nah. <i>ichan</i> "his/her home" (<i>chantli</i> "home" + <i>i-</i>)[10,22]	the dwelling, her house	C	<i>nídach ichan</i> "far within her house", literally Atziri's royal temple at Lira Vaal, where the Commander stands. The root <i>chantli</i> survives in modern Mexican/Central-American Spanish <i>chante</i> "house, home" (see §13).
ik'bala	Maya	Maya <i>ik'</i> + <i>ba'</i> [7]	spirit-self, soul	C	
ikba'yucane	Maya	Maya <i>ik'</i> + <i>ba'</i> + <i>-ane</i> hon.[7]	the deathless-spirited ones	C	
Ik'eche	Maya	Maya <i>ik'</i> "spirit/breath" + <i>-ech</i> "you" + <i>-e</i> voc.[7]	"O spirit / thou art breath"	C	
ik'el	Maya	Maya <i>ik'</i> + <i>-el</i> [7]	the spirit, the unseen	C+L	
itsok / itzil	Maya	Maya <i>iitz</i> "sap, essence"[7]	blood, essence	C+L*	Latitude-dependent; strong adversarial competitor: Nah. <i>itztli</i> "obsidian, sacrificial blade" (thematically apt in a sacrifice corpus, clean form). Logged as a live co-reading in §10.12. <i>le'itzil</i> = "the essence."
Ixchel	onomastic	Yucatec <i>Ix-</i> (fem./agentive) + <i>Chel</i> "rainbow" (the goddess Ix Chel)[56]	the Godstealer (a Vaal citizen; later the Trialmaster)	H+L	Attested Maya theonym reused for a Vaal figure; a Yucatec (not Nahuatl) name. See §12.6. Hardened per HARDENING_PROTOCOL.md (Gate 1 pass: <i>Ix-</i> + <i>Chel</i> reconstructs strictly, no residue; Gate 2 pass: no cross-language competitor).
jare	K'iche'	K'iche' <i>are'</i> "he, she, it; it is, that is" (focus) / <i>ri</i> "the"[47]	connective: "thus, and so"	C	The /r/ marks Highland Maya; grammatical glue, not a content word. Now confirmed in a dedicated K'iche' dictionary[47] (Christenson lists <i>are'</i> as the focus pronoun and <i>ri</i> as the article), upgrading the earlier phonology-only note[17]. The apostrophe form <i>jare'</i> is attested in-game (Text 5). See §10.2.
k'áak' (kahk)	Maya	Maya <i>k'áak'</i> "fire"[7][36]	fire	H	Mektul's cry <i>kahk-tche</i> (Text 5); distinct from <i>ek'</i> "star, dark."
Kextal	Maya	Maya <i>k'ex</i> "substitution → transform"[7]	transformation, ritual renewal	C	Rendered "transform us."
ki' (Kí')	Maya	Maya <i>ki'</i> [7]	good, delicious, sweet	H	A blood-drinker's relish: "Sweet!"
kifba	Maya	Maya <i>k'i'ik'</i> "blood" + <i>ba'</i> [7]	the heart, lifeblood	C+L	Distinct from <i>itsok's iitz</i> .
kíimil	Maya	Maya <i>kíimil</i> [7]	death, the dead	L	

TOKEN	LAYER	SOURCE	GLOSS	CONF.	NOTES
kilya	Nahuatl	Nah. <i>quil-</i> "green, verdant" + inchoative <i>-ya</i> "to come to be"[49]; in Text 1 fronted by optative <i>mā</i> (see <i>ma</i>)	evergreen, everlasting (contextual "undying")	C+L	Re-analysis. The earlier <i>ma</i> "not" + <i>kīim</i> "death" parse fails: the surface holds no <i>kīim</i> (<i>kīim</i> ≠ <i>ki</i>) and it double-counted the separate word <i>ma</i> . Here <i>quil</i> + <i>-ya</i> segments exactly, both pieces attested (<i>quiltic</i> "green"[49]; the inchoative <i>-ya</i> of <i>hueyiya</i> "to grow"[46]), so the root and phonology are committed (C) and the step from "green / evergreen" to "undying" is the lore-anchored gloss (L). <i>mā kilya Zerphi</i> = "may you be evergreen, like Zerphi." Cf. <i>celiya</i> "to sprout, turn green again, revive"[49], the right sense but a /s/-onset near-miss.
ko'janti	Maya	Maya <i>ko'</i> "come" + <i>han-</i> "eat" + <i>-ti</i> [7]	come, devour	L	Addressed to Atziri. The <i>-ti</i> here is best read as the Yucatec relational <i>ti'</i> (contrast the Nah. inchoative <i>-ti</i> in <i>pochiti</i> , §10.5).
kujkuali	Nahuatl	Nah. <i>cualli</i> [10]	good, worthy	H+L	
kutsen	Maya	Maya <i>kutz</i> "sacrificial bird"[7]	the offering	C+L*	Latitude-dependent; adversarial competitor: K'iche' <i>kotz'i'j</i> "flower, candle" (also an offered thing).
kux / kuxkal / kuxte'	Maya	Maya <i>kuxtal</i> [7]	life, the living	L	
k'ux (quxzeh)	Maya	Maya <i>k'ux</i> "to bite, gnaw; rancor"[7][36]	bite!, be fierce	H	Ejective /k'/ (corpus <i>q</i>), distinct from <i>kux</i> "life"; in <i>quxzeh</i> (Text 4) the <i>-zeh</i> coda is a non-morphemic tail (cf. <i>Otsuks -s</i>). See §10.11.
le / le'	Maya	Yucatec article <i>le...</i> <i>o</i> '[7]	the	L	
líimek	Maya	Maya <i>lu'um</i> "earth"[7]	those in the earth, the dead	L	Replaced an earlier Tagalog guess.
ma	Maya + Nahuatl	Nah. <i>mā</i> / Maya <i>ma</i> '[7,10]	optative "may"; negation "not"	H	
máax	Maya	Yucatec <i>máax</i> "who?"[7]	who (interrogative)	H	Hardened (§17); reconstructs strictly (5/5 blind) and survives the adversarial pass. New in Text 4 (§7); ruled out a "crush" reading in §10.4.
Ma'oxe	Maya	Maya <i>ma'</i> "without" + <i>xok</i> "count"[7]	the countless / numberless spirits	L	"Without-count" → innumerable; Atziri's faithful. See §10.1.
mucane	Maya	Maya <i>muk'</i> "strength" + <i>-ane</i> [7]	the mighty, the enduring	C*	Latitude-dependent; adversarial competitor: K'iche' <i>muq</i> "to bury, hide" (partly already noted as the shadow sense <i>muk</i> "bury").
mujuk'	Maya	Maya <i>muk'</i> "strength, force"[7]	strength, might	C	Confirmed by " <i>u mujuk' le mucane</i> " = "the might of the mighty."
náach (niáach)	Maya	Maya <i>náach</i> [7]	far, distant	H	Hardened (§17); reconstructs strictly (5/5 blind) and survives the adversarial pass.
nochira	Nahuatl	Nah. <i>nochi</i> / <i>mochi</i> [10]	all, everything	C	

TOKEN	LAYER	SOURCE	GLOSS	CONF.	NOTES
Panquetzaliztli	Nahuatl	Nah. <i>pan(tli)</i> "banner" + <i>quetza</i> "raise" + <i>-liztli</i> [10] [55]	"the raising of banners" (15th veintena; a Vaal mace)	H+L	Attested Aztec war-festival (Huitzilopochtli); fully Nahuatl, no stylization. See §12.5. Hardened per HARDENING_PROTOCOL.md (Gate 1 pass: exact attested compound, no residue; Gate 2 pass).
pochiti	Maya + Nahuatl	Maya <i>poch</i> "hungry, gluttonous"[8] + Nah. inchoative - <i>ti</i> [14]	the Hungering One (Atziri's name for the Beast)	C+L*	Latitude-dependent (cross-graft); adversarial competitor: Nawat <i>puchini</i> "it bursts/frays"; lore still favors <i>poch</i> . Standalone word[6]; <i>ch</i> = /tʃ/; a <i>pōchōtl</i> "ceiba" co-reading is logged in §10.5.
pul / puul	Maya	Maya <i>pul</i> / <i>puul</i> "to throw, cast, hurl"[36]	throw, cast, cast down	H	Hardened (§17): Gate 1 pass (Yucatec <i>pul/puul</i> reconstructs strictly, no residue), Gate 2 survives (K'iche' <i>pul-</i> "overflow" is a same-family echo, not a genuine competitor). Wordlist <i>puul</i> = <i>arrojar</i> , <i>echar</i> , <i>lanzar</i> , <i>tirar</i> ; <i>pul kaabal</i> "cast down." In <i>upulché</i> (Text 5, Zantico) = <i>u-pul-ché</i> "cast down" (the <i>-ché</i> tail soft, §10.7).
qexcan	Maya + Nahuatl	Maya <i>k'ex</i> "transform" + <i>-can</i> (Nah. locative "place of")[7,10]	place of transformation	C+L	The temple as crucible. A Maya root + Nahuatl suffix cross-graft.
Quecholli	Nahuatl	Nah. <i>quecholli</i> (roseate spoonbill; the 14th veintena) [10][54]	precious-feather bird / the weapon-month (a Vaal mace)	H+L	Attested Aztec month of weapon-making (Mixcoatl); fully Nahuatl. See §12.5. Hardened per HARDENING_PROTOCOL.md (Gate 1 pass: exact attested lexeme, no residue; Gate 2 pass).
sakilja	Maya	Maya <i>sak</i> "white, pure" + <i>ha'</i> "water"[7]	the sacred white draught	C+L	Cf. ceremonial <i>sak ha'</i> ; blood-metaphor <i>sak juy ch'ich'</i> "white stirred blood."
ta'nuk	Maya	Maya <i>nuk</i> / <i>nojoch</i> "big, great"[7]	the great one	C	<i>nuk</i> committed; <i>ta'</i> - element soft.
tala	Maya	Maya <i>taal</i> "to come"[7][36]	come, comes	C*	Latitude-dependent; adversarial competitor: Nawat/Nah. <i>ta:l</i> "earth, land" (form-identical, different sense; context favors "come"). Free form of the verb (cf. bound <i>a'nul</i> , §10.9). Text 5.
te	Maya	Maya <i>ti'</i> / <i>te'</i> [7]	of / to (relational)	C*	Latitude-dependent; adversarial competitor: Nah. <i>teŋ</i> "stone" (content word vs the relational; syntax favors the relational).
Teoyuxtlane	Nahuatl	Nah. <i>teōtl</i> "god" + <i>Yux</i> (Yutsal) + <i>-tlān</i> "place" + voc. - <i>e</i> [10,13]	place of divine Utzaal	C	
Ti	Maya	Maya <i>ti'</i> [7]	to / for you (dative)	H	Fixes Atziri as recipient.
til	Maya	Maya <i>til</i> [7]	to burn, kindle	C	Death-as-fire.
tlapēc / yorlapēk	Nahuatl	Nah. <i>tlapechtli</i> "platform, scaffold, altar-bed"[10]	the (sacrificial) altar-platform	C+L	<i>yo-</i> prefix unexplained (poss. Nah. <i>yōl-</i> "heart"?).
Tlaxye'	Nahuatl	Nah. <i>tlāl-</i> / <i>tlācah</i> [10]	land, people	L	Replaced an earlier Quechua guess.

TOKEN	LAYER	SOURCE	GLOSS	CONF.	NOTES
tlayeb	Nahuatl	Nah. <i>tlā-</i> + <i>tlayohua</i> "night"[10]	the (sacred) dark	C+L	
tul	Maya	Maya <i>tul</i> "decline"[7]	wane, dwindle	H	Decay-as-fire.
Tzokan'te / tzok	Maya	Maya <i>ts'ook</i> [7]	end, the last	L	
u	Maya	Maya <i>u-</i> (Set A 3rd person)[7]	his, her, its; the	H	<i>u mujuk'</i> "his/the might."
uch' / pu uch'	Maya	Maya <i>uk'</i> "to drink", or <i>puuch'</i> "to crush"[7]	drink / crush	H	Ambiguous: <i>uk'</i> fits the draught theme; <i>puuch'</i> fits his "Crush!" barks.
u'tra	Spanish	Rom. Spanish <i>otra</i> [21]	another	C	One word (not <i>u</i> + <i>tra</i>); the /r/ flags the Romance loan.
Vaal	onomastic	(name)[5]	the people, the empire	C+L	
waaj (waja)	Maya	Maya <i>waaj</i> [7]	bread; altar-offering	H+L	Food offered on Janal Pixán altars, i.e. an offering.
Xatlene	Nahuatl	Nah. <i>xōtla</i> "to kindle, blaze, glow"[10]	kindle! / the kindler	L	Text 2; <i>x</i> = /ʃ/, <i>-ne</i> epithet-former (cf. <i>mucane</i> , <i>yuquia</i>); root verified, one soft step (the <i>o</i> → <i>a</i> vowel).
xefe	Spanish	Rom. Spanish <i>jefe</i> (archaic <i>xefe</i>)[21]	chief, commander	H	Romance loan; the in-game title is <i>Royal Commander</i> .
xi	Nahuatl	Nah. <i>xi</i> -[10]	imperative "do! / make!"	H	
xu'te	Maya	Maya <i>xul</i> "end" + <i>te</i> '[7]	the end	C	
yax / Yaxe	Maya	Maya <i>yax</i> [7]	first, new, green	L	
yuquia	Nahuatl	Nah. <i>yocoya</i> "create, devise"[10]	the made, the wrought	L	Cf. <i>Moyocoyatzin</i> "self-creator." Replaced an earlier Quechua guess. The <i>u</i> is pronounced /o/ (Text 2 audio), the expected Nahuatl reflex (no /u/; §2.3).
Yutsal / yutsal	Maya	Yucatec <i>γ-</i> (3 poss) + <i>uts</i> "good" + <i>-al</i> ; attested as <i>yutsal</i> "(its) goodness, all-good" (Cordemex) [51]	the good (place); Utzaal	C+L	Root <i>uts</i> "good" attested (Cordemex <i>uts</i> "cosa buena"; modern <i>uts</i> "bien," <i>utsil</i> "goodness") [51]. Lowercase <i>yutsal</i> is the common noun "the good, the good place" (translated as such in §5); capitalized <i>Yutsal</i> is the same root as the proper name of the city Utzaal (Doryani's seat, not the capital; kept in the §4 translation), so <i>Utzaal</i> = <i>utz</i> "good" + <i>-aal</i> "the good place." The <i>γ-</i> (Set A, pre-vocalic) is Yucatec-specific, K'iche' would use <i>r-</i> , though <i>utz</i> "good" is also basic K'iche'. Clipped to <i>Yux</i> in <i>Teoyuxtlane</i> .
Zerphi	onomastic	(name)[5]	unaging Vaal noble	C	





10. APPENDIX: RECORDED ALTERNATES & OPEN QUESTIONS

Every candidate considered, kept for future reference.

10.1 MA'OXE: MA' "WITHOUT" + A SECOND ROOT

PARSE	MEANING	STATUS
ma' + xok "count"	the countless / numberless spirits	★ chosen, cleanest phonetics (the <i>x</i> lands exactly) and best theme
ma' + óol "spirit, will"	the spent / will-less spirit	alt, strongest single root (óol is everyday Yucatec), but needs an <i>l→x</i> drift
ma' + óoxol "heat" (?)	the cold spirit	△ root óoxol unverified
máax + -e'	(would read "whose spirit?")	✗ máax = "who?", not "crush"
ma'ax / mix "none"	"not even a spirit"	△ the real "none" is <i>mix</i> ; conflated

10.2 JARE: ANOMALOUS /R/ MARKS A NON-YUCATEC/NAHUATL SOURCE

PARSE	MEANING	STATUS
K'iche' focus/ demonstrative particle (are' / ri family)	connective: "thus, and so"	★ chosen, explains the /r/ (K'iche' keeps it), fits the connective slot, no semantic stretch. The exact form <i>jare'</i> now appears in-game (Text 5, the Kamasan Smith), confirming the apostrophe form; the category, a K'iche' connective, remains the answer.
K'iche' <i>jarem</i> / <i>jarik</i>	decay, wearing-out	alt, phonologically fine, root unverified
Yucatec <i>jaal</i> + <i>l→r</i>	to the edge / limit	alt, real word, but the <i>l→r</i> swap is a stretch and leaves the /r/ unexplained
Nahuatl <i>xalli</i> (<i>x→j</i> , <i>l→r</i>)	dust, ash	△ low, requires two corruptions

Xinkan / Aztecan check. The /r/ is not Aztecan: Nahuatl and Nawat have no native /r/, and Nahuatl's demonstratives (*in* / *inin* / *inon*) carry none.[27] Xinka *does* have a tap /r/, but its attested demonstrative is the third-person *nah*, no *are'*-type form is on record.[26] So K'iche' *are'* / *ri* remains the one attested fit, and the /r/ still marks the Highland-Maya seam.

10.3 OTHER RESOLVED HOLDOUTS

These weak or cross-family guesses were replaced with attested Maya/Nahuatl roots, keeping the whole text inside its geographic palette:

TOKEN	REJECTED GUESS	ADOPTED ROOT
anab	Taíno <i>naboría</i>	Classic Maya <i>anaab</i> "courtier"
Tlaxye'	Quechua <i>llaqta</i>	Nah. <i>tlācah</i> "people"
líimek	Tagalog <i>limot</i>	Maya <i>lu'um</i> "earth"

TOKEN	REJECTED GUESS	ADOPTED ROOT
yuquia	Quechua <i>yuya</i> "memory"	Nah. <i>yocoya</i> "create"

10.4 TEXT 3: PROPOSED IN-PALETTE PARSES (SOFT, NOT COMMITTED)

Why these sit here and not in §9. Membership in the committed lexicon turns on the *whole token's* security, not on whether a dictionary holds the root, and most roots below are in fact attested. What keeps each token here is one unresolved step in its derivation: an irregular sound-change outside the established correspondences (the *a→u* of *buxa*, the *o→u* of *Eche lu* and *puxe*), a truncation (*inib*), a segmentation reasoned out rather than looked up (*Donuks*, after no Vaal name matched), imperfect phonetics (*fukuur*, *soxsal*), or two live candidates (*te'moxti*, *daka*, *pu*). "Strong" therefore means "verified root, best-available parse," not "settled." And unlike the committed tokens, which map by regular rule and are often cross-confirmed by recurrence or the game files, these appear once, in Text 3, with nothing yet to check them against. A cleaner phonological account or a second attestation would graduate any of them; *nochbe* (built only from the committed *nochi* + *ba'*) and *inib* (*in* + the committed *'Ibil*) are nearest the line.

A lore check found no Vaal name matching *Donuks* or *puyao* (checked against Kamasa, Kopec, Arakaali, Kishara, Xibaqua, Tetzlapokal, Ketzuli, Napuatzi, Doryani, Zolin/Zelina), so these are decomposed rather than looked up. Notably, three resolve toward *the drink*, reinforcing the rite as a communal draught.

TOKEN	PROPOSED PARSE	READING	STRENGTH
Donuks	Maya <i>to'on</i> "we/us" + <i>uk'</i> "to drink"	"we who drink / our draught"	strong, both roots solid; <i>-uks</i> coda matches <i>Gyan'uks</i> ; the initial <i>t→d</i> now read as post-vocalic voicing after <i>atla</i> (proposed environment), consistent with the commander's Spanish-influenced dialect
puxe	Maya <i>pox</i> ("posh"), ceremonial corn/cane liquor (Highland Maya)[7]	the sacred drink	strong, fits the K'iche'/Highland seam and the drink theme
inib (Kí' inib)	Maya <i>in</i> "my" + <i>'Ibil</i> "flesh"	"my flesh" → "Sweet, my flesh!"	strong, both roots already verified; minor <i>-ibil→-ib</i> contraction
nochbe (Eche lu nochbe)	Nah. <i>nochi</i> "all" + Maya <i>ba'</i> "self"	"all of ourselves"	strong, both roots verified; better theme than <i>noh beh</i> "road" (now an alt)
buxa (u'tra buxa)	Maya <i>ba'ax</i> "what, what-thing" (interrogative; ends in <i>x = /ʃ/</i>)[7]	"another such thing / another one", the commander dismissing the intruder	strong, verified root, preserves <i>x = /ʃ/</i> (unlike <i>búuk</i>), terminal-vowel stylization matches <i>Ma'oxe</i> / <i>puxe</i> ; soft point is the <i>a→u</i> vowel drift, now read as conditioned , the rounded /u/ of the preceding loan <i>u'tra</i> plausibly triggers the rounding (a proposed environment, not a free drift)

TOKEN	PROPOSED PARSE	READING	STRENGTH
Eche lu (Eche lu nochbe)	Spanish <i>échelo</i> = <i>eche</i> (formal imper. of <i>echar</i> "throw, cast, pour out") + <i>lo</i> "it"[21]	"pour it out / cast it out, all of ourselves!"	strong, verified root; the <i>ch</i> /tʃ/ is etymological (Lat. <i>iacitare</i> → <i>ch</i>), so it supplies the very sound <i>he'ela'</i> lacked; only drift is <i>o</i> → <i>u</i> ; fits the Commander's Spanish-martial seam and imperative register. In-palette alt: Maya <i>-ech</i> "you" + <i>le</i> "the" → "to you, the all of ourselves", sound, but leans on a bound suffix used freely plus a larger <i>e</i> → <i>u</i> shift. Note that <i>he'ela'</i> / <i>he'la'</i> "here it is, tenga" is itself attested in living Yucatec (Belize)[30], so it is a genuine alternative, but at two syllables and lacking /tʃ/ it still fits <i>Eche lu</i> less closely than <i>échelo</i>
daka	Spanish <i>daca</i> "give it here" (RAE <i>Autoridades</i> : a defective imperative, contraction of <i>da acá</i>) [20]	"give / bring [the drink]", imperative, fitting <i>Xi</i> ... <i>puxe</i>	source well-diagnosed, lexical choice soft, the /d/ is loan-only across the palette, so (like /f/) it marks a Romance loan, and the Commander's speech already carries <i>xefe</i> , <i>u'tra</i> ; <i>daca</i> is the best Spanish fit, though not the only conceivable imperative. The Maya alternative <i>taak</i> "to want" needs an unattested <i>d</i> → <i>t</i> respelling and reads weaker, the bigger leap, not the smaller one. Cross-checked against the zone's other families (§13): Aztec supplies the right <i>meaning</i> , <i>maca</i> / <i>maka</i> "to give," which would suit the <i>xi</i> -imperative, but its /m/ onset cannot yield the form <i>daka</i> ; Xinka has a native /d/ yet no attested "give" verb resembling <i>daka</i> . So <i>daca</i> keeps the form, now with its "give!" sense independently echoed by Nahuan <i>maca</i> [27]
fukuur	Romance loan, forced by form. Read form-first, not from any assumed meaning: /f/ and /r/ are both loan-only in every palette language, Yucatec, Classical Nahuatl, K'iche', and now Nawat (Campbell: all <i>f</i> -words tagged Spanish, /r/ only in loans)[53], so a token carrying both cannot be assembled from native roots in any of them. The attested Spanish form matching the skeleton f-u-[k/g]-u-r is the <i>fulg</i> - family, <i>fulgor</i> "radiance" / <i>fulgurar</i> "to flash" (Lat. <i>fulgur</i>)[21]	"the radiance / flash!" (the sense is read off the form, not assumed)	loan diagnosis firm (two loan-only phonemes co-occurring, the only such token in the corpus); the exact lexeme within the <i>fulg</i> - family stays soft. Fire imagery from <i>neighbouring</i> tokens (<i>A'te fukuur</i> !; the English bark "Flaming Glory!") is used only as context, not as the search key.
soxsal (ko'soxsal)	Nah. <i>xoxoc</i> - "raw, fresh, green" (<i>xoxoctic</i> / <i>xoxouhqui</i>)	"the raw/fresh [blood]!"	moderate, real root; <i>xoxoc</i> → <i>soxsal</i> imperfect (alt: <i>sáasal</i> "light"; <i>sotz'</i> "bat")
puyao / Na' puyao	Nah. <i>poyahua</i> "to darken, redden"	"the reddened (offering)"	soft, common noun, not a name (" <i>u'te puyao</i> " " <i>a'te itsok</i> ")
te'moxti	<i>te'</i> "of" + Nah. <i>mochi</i> "all" or Maya <i>mux</i> "grind, crush"	"of all" / "of the grinding"	moderate, two viable candidates
pu (pu uch')	proclitic intensifier, or part of <i>puuch'</i> "crush"	"drink deep" / "crush"	soft

Rejected on verification: *fukuur* ← Maya *púukul* "destruction" (root unattested + ad-hoc *p*→*f*, and no native source for the final /r/; both /f/ and /r/ are loan-only across the palette, so a Romance source is forced, not merely preferred). *puyao* ← Maya *puy* "slice" + *ha'* "water" (*puy* "cut" unattested, the real Yucatec cut-verbs are *xot*, *ch'ak*, *kup*).

10.5 POCHITI: ATZIRI'S NAME FOR THE BEAST

Not from Texts 1–3: a standalone Vaal word Atziri speaks in her PoE2 encounter.[6] She has no other Vaal lines there; this is her one word for the Beast, the primordial entity she communed with, triggering the Cataclysm. Two attested roots fit, and the lore supports both at once.

PARSE	READING	STATUS
Maya <i>poch</i> "gluttonous, hungry, craving" + Nah. inchoative <i>-ti</i> (ligature <i>-i-</i>)[8,14]	"the Hungering One / the Devourer"	★ lead, root verified to RAE level (<i>poch</i> 'goloso, hambriento'; Yucatec <i>poch</i> "antojo, deseo"); phonology exact (<i>ch</i> = /tʃ/); fits the Beast that Atziri sustains with sacrifice
Nah. <i>pōchōtl</i> "ceiba / silk-cotton tree"[11,12] (UNAM GDN: "métaphor., protecteur"; the <i>āhuēhuētl</i> + <i>pōchōtl</i> shade = a ruler's authority; the Maya <i>axis mundi</i> world-tree linking Xibalba, earth, sky)	"the sheltering tree / Great Protector"	○ co-reading, root verified, but it needs an unattested <i>-ōtl</i> → <i>-iti</i> step where <i>poch</i> + <i>-ti</i> segments cleanly; the lack of in-game support for a protector/"Mother" Beast (see note) is a secondary strike. Weaker than the <i>poch</i> reading on form first, and on lore second

Correction. An earlier version read these two roots as a deliberate duality, devourer *and* nurturing "Mother", on the claim that an Aggorat cult venerated the Beast as "Mother." That claim is unsupported: Aggorat is simply a Vaal city of altars and zealots, with nothing tying the Beast to a "Mother" cult, and the "mother / brood / family" theme in present-day Act 3 belongs to a separate boss, the Queen of Filth, whose fight spawns "Younglings" and who calls the monsters her children, not to the Beast. So *poch* "the Hungering One" is the reading that wins on form, it segments as *poch* + *-ti* with no unattested step, and the lore (the devourer Atziri feeds) corroborates it; *pōchōtl* "ceiba / protector-tree" survives only as a phonological alternative that needs the unattested *-ōtl* → *-iti* drift and, separately, lacks thematic support. Because in-game lore can be revised in a later patch while a morphological fit cannot, the form argument is treated as the load-bearing one and the lore as corroboration.

The *-iti* tail, worked out. Because the meaning is fixed here ("the one that is hunger"), the suffix's job is given and only its source was open, and it resolves cleanly inside the palette. It is Nahuatl *-ti*, the denominal inchoative "to become / be X" (attested: *tlācati* "to be born," *teti* "to harden into stone," *cuauhti* "to stiffen like wood"), joined by the standard Nahuatl ligature *-i-* (the same *-ti-* seen in *tlal-ti-cpac*, *Tenoch-ti-tlan*). So *poch* + *-i-* + *-ti* = "(that which) is / becomes hunger", a Maya root inflected with Nahuatl morphology, exactly the cross-graft already on record in *qexcan* (Maya *k'ex* + Nah. *-can*). The one soft step is reading the resulting stative as a personified name, which is how the conlang builds epithets throughout (*Xatlene* "the kindler," *yuquia* "the made"). Note the surface *-ti* also ends *ko'janti*, but there it is better read as the Yucatec relational *ti* "to / at", so the two are homophones, not one morpheme.

(Now carried in the §9 lexicon; the two readings and their lore basis are kept here.)

Clarity it gives elsewhere. *pochiti* completes a three-way minimal contrast of look-alike roots the orthography deliberately keeps apart: **poch** /potʃ/ "hunger" (*ch* = /tʃ/) vs **pox** /poʃ/ "the *posh* ceremonial drink" (*x* = /ʃ/, the soft reading of *puxe*) vs **puuch'** /pu:tʃ'/ "crush" (the *uch'* / *pu uch'* alternate). Three sounds, three meanings, all attested, independent confirmation that the conlang's *ch* / *x* / *ch'* spellings track real Maya phonemic distinctions, which retroactively supports the *x* = /ʃ/ rule used for *buxa*. The *pōchōtl* co-reading is itself corroborated by a living reflex: *pochote* (the ceiba) and colloquial *pochotón* "husky, thick-built" (see §13).

10.6 EZTLI PILLI: A VAAL TOME OF FORBIDDEN BLOOD-KNOWLEDGE

Like pochiti, a standalone Vaal name, not from Texts 1–3: the title of an in-game tome of forbidden, volatile knowledge, deemed too dangerous to use.[6] Pure Classical Nahuatl, and unusually un-stylised, two full roots with their absolutes intact.

- **eztli** "blood"[10], well attested (Sahagún, *Florentine Codex*). The *z* is Nahuatl /s/, not the alien /z/, so it stays inside the palette.
- **pilli** "noble, lord; prince; child"[10], as in *Xōchipilli* "flower prince," *Piltzintecuhli* "young prince."

So *Eztli Pilli* = "**Blood Noble / Blood Prince.**" The pairing is not merely plausible but **attested as a collocation**: Classical Nahuatl *huey pilli, cualli eztli* "a great noble of good blood." [10] A forbidden grimoire of Vaal blood-magic titled "the Blood Prince" sits squarely in the corpus's blood-and-lineage register. Confidence: roots **C** (verified, the collocation itself attested), reading **C+L**.

This is the cleanest non-text token yet, both roots full and unaltered, and a second Vaal "tome" reference beside Zelina's *Tome of Guatelitzi* (§10.8); unlike that one, *Eztli Pilli* parses without residue. (*Now carried in the §9 lexicon.*)

10.7 STILL OPEN (KEPT OPEN, NOT FORCED)

TOKEN	NOTE
ukto (Ela tlayeb ukto)	Soft, leading reading Nahuatl <i>ocotl</i> / <i>ocutl</i> "pine torch, resinous firebrand" (root <i>oco-</i> / <i>ocu-</i> "pine"; <i>ococuahuitl</i> "pine wood," <i>ocototon</i> "pine splinters") [52]: the light that burns in the dark, matching <i>Ela</i> "burns." The variant <i>ocutl</i> is attested with /u/ and paired with the burning verb in the Florentine Codex, so the corpus /u/ is the source vowel; the final <i>-tl</i> cluster reduces and reorders to <i>-to</i> , a stylization, so it stays soft. Displaces the earlier "drink" guesses (<i>uk'</i> , <i>octli</i>), which a burning torch rules out. Text 4 catechism. Living-Nawat corroboration, reached form-first (by following the <i>-uku-</i> shape, not the gloss): Campbell lists Pipil <i>uku-t</i> "pine, pine kindling, torch pine" (pl. <i>uhukut</i> / <i>ohokot</i>), from Proto-Nahua *oko-, CN <i>ocotl</i> , an independent reflex of the same torch-pine root in the geographically apt Central American branch. [53]
ta' (ta' en)	Maya <i>ti'</i> "to/for" or <i>taak</i> "to want" (needs <i>k</i> →glottal); "no more for me" / "I want no more." A third Yucatec homophone, <i>ta'</i> "excrement" [30], is attested but contextually impossible in <i>ta' en</i> / <i>ta'nuk</i> , so it is excluded
upulché (Ti' ek'le upulché)	<i>u-</i> (3rd person) + <i>pul</i> "to throw, cast" (§9) + <i>-ché</i> ; "into the dark, cast down" (cf. <i>pul kaabal</i> "cast down"). The root <i>pul</i> and the opener <i>Ti ek'</i> are committed; the <i>-ché</i> tail is soft, readable as <i>che'</i> "wood, stave" but possibly lexicalized, and with <i>u-</i> third person the literal is "it casts," the object unexpressed. Text 5, Zantico (the Kamasan Smith).

10.8 GUATELITZI: IDENTIFIED AS THE ARCHITECT OF FLESH

Zelina's enrage bark names a Vaal tome, "*Tome of Guatelitzi*." This was earlier logged as an unattested stem and reconstructed (below) as a *cuauh-* "tree" compound. **That reconstruction is now withdrawn: *Guatelitzi* is an attested in-game proper name.** *Guatelitzi, Architect of Flesh* is one of the Vaal Architects of the **Temple of Atzoatl** (PoE1 Incursion, v3.3.0; core v3.5.0). [38] His incursion-room line runs *Pools of Restoration, Sanctum of Vitality, Sanctum of Immortality* (room flavour: "*Flesh rending for life unending*"), and his signature item modifiers are all maximum Life, Energy Shield, and Life / Energy-Shield regeneration: an architect whose craft is flesh and unending life, an immortality experimenter. (Identification contributed by *Tenebris-Umbra*.)

This resolves the token cleanly and on theme. The PoE2 Vaal temple reuses Atzoatl's room-grid and lore, and the Vaal there take the player for the old "demon of Atzoatl" (§11). A Blood Priestess invoking the "*Tome of Guatelitzi*" is naming that flesh-and-immortality architect: a grimoire attributed to the Vaal who chased unending life through

flesh fits Zelina's blood-magic register exactly, and it pairs with the other named Vaal tome, *Eztili Pilli* "the Blood Prince" (§10.6). So *Guatelitzi* is **onomastic**, a known character name, not a common-noun phrase to be parsed; it is now carried as a confirmed name in §9.

The earlier etymological reconstruction (superseded, kept for the record). Before the lore identification above, the name was reverse-engineered from its phonology; that analysis is preserved here as the secondary account it has become. Like the other Atzoatl architects (*Ahuana*, *Atmohua*, *Cholotl*, *Citaqualotl*, *Estazunti*, *Hayoxi*, *Jiquani*, *Matatl*, *Opiloti*, *Paquate*), *Guatelitzi* is Nahuatl-flavoured and ends in the reverential *-tzi(n)* seen elsewhere in the corpus (*Atziri*, *Napuatzi*), so the *Guate-* onset was read as a hidden Nahuatl /k^waw/ root:

- **Guatemala from Nahuatl Cuauhtēmallān "place of many trees"**[25], *cuahui(tl)* "tree" + *tema* "to fill / abound" + *-tlan / -lan* "place of." The name is the Nahuatl calque of K'iche' *k'iche'* "many trees / forest," the macro-name the Highland Maya gave their own land. Spanish regularly renders Nahuatl /k^waw/ as *gua-* / *cua-* (*Cuauhtēmoc* → *Guatemoc*), so the *Guate-* of both *Guatemala* and *Guatelitzi* was taken to resolve to Nahuatl **cuauh-** "tree."
- **-tzi** = the Nahuatl reverential *-tzin*, with the final *-n* routinely dropped in the corpus (cf. *Atziri*, *Napuatzi*).
- **-teli-** / **-tel-** (medial) was the soft, unresolved element. The cleanest fit mirrored the *Cuauhtē-* of *Cuauhtēmallān*: *cuauh-* "tree" + the *-tē-* of *tema* "to fill / abound," an "abounding-tree" stem, with *-li-* a connective before *-tzin*. Plausible, never nailed down.

That yielded *Guatelitzi* ≈ Nahuatl *cuauh-* "tree" + reverential *-tzin* → "*the revered tree / honoured forest*," a reading that sat neatly on the ceiba theme already in play (*pōchōtl*, §10.5) and on the K'iche' "many trees" identity of the region the language is built from, a reasonable parse for what then looked like an unattested coinage. It is set aside now only because the referent is known: *Guatelitzi* is a person's name, not a descriptive phrase, so the "tree" gloss falls away even though the Nahuatl-flavoured phonology that suggested it still holds.

Xinkan checked, then set aside. Xinka, a language isolate of southeastern Guatemala, was tested as an alternative source while the token was still open; it supplies nothing (attested Xinka basic vocabulary shows no *guate*-like root: "water" *uy*, "fire" *ura*, "sun" *parri*, "black" *tz'uona*).[26] The question is moot now that the token is a known proper name, but the cross-check is kept for method.

10.9 TEXT 4: TOKEN ANALYSIS & ROOT-SEARCH

The text itself is §7; this section holds the token-by-token analysis and the root-search behind the soft readings.

TOKEN	PARSE	READING	CONFIDENCE
Máax	Yucatec <i>máax</i> "who?"[7]	who	★ attested (now in §9)
ka	Yucatec dependent / subjunctive <i>ka(j)</i> [7]	"that / when (you)..."	○ real particle; function here soft
ti	Maya <i>ti'</i> "to / at"[7]	to	★ committed (§9)
U'te	Maya presentative (<i>u</i>) <i>yan / at</i> [7]	"behold / here stand"	★ committed (§9, <i>A'te / U'Te / Yatle</i>)
mucane	Maya <i>muk'</i> "strength" + <i>-ane</i> [7]	"the mighty"	★ committed (§9)
en	Maya <i>-en</i> "me / I"[7]	me	★ committed (§9)

TOKEN	PARSE	READING	CONFIDENCE
a'tul	Set A 2nd-person a- "you / your" (confirmed, <i>tuyo</i> "your" = <i>a</i> [36]) + <i>taal</i> "to come"[7][36]; soft point is the <i>taal</i> → <i>-tul</i> raising (aa→u). Alt: animate classifier <i>-tíul</i> "one [being]"	"(that) you come"	○ soft, the prefix is attested; the <i>taal</i> → <i>tul</i> drift is the open step
cheyel	Nah. <i>chiya</i> / <i>chiā</i> "to watch, wait, lie in wait"[33] + Maya status <i>-el</i> (<i>chīya</i> → <i>chey-</i>); the Yucatec watch/sentinel field was searched and set aside on phonology , <i>ch'úuk</i> "lie in wait, ambush," <i>cha'an</i> "behold, watch," <i>pa'at</i> "wait," <i>kanan</i> "guard"[36] (all apt in sense, none a path to <i>cheyel</i>); also set aside: <i>chéel</i> "rainbow," <i>che'eh</i> "laughter," <i>che'+-el</i> (a noun), <i>ch'i'ibal</i> "lineage"	"to keep watch / stand sentinel"	○ soft-leading, attested sentinel verb, on-theme; the native field offering no closer form is what justifies the Nahuatl graft
Axba	Yuc. <i>ba'ax</i> "what" (confirmed[36]) by metathesis <i>ba'ax</i> → <i>axba</i> , an exclamation "What!?", parallel to <i>buxa</i> from the same root (§10.4); <i>aax</i> "wart" and <i>ba'ate'el</i> "fight" are false friends	"What!?"	○ soft, root attested; the metathesis is the open step
Kíibsa'	Maya <i>kíim</i> "die" + causative <i>-s</i> [28] + deictic <i>-a'</i> (Yuc. <i>kíims</i> "to kill")	"kill (it)!"	○ soft, well-framed, but needs an <i>m</i> → <i>b</i> spelling
ta'	open token (<i>ti'</i> vs <i>taak</i> ; here perhaps "to / at")	"to / before"	open, same item flagged in §10.7
Otsuks	Yucatec <i>tsuk</i> "cluster, company" (Cordemex)[31] framed by the Nah. exclamatory/presentative <i>o</i> "oh! / behold!" (Molina: <i>he aquí</i>)[34], a Nahuatl-particle + Maya-root graft (cf. <i>qexcan</i> , <i>pochiti</i>); only the <i>-s</i> coda open	"Behold, the company! / Company, fall in!"	○ soft, root and onset particle now attested; only the <i>-s</i> coda open
quxzeh	Maya <i>k'ux</i> "to bite, gnaw; ache; rancor"[33] (modern <i>k'uux</i> "morder; odiar; rencor"[36]), the corpus <i>q</i> marks ejective /k'/ (cf. <i>qexcan</i> ← <i>k'ex</i>), so the root is <i>k'ux</i> , not plain <i>kux</i> "life"; <i>-zeh</i> tail still not an attested suffix	"bite! / be fierce!"	● root closed: <i>k'ux</i> "to bite" attested in Cordemex and the modern wordlist[7][36], now in §9; the <i>-zeh</i> coda is a non-morphemic tail (cf. <i>Otsuks</i> <i>-s</i>). See §10.11

Root-search, Otsuks, -zeh (recurrence vs. attestation). The new captures settle one question and leave another open. *Otsuks* now **recurs as a fixed opener**, the same form across Tizoc, Cotan, and Axilo, and again in *Otsuks*, *quxzeh!*, so by position and function it is unmistakably a lexicalised drill-command. The first survey of the *soldier / warrior / men* field found no phonological fit, Yucatec *holkan* "soldier," Nahuatl *yāōquīzqui* "soldier," *tiacāuh* "valiant man," K'iche' *achi* "man," *ajlab'al* "warrior" all sit far from /*otsuks*/[29]; the Nahuatl **Otōntin** warrior society (the Otomi shock-troops who fought beside the *cuāchicqueh* behind *Cuachic*, §5, §11) is the closest in *kind*, but *otōntin* → *otsuks* has no regular sound path, so it was noted, not adopted.[29] The breakthrough is in the **core layer**: Yucatec **tsuk** "group, cluster, bunch", Cordemex glosses *tsuk* (*tzuc*) as *grupo de árboles pequeños, montecillo* ("a clump of small trees, a thicket"), beside "crest, tuft" and "animal's crop"; and, **now verified directly in the Cordemex text to hand** (the dictionary copy added to the project), it also appears under *much'* as a **numeral classifier for clusters/heaps**, *hun tsuk* "one cluster," *ka' tsuk* "two," *tsuk-en-tsuk* "in clusters." [31] A modern peninsular wordlist independently lists *tsúuk* beside *múuch'* for *montón* "heap"[36], so the root is attested across both colonial and present-day Yucatec. That classifier use is the clincher: a form whose grammatical job is to *count groups* extends to a body of men with no stretch at all, so *Otsuks!* reads as a unit-call, "Company! / Fall in!", native Yucatec, not a loan. What stays open now is only the coda. The *o-* onset, earlier set aside as a phantom affix, is in fact an **attested Nahuatl particle**: Molina (1571) records *o* as an exclamatory interjection ("del que está afligido y hace exclamación") and as the adverb *he aquí* "behold / here it is." [34] That makes *Otsuks* a Nahuatl-particle + Maya-root graft on the established *qexcan* / *pochiti* pattern, *o* "behold!" + *tsuk* "company", and its presentative sense rhymes with the Vaal presentative *U'te* / *A'te* already in the corpus, so *Otsuks!* reads naturally as "Behold, the

company!" / "Company, fall in!" The lone open element is the *-s* coda, best taken as the same stylised romanization seen in *Donuks* and *Gyan'uks* (a closing *-s/-ks* with no morpheme behind it) rather than a Yucatec affix. So *Otsuks* moves from "no attested root" to **attested root, unresolved morphology**, a soft reading, no longer rootless but not yet committed. Its recurrence as a fixed opener also kills the "-uks suffix" idea independently: the coda carries no meaning, it just closes a fixed command; a non-lexical drill-cadence (cf. English "Hut!") is now only a fallback. (*-uks* stays a romanization artifact, *Donuks* ← *uk'* "drink", *Gyan'uks* ← the *-cuic* of *yancuic* "new", not a morpheme.) *-zeh* likewise stays unidentified: Yucatec's causative is *-s / -kun(s)*, not *-zeh*, and Nahuatl has no *-zeh* (agentive *-keh* / *-queh*, reverential *-tzin*),[28] so the root carries the sense, now read as Yucatec **k'ux "bite, gnaw, sting; rancor"** rather than *kux* "life," since the corpus's *q* marks the ejective /k'/ (cf. *qexcan* ← *k'ex*), giving "*Otsuks, quxzeh*!" the sharper drill sense "Company, bite! / be fierce!"[33][35], while the *-zeh* tail is left open. (A neat Spanish reflexive *-se*, as in *cuídese*, would fit *kux* "live" → "keep yourselves alive," but it cannot ride the better-supported *k'ux* root, so it is logged as a competing reading, not adopted.) Neither *Otsuks* nor the soft tokens above are admitted to §9 until a cleaner root or a parallel attestation appears.

Dictionary pass on the three soft tokens (full Cordemex [31,32] plus the reachable colonial and modern dictionaries): **cheyel** has no Yucatec headword that fits the *Máax ka ti* __? "who are you to __?" frame: the native watch/sentinel verbs, *ch'úuk* "lie in wait," *cha'an* "behold," *pa'at* "wait," *kanan* "guard"[36], fit the sense but offer no path to the *form* "cheyel," and the noun candidates (*chéel* "rainbow," *che'eh* "laughter") fit neither. With the Yucatec field exhausted, the frame is best filled by **Nahuatl *chiya* "to watch / wait / lie in wait"** + Maya *-el*[33], a sentinel verb suited to a sentry challenge, a soft-leading cross-graft (the mirror of *pochiti*), not committed. **a'tul's** *a-* + *taal* "to come" reading remains the best sentry-challenge fit, with the animate classifier *-tíul* noted only as a weaker alternative. **Axba** is most likely onomastic, a sergeant-style exclamation or name rather than a common-noun lookup; the only near-match, *haxba* "to twist oneself" (< *hax* "to drill/twist"), is a false friend. All three are held open, not forced.

10.10 MY LITTLE WORD LAND CROSS-REFERENCE PASS

A scan of the My Little Word Land Nahuatl word-list[46] (an informal source, used here for corroboration only, not promotion) against the corpus. Every root below is independently attested in the primary Nahuatl record[10], so the list adds living-Nahuatl support but nothing is committed on its strength alone. No token was promoted to §9 as a result.

VAAL TOKEN / NAME	CORPUS STATUS	MATCHING LIST ENTRY	WHAT IT SUPPORTS
Atzoatl	§12.2	<i>atzoatl</i> "dirty water"	the <i>ā(tl)</i> + <i>tzoatl</i> reading, direct
Atziri	committed (§4, §9)	<i>atzintli</i> "water [H.]"	the reverential <i>-tzin</i> (<i>ātl</i> + <i>-tzin</i>)
Xatlene	§9 committed	<i>xotlatoc</i> "it is burning" (<i>xōtla</i>)	<i>xōtla</i> "to kindle, burn"
Eztl Pilli	§10.6	<i>piltzintli</i> "child," <i>nopilhuan</i> "my children"	the <i>-pilli</i> "noble, child" tail
Kuetzakala	§12.3	<i>cali</i> "house" (<i>calli</i>)	the <i>-kala</i> ~ <i>calli</i> "house" tail
Ahuatodli, <i>quiquate</i>	Text 5, soft	<i>cua</i> "I eat," <i>quicuahtoc</i> "he is biting him"	<i>qui-</i> (3sg obj) + <i>cua</i> "eat"
Ahuatodli, <i>tatlat</i>	Text 5, soft	<i>tlatla</i> "burn" (<i>tlatlatizque</i>), <i>tlahtoā</i> "speak" (<i>omitlahto</i>)	both candidate roots; the reading stays ambiguous
Ahuatodli, <i>xictep</i>	Text 5, soft	<i>xi-</i> imperatives, <i>tepetl</i> "hill"	the <i>xi-</i> "do!" + <i>tepe-</i> parse

The list also confirms the broader Nahuatl frame (*atl* "water," *calli* "house," *coatl* "snake," *iletl* "fire," *miqui* "to die," *tlamanalli* "offering"), consistent with the corpus's Nahuatl layer but adding no new committed token.

Soft-token sweep. Running every still-open token against the full list[46] for any root that would close it. The honest result: none closes. Two Ahuatotli tokens gain partial Nahuatl support; the rest find no match, and the Maya softs are out of scope for a Nahuatl source.

SOFT TOKEN	LAYER	LIST RESULT	OUTCOME
Ahuatotli <i>huatat</i>	Nahuatl	<i>cuauh-</i> "tree, eagle" present (<i>cuahuil</i> , <i>cuauhtli</i>)	partial; the reduplicated <i>-tat</i> is unexplained, stays soft
Ahuatotli <i>cutlotl</i>	Nahuatl	<i>cuitla-</i> present (<i>icuitlapan</i> "behind it")	partial; the fit is loose, stays soft
Ahuatotli <i>xomaplat</i>	Nahuatl	only <i>tlapal-</i> "help, colour" embedded (<i>tlapalehuia</i> , <i>imiazcatlapalhuan</i>); no <i>xoma-</i>	no closure, stays open
Ahuatotli <i>zahua moti</i>	Nahuatl	no match	stays open
Ahuatotli <i>azcado</i>	Nahuatl	no <i>azcatl</i> "ant" headword in this list	unchanged; the ant reading rests on other Nahuatl sources, not this one
Xibagua <i>-aqua</i>	soft	no native <i>aqua</i> ; the exact form match is Latin <i>aqua</i> "water"[59] (Spanish <i>agua</i> ruled out by regular <i>qu > gu</i>); a cross-graft onto the Maya root, and out-of-palette	logged soft, "water" (§12.1)
Mektul <i>ipkaat</i> , <i>ka'tse</i> , <i>koriek</i> , <i>xecwa</i>	Yucatec / K'iche'	out of scope	Maya tokens; to be checked against Cordemex, not a Nahuatl list
Long-standing Maya softs (<i>cheyel</i> , <i>a'tul</i> , <i>Axba</i> , <i>quxzeh</i> , <i>Otsuks</i> , <i>ta'</i> , <i>fukuur</i> , <i>Ma'oxe</i>)	Yucatec	out of scope	Maya; remain on the Cordemex track

Net effect: no promotion, no closure. Being a Nahuatl pool, the list can only speak to the Nahuatl-side softs (the Ahuatotli set), and there it corroborates two roots without proving either. The Maya-side softs, which are the bulk of the open list, still need the Cordemex and modern-wordlist track.

10.11 DICTIONARY CLOSURE PASS

A targeted pass over the still-soft tokens, Cordemex first[31][32], then the modern peninsular wordlist[36] where Cordemex gave nothing. Two tokens close; the rest are held, each with its reason.

TOKEN	CORDEMEX	MODERN WORDLIST	OUTCOME
kahk-tche (Mektul)	<i>k'áak'</i> "fire," <i>che'</i> "tree, wood"	<i>fuego</i> / <i>lumbre</i> = <i>k'áak'</i> ; <i>árbol</i> / <i>madera</i> = <i>che'</i>	closed: <i>k'áak' che'</i> "fire-wood, firebrand"; both roots now in §9
quxzeh (Text 4)	<i>k'ux</i> "to bite, gnaw; rancor"	<i>morder</i> = <i>k'nux</i>	closed at root: <i>k'ux</i> "bite" now in §9; <i>-zeh</i> a non-morphemic coda (cf. <i>Otsuks -s</i>)
xecwa (Mektul)	<i>xek</i> only in <i>hekeb xek</i> "saddle"; no verb fit	<i>cortar</i> = <i>xoot</i> ; <i>atacar</i> = <i>k'óoch</i> ; no <i>xec-</i>	open: <i>-wa</i> poss. <i>waaj</i> "offering"; <i>xec-</i> unresolved

TOKEN	CORDEMEX	MODERN WORDLIST	OUTCOME
ipkaat (Mektul)	no clean <i>ip-</i> root	<i>k'áat</i> "ask, want" confirmed; no <i>ip-</i>	partial: <i>-kaat</i> = <i>k'áat</i> "ask, want"; <i>ip-</i> (poss. in "I") unresolved
ka'tse-ik (Mektul)	<i>tse'ek</i> "sermon, to punish" (poor fit)	<i>tse</i> only in <i>tséel</i> "side"	partial: <i>ka'</i> "two" + <i>ik'</i> "wind/spirit" hold; <i>tse</i> unresolved
ukto (Text 4)	Nah. <i>ocotl</i> / <i>ocutl</i> "pine torch" (root <i>oco-</i> / <i>ocu-</i>)	<i>antorcha</i> (torch), the thing that burns	soft: leading reading <i>ocotl</i> "torch," burns in the dark; <i>-tl</i> → <i>-to</i> stylization[52]
ta' (Text 3)	<i>ti'</i> "to/for"; <i>taak'</i> "want"	<i>para</i> = <i>tia'a'</i> ; <i>querer</i> = <i>k'áat</i> , <i>óot</i>	open: two readings logged, neither decisive
koriek (Mektul)	not Yucatec (the <i>r</i> marks K'iche')	Christenson[47]: re-cut <i>ko-</i> "strong, hard" + inchoative (cf. <i>kowirik</i>); also <i>kor-</i> "loosen," <i>k'or-</i> "knock"	soft: leading reading "harden / grow strong"; form not exact, held open
cheyel, Axba, a'tul (Text 4)	exhausted earlier in §10.9	no new fit	open: held as before (Nah. graft / metathesis / <i>a-</i> + <i>taal</i>)

Result: two closures (*kahk-tche*, *quxzeh*) and three new committed roots in §9 (*che'*, *k'áak'*, *k'ux*). The rest resist the Yucatec record: most are partial (one attested root plus an unresolved fragment), and *koriek* needs the K'iche' track, which these two dictionaries cannot supply.

K'iche' follow-up (Christenson[47]). The two K'iche'-seam items were then run against a dedicated K'iche' dictionary, the project's first (Christenson, compiled from field work in Momostenango and Totonicapán). *jare* is confirmed and upgraded off the phonology-only note[17]: Christenson lists *are'* as the focus pronoun "he, she, it; it is, that is" and *ri* as the article "the," precisely the *are'* / *ri* family already proposed, so *jare* now rests on a real K'iche' source. *koriek* stays open, but a second pass that does not assume the segmentation *kor-* improves the candidate. The *r* is consistent with K'iche' (which keeps the tap *r* that Yucatec lost and Nahuatl never had). Re-cut as *ko-* + *-(i)rik*, it lands on the strength field: *ko* "hard, strong, firm" with the inchoative *-(i)rik* "to become X," whose attested realisation is *kowirik* "to become strong, to become hard, to fortify" (also "to freeze")[47]. A boss shouting *koriek!* as a harden-or-strengthen cry fits an attack pool far better than "loosen," so *ko-* "strong" + inchoative is logged as the leading reading, held soft: the dictionary's surface form is *kowirik* (medial *-w-*), and *koriek* would need that *-w-* lost and a *-rik* to *-riek* vowel glide, neither yet shown. The earlier *kor-* "to loosen" (*korobaj*, *koronik*) and a third candidate *k'or-* "to knock, tap" (*k'ork'a'*, *k'orok'ot*) are kept as weaker alternatives, the latter doubly so since its ejective /k'/ should surface in the corpus as *q-* (*qoriek*), not the plain *k-* seen here. The K'iche' sourcing gap is now filled for *jare*; *koriek* is a soft K'iche' token with a candidate parse, the gloss open while the dictionary facts (*ko*, *kowirik*, *kor-*, *k'or-*) stay fixed.

10.12 ADVERSARIAL COMPETITORS (LOGGED FROM THE HARDENING PASS, §17)

The adversarial pass (a blind decoder given only the surface form, hunting the best root in a *different* source language) surfaced genuine cross-language competitors for seven committed tokens. None overturns the

committed reading, which still wins on context and neighboring tokens, but each shows the committed reading is not uniquely forced by form alone, so it is logged here and flagged with a star in §9.

- **itsok / itzil** committed Maya *iitz* "essence, blood" vs **Nahuatl** *itzli* "**obsidian, sacrificial blade**." This is the strongest of the set: obsidian-as-sacrificial-blade is thematically as apt as "blood/essence" in this corpus, the form fit is clean, and Nahuatl is a core palette language. Treated as a live co-reading, not dismissed.
- **tala** Maya *taal* "to come" vs Nawat/Nah. *ta:l* "earth, land" (form-identical, different sense).
- **pochiti** Maya *poch* "hungry" vs Nawat *puchini* "it bursts, frays."
- **kutsen** Maya *kutz* "sacrificial bird" vs K'iche' *kotz'i'j* "flower, candle" (both plausible as an offered thing).
- **A'te / U'Te** Maya presentative *at/yan* "behold" vs Nah. *ahtle* "nothing" (form-close, opposite sense).
- **te** Maya relational *ti'/te'* vs Nah. *tetl* "stone."
- **mucane** Maya *muk'* "strength" vs K'iche' *muq* "bury" (already carried as the shadow sense).

Four other apparent competitors (*ma*, *qexcan*, *sakilja*, *Yutsal*) were set aside as cognates: the "competing" K'iche'/Maya form is the *same* root in a sister language (e.g. Maya *sak* = K'iche' *saq* "white"), which is corroboration, not competition. Full method and the survival tally are in `ADVERSARIAL_RESULTS.md`; the summary is in §17.



11. LORE ANCHORS

- **Atziri**, the vain Vaal queen who sought immortality and eternal beauty ("grace made eternal"), demanding human sacrifice *to* her as "an act of divinity... to transform the world." Her communion with the Beast triggered the Cataclysm that destroyed the Vaal overnight; she lingers afterward in a nightmare realm.
- **The Beast** (*pochiti*), the primordial entity Atziri communed with; her attempt to merge with it triggered the Cataclysm. In her PoE2 encounter she speaks no Vaal except this single word for it: *pochiti*, best read as "the Hungering One" (Maya *poch* "gluttonous, hungry"), the devourer she feeds. (A Nahuatl *pōchōtl* "ceiba / protector-tree" reading is phonologically possible but has **no** in-game support; see §10.5.)
- **Utzaal (Yutsal)**, Doryani's seat of power and research stronghold (where he pursued immortality, corruption, and time), reachable in PoE2 by travelling to the moments before the Cataclysm. **Not** the capital.
- **Capital / royal seat, Lira Vaal**, Atziri's royal temple, was the Vaal capital for much of the empire's history; after its temple burned, **Azala Vaal** (now buried below Sarn) was named the new capital. Atziri's seat is Lira Vaal, where her Temple, and its Commander, appear in PoE2 0.5.0. Utzaal was Doryani's separate domain.
- **Zerphi**, a noble whose autopsy revealed a twenty-year-old's body at age 168, prompting Atziri to set Doryani researching his longevity. "Undying" in legend, though he died.
- **Cuachic Vault**, a PoE2 0.5.0 interlude area holding surviving Vaal. Its bosses are the sibling Blood Priestess **Zelina** (who sings the litany's call) and Blood Priest **Zolin** (the response). Doryani's pass-phrase, "the time of the sparrow has come," is rejected on sight: Zolin emerges with "What nonsense do you speak?" and Zelina with "What do you ramble about?", and they attack rather than parley.
- **Cuachic = the Shorn Ones**. *Cuachic* is attested Classical Nahuatl. The *cuāchicqueh* ("shorn ones") were the most prestigious Aztec warrior society: heads shaved but for a braid over the left ear, faces painted half blue and half red, sworn never to step backward in battle on pain of death. So the Blood Priests are an elite Vaal warrior-priest order, which is why they fight to the death rather than heed a passphrase, and why Zolin's barks run to "We stand the test of time," "No sacrifice is too great," and "Cuachic manœuvre!" It plants a second attested Nahuatl military term beside the *xeŋe* ("commander") register of Text 3.
- **The sibling blood-pact (from the English barks)**. Zolin sustains himself by draining Zelina, "We live for one another," "Loyal transfusion," "Pact... of blood", and each rages at the other's fall ("I will avenge you, sister!"; "No! Zolin! You will suffer for this!"). Zelina brands the player a "foul Outlander." Her enrage line names a Vaal tome, the **"Tome of Guatelitzi"**, now identified as *Guatelitzi, Architect of Flesh*, the Vaal flesh-and-immortality architect of the Temple of Atzoatl (§10.8).
- **Eztli Pilli**, an in-game Vaal tome of forbidden, volatile knowledge, "too dangerous to use." Its title is Classical Nahuatl *eztli* "blood" + *pilli* "noble, prince" = "the Blood Prince" (see §10.6), a fitting name for a grimoire of Vaal blood-magic, and a second Vaal *tome* alongside the *Tome of Guatelitzi*.
- **The invader / "Demon of Atzoatl"**, the **Temple of Atzoatl** was the PoE1 Incursion temple, repeatedly raided by the Exile ("Godslayer") ~20 years before PoE2 to kill its architects. In PoE2 the Vaal mistake the player for that demon. The Commander's *u'tra buxa*, now read as *u'tra ba'ax* "another such thing / another one" (see §10.4), most likely points at this returning intruder, though the temple he defends is Atziri's at Lira Vaal, not Atzoatl.

- **Naming consistency**, the surrounding Vaal proper names (Cuachic, Ketzuli, Napuatzi, Atzoatl) are uniformly Nahuatl-flavored, consistent with the Nahuatl layer identified above.





12. VAAL AND VAAL-RELATED FIGURES

The same attested-root method, applied to the proper names of Vaal and Vaal-adjacent figures. The useful result is the sorting: a few names carry live Maya or Nahuatl roots, a few are opaque to the palette, and a few are clearly borrowed from outside it. Forcing Nahuatl onto every Vaal name would be a mistake; this section shows where it fits and where it does not.

12.1 XIBAQUA, THE FIRST VAAL

Lore. *Xibacqua* was a historical Vaal figure whose remains were held to be the **progenitor of all Vaal**, "born from the flesh of ancient Vaal gods." The origin myth runs: "*Xibacqua's treachery was met with divine fury. One by one, the gods reclaimed their flesh, until all that remained was a droplet of pure light: the first Vaal.*"[39]

Root, Xib-. Two attested Maya readings, both on theme, are logged rather than chosen: – Yucatec *xiib* "male, man"[36], i.e. "the man / the first man", a plain fit for a progenitor. – K'iche' *xib'* "fear," the root of **Xibalba**, the Maya underworld ("place of fear")[17], fitting a being drawn from the gods' flesh, and resonant with the Vaal fear-cosmology (cf. *Yugul*, who held fear the "bedrock of the cosmos," §12.3). It also lands on the **K'iche' seam** already opened by *jare* (§10.2), the one place Highland Maya surfaces.

Tail, -aqua. A closer look sharpens this. It reads as "water," but the cleanest match is not native. Classical Nahuatl has no bare *aqua*: its water root is *ā(tl)* (the *atl* of *Atziri*, *Atzoatl*), which needs a following element the tail does not supply, and the nearest verb, *āquia* "to insert, submerge, drive in, transplant" (Molina)[59], is /a-ki-a/, a near-miss on /a-kwa/ that a strict reading cannot bridge. K'iche' offers nothing (its *aq* is "pig," *aqan* "leg, altitude"; water is *ja'*), and Yucatec has *akal* "pond" and *aka'an* "pooled water" but no *aqua*. The exact form match is **Latin aqua "water"** (/akwa/)[59]; the Spanish reflex is ruled out, since Latin *qu* regularly became Spanish *gu* (*aqua* to *agua*) and the Vaal tail keeps the /k/, not /g/. So if *-aqua* is a water-word it is the Latin form, not the Spanish, making *Xibacqua* "fear-water" or "the well of fear," resonant with the Vaal water-and-blood cosmology and with Xibacqua as the progenitor "droplet." This stays **soft, not committed**: it is a cross-graft (a Maya root plus a Latin element) and Latin sits outside the Mesoamerican palette, its only other appearance being the *Omnitect* (§12.4). **Status:** *Xib-* attested (two Maya readings kept); *-aqua* soft, best read as Latin *aqua* "water," with the cross-graft caveat.

12.2 ATZOATL, THE TEMPLE

Lore. The **Temple of Atzoatl**, a Vaal treasure-temple of many pursuits (immortality, corruption, weather-control, sacrifice), built late in Atziri's reign and made her final seat; the present-day Vaal take the player for its "demon" (§11).[40]

Root. Nahuatl, built on *ātl* "water"[10] (the root behind *Atziri*, *ātl* + reverential *-tzin*, a form the same list independently carries as *atzintli* "water [H.]") compounded with *tzoatl* "dirty water, slops." *Tzoatl* is itself an attested Nahuatl word: Molina 1571 glosses it "lauazas, o lauaduras" (dishwater, washings), recorded in the Wired Humanities dictionary[10]. *Atzoatl* therefore reads as "dirty / fouled water," apt for a temple given over to corruption and blood-sacrifice (§11). This **resolves the medial -tzo-** left open before: it is not a weak *tzotl* "grime" guess but the attested compound *a(tl)* + *tzoatl*. The same gloss is listed directly in a community Nahuatl word-list, which renders *atzoatl* as "dirty water"[46]; that list is an informal primary source, recorded here only because

Molina independently confirms it via [10]. **Status:** *ātl* + *tzoatl* "dirty water" attested; medial resolved. Consistent with the Nahuatl place-name layer (*-tlān* in *Teoyuxtlane*, the *cua-* of *Cuachic*).

12.3 VAAL GODS

The palette test, name by name. Only two carry live Nahuatl roots; the rest are opaque or out-of-palette, and are marked as such rather than forced.

FIGURE	LORE	NAME ANALYSIS	VERDICT
Kuetzakala	a Vaal-related divine name; reported (PoE2, via Doryani) as a "goddess of the siege," not independently verified here	Nahuatl <i>quetzal(li)</i> "quetzal-plume; precious"[10] (root of <i>Quetzalcoatl</i>) + <i>-kala</i> poss. <i>calli</i> "house, structure"[10]	Nahuatl root attested; tail soft
Yaomac	ancient Vaal god, three serpent heads, shepherd of Vaal souls[41]	Nahuatl <i>yāō(tl)</i> "war, enemy" + <i>-mac</i> "in the hands of" (locative of <i>māitl</i> "hand")[10] → "in the hands of war"	clean Nahuatl parse
Kamasa	a Vaal god; the Kamasan Smith (Text 5) is "of Kamasa"[42]	poss. K'iche' <i>Camazotz</i> , the Popol Vuh death-bat (<i>kame</i> "death" + <i>sotz'</i> "bat")[45]	Maya / K'iche' lead, soft at the derivation
Kopec	a Vaal god, little known, with a temple[42]	no transparent root; a <i>-pec</i> ~ Nahuatl <i>-tepec</i> "place of" link drops the <i>te-</i> and stays weak; the "sun god / Xipe Totec" tie is thematic only	opaque (weak guess logged)
Yugul	"Reflection of Terror," a fear-scholar who ascended to godhood[41]	reading the medial /g/ as a stylised consonant (the corpus's /g/ convention, §2.4): Yucatec <i>yuk</i> "to shake, agitate, set trembling" (<i>agitar</i> , <i>menear</i> [36]), the bodily sign of terror, tail <i>-ul</i> / <i>-kul</i> open; an alternative folds the medial into Maya <i>k'u</i> "god, sacred" (Cordemex <i>maban k'u</i> "atheist," <i>k'u na</i> "temple"[31]; modern <i>k'uh</i> [36]), <i>yu(k)</i> + <i>k'u(l)</i> "the divine one," fitting an ascended god. The earlier "fear" gloss is set aside (Yucatec "fear" = <i>sahkil</i> [36])	○ soft (was opaque): root candidate(s) attested, segmentation open
Apep	a serpent-deity of Vaal chaos / poison (items <i>Apep's Supremacy</i> , <i>Apep's Slumber</i>)	Egyptian <i>Apep</i> / <i>Apophis</i> , the chaos-serpent[45]	borrowed from outside the palette
Ralakesh	"Master of a Million Faces," a Vaal god of subjugation who controlled his subjects' minds and governed humanity through its animal instincts; mind-domination is his attested core (not wholly unique, Kitava likewise corrupts minds)[41]	Sanskrit <i>Rākṣasa</i> , the shape-shifting illusion-demons, fitting the "million faces"[45]	outside the palette
Arakaali	"Spinner of Shadows," an ancient (pre-Vaal) spider goddess	a blend of Greek <i>Arachne</i> "spider" and Hindu <i>Kālī</i> (death, time)[45]	outside the palette

The sorting is the point, and a fresh comparative pass, cross-checked here against the dictionaries, sharpened it. *Yaomac* now parses cleanly, *yāō-* "war" + *-mac* "in the hands of," and *Kuetzakala* sits firmly on *quetzal-*. *Kamasa* gains a real Maya lead in *Camazotz*, the Popol Vuh death-bat, which also ties it to the Xibalba theme behind *Xibagua* (§12.1). *Kopec* stays opaque (its *-pec* drops the *te-* of *-tepec*, and the link stays weak). *Yugul* now carries a soft reading: with the medial /g/ read as a stylised consonant (§2.4), it resolves to Yucatec *yuk* "to shake, set trembling"[36], the bodily sign of terror, with the tail *-ul* / *-kul* open, or alternatively folds the medial into Maya

k'u "god"[31][36] for an ascended-god reading; the earlier "fear" gloss is set aside, since Yucatec "fear" is *sahkil*[36]. The three outliers pin down to specific foreign sources, Egyptian *Apophis*, Sanskrit *Rākṣasa*, and a Greek *Arachne* plus Hindu *Kālī* blend, the same caution the /d/ and /r/ tests enforce (§2.4).

12.4 THE ARCHITECTS OF ATZOATL

The Architects of Atzoatl were the Vaal master-builders and specialists, each governing one domain of the Temple of Atzoatl[43]. Their names are uniformly Nahuatl-styled, the *-tl* / *-tli* / *-atl* absolute endings dominate, with a handful of Maya-rooted exceptions; only *Guatelitzi* (Flesh, §10.8) has been carried to a committed reading. The roles below are the in-game, lore-attested functions; conjectural readings of the lore are omitted. The name-root column uses the same standard as the rest of the corpus: an attested root with a workable phonology is marked ○, a name with no isolable root is marked opaque, and a fully worked name is marked ●. Per-architect wiki pages branch from the *Incursion room* roster hub[43]; the lore was compiled by the community in a single reference post[48].

ARCHITECT	DOMAIN	ROLE (LORE-ATTESTED)	NAME ROOTS
Ahuana	Ceremonies	Oversaw the Vaal's mass human sacrifices; in Incursion, sacrifices unique items to create others.	○ poss. Nahuatl <i>ahua-</i> (<i>ahuatl</i> "oak, thorn")[46]; <i>-na</i> tail open; soft-to-opaque
Guatelitzi	Flesh	Directed life-force transferral and the study of immortality; also produced powerful prosthetics.	● Nahuatl <i>cuauh-</i> "tree, eagle" (<i>Gua-</i> = /kʷ/) + reverential <i>-tzin</i> (§10.8)[38]
Atmohua	Iron	Crafted the Vaal's armours.	<i>at(l)-</i> "water"? + <i>-mohua</i> open; opaque
Cholotl	War	A military commander; gathered the most war-driven Vaal.	○ <i>-tl</i> absolute + <i>choloa</i> "to flee, leap"[10] candidate
Ticaba	Arena	Held gladiator bouts and ritual combat.	opaque
Azcapa	Guild	A jeweller who made gold trinkets and baubles for Atziri.	○ Nahuatl <i>āzcatl</i> "ant"[10] + locative <i>-pa(n)</i> "at" (cf. <i>Azcapotzalco</i>)
Estazunti	Vault	Stored and catalogued the gifts sent to Atziri.	opaque (only a <i>-tzin</i> -like tail)
Juatalotli	Hoard	Gathered and preserved Vaal relics and culture.	○ <i>Jua-/Gua-</i> onset = Nahuatl / kʷ/ (§2.4) + <i>-otli</i> absolute; root open
Puhuarte	Forge	A smith; his fire-craft is later weaponised by the Omnitect.	opaque
Zantipi	Concealment	Built strongboxes and hidden passages.	opaque
Hayoxi	Destruction	Made explosives.	○ Yucatec <i>haay</i> "to raze, level"[31] (a Maya root in a Nahuatl-styled set); <i>-oxi</i> open
Matatl	Fortifications	Master trap-builder and architect.	○ Nahuatl <i>matlatl</i> "net, snare"[10] (<i>-tl</i> absolute), apt for traps
Xopec	Power	Studied electricity.	opaque (<i>-pec</i> parallels <i>Kopec</i> ; the <i>-tepec</i> link stays weak, §12.3)

ARCHITECT	DOMAIN	ROLE (LORE-ATTESTED)	NAME ROOTS
Paquate	Corruption	Master of concentrating corruption; "Paquate's Mechanism" is a reusable corruption device (a corruption altar that replicates a Vaal Orb), housed within Jiquani's Machinarium/Sanctum rather than driving it.	<i>paqui</i> "to be glad"[46] candidate + <i>-te</i> open; opaque-leaning
Zalatl	Thaumaturgy	Oversaw virtue-gem work; a student of Doryani.	○ <i>-tl</i> absolutive; root (<i>zal-</i> / <i>tzalan?</i>) open
Citaqualotl	Swarm	Created metallic insect monsters.	○ <i>-tl</i> absolutive + <i>cua/qua</i> "to eat" (<i>cualo</i> "be eaten")[10] candidate
Jiquani	Industry	Built soul-core constructs at the Utzaal machinarium; his constructs helped excavate the Cuachic Vault.	opaque (/g/-class <i>Ji-</i> onset)
Opiloti	Strife	Researched Timeless Monoliths ("Obelisks Beyond Time").	opaque (<i>-ti</i> tail)
Tacati	Toxins	Worked plant- and snake-based poisons; venerates the serpent Apep.	opaque (<i>-ti</i> tail; <i>tlāca-?</i> open)
Topotante	Storms	Built "spires" that altered the weather.	opaque
Zilquapa	Breach	Studied or worshipped the Breach (Cult of the Purple Flame).	opaque (poss. <i>-qua-</i> /k"/ + locative <i>-pa</i>)
Quipolatl	Nexus	Proposed and designed the Temple of Atzoatl.	○ <i>-atl</i> absolutive (shared with <i>Atzoatl</i>); <i>poloa</i> "to lose, destroy"[10] candidate
Tzamoto	Torments	Imprisoned and tortured those who disobeyed Atziri.	○ <i>tzacua</i> "to close, imprison"[10] candidate, apt for torment; <i>-moto</i> tail (cf. <i>Uromoti</i>)
Uromoti	Expansion	Ran exploration and city-planning; built the Cuachic Vault shelter with Jiquani; drew maps of fictional lands.	opaque (<i>-moti</i> tail, cf. <i>Tzamoto</i>)
Xipocado	Royal Architect	Rebuilt Lira Vaal; mini-boss of <i>Fate of the Vaal</i> ; spied on Doryani.	○ Nahuatl <i>Xipe</i> (Totec, "the Flayed One")[10] candidate; <i>-cado</i> open

Four further figures sit beside the temple roster:

FIGURE	DOMAIN	ROLE (LORE-ATTESTED)	NAME ROOTS
Vaal Omnitect	Construct atop Atzoatl	A construct in the temple's top chamber.	not Mesoamerican: Latin <i>omni-</i> "all" + <i>-tect</i> (from <i>architect</i>), "all-builder"
Ketzuli	Time / High Priest of the Sun	Oversaw the solar-powered time machine at Utzaal; later turned Undying.	○ Nahuatl <i>quetzal(li)</i> "quetzal-plume, precious"[10] (cf. <i>Kuetzakala</i> , §12.3); tail open
Mahuxotl	Banished Architect	Known only from his unique shield (<i>Mahuxotl's Machination</i>): the architect who tried to fuse every dark Vaal secret at once.	○ Nahuatl <i>māhua</i> "to spread, thicken"[10] + <i>xolotl</i> "monster, deformity; servant"[10] (contracted <i>-xotl</i> ; cf. the Commander <i>Xolotl</i> , §6), a "spreading aberration"; derivation soft
Ahuatodli, the Blind	Vaal Outpost Delve boss	Dressed as an architect, inhabits "The Grand Architect's Temple," speaks short Vaal lines (Text 5).	○ Nahuatl <i>ahua-</i> "oak, thorn"[46] + <i>tototl</i> "bird"[10] (§8)

12.5 VAAL WEAPON-NAMES: THE VEINTENA MACES

Two Vaal war-maces are named, in fully attested Classical Nahuatl, after the twenty-day month-festivals (*veintenas*) of the Aztec calendar, and both carry an inscription attributed to **Doryani of the Vaal**. This is deliberate, un-stylised use of real Nahuatl, and specifically of the war-and-weapon festivals.

- **Quecholli** (PoE2 unique mace)[54]. *Quecholli* is the 14th veintena, the festival of **Mixcoatl** given over to weapon-making and the hunt; the word is a bird-name (the roseate spoonbill / a precious red-feather bird). A Vaal weapon named after the weapon-making month is exact.
- **Panquetzaliztli** (PoE1 mace, the fated form of *Quecholli*)[55]. *Panquetzaliztli* is the 15th veintena, "**the raising of banners**," the festival of the war-god **Huitzilopochtli**: *pan(tli)* "banner, flag" + *quetza* "to raise, erect" + *-liztli* (action-noun). The *quetza* root is the one already in the corpus behind *Ketzuli* and *Kuetzakala* (and *Quetzalcoatl*).

Both are onomastic and fully Nahuatl-attested with no stylization, so they are carried in §9 as **hardened** name-tokens (H+L): each passes Gate 1 (the surface is the attested lexeme, no residue) and Gate 2 (no cross-language competitor), per `HARDENING_PROTOCOL.md`. Their value is corroborative: two independent, un-stylised Nahuatl data-points confirming the developers reach for real Classical Nahuatl, not merely Nahuatl-shaped invention. (These two names, with *Ixchel* below, were hardened after the null-model battery; the null-tested 62-token figures of §17 stay the headline while these are reported alongside as an enlarged base in §17.7, with a full re-run of the battery on the enlarged corpus as the versioning trigger, §14.)

12.6 IXCHEL, THE GODSTEALER

Ixchel is named on the wiki as a former **Vaal citizen**, rescued during the Empire's collapse by the Order of the Djinn and set to steal the three-headed god *Yaomac* (whence the title "**Godstealer**"); he later became the **Trialmaster** of the Trials of Chaos[56]. The name is the attested Maya goddess **Ix Chel**: Yucatec *Ix-* (the feminine / agentive prefix) + *Chel* "rainbow," the moon, weaving, and midwifery goddess. A real Maya theonym reused for a Vaal figure, and a Yucatec name (not Nahuatl), sitting with the thin Maya-name presence already noted (*Kamasa* via *Camazotz*, *Xibaqua* via *xib'*). "Godstealer" is the English epithet, not Vaal. Carried in §9 as a **hardened** onomastic name (H+L): *Ix-* + *Chel* reconstructs strictly (Gate 1) with no cross-language competitor (Gate 2), per `HARDENING_PROTOCOL.md`; the *Ix-* prefix is itself an attested Yucatec morpheme.

12.7 LATER VAAL ARTISANS (THE STRONGBOX ARCHITECTS)

Two further Vaal **Architect** bosses appear on unique strongboxes, distinct from the Atzoatl sixteen; both are Nahuatl-styled and parse to attested roots but not as whole names:

- **Ixtolatl, Artisan of Sacrifice**[57] (boss of *Gifts of the Sacrificed*). *ix(tli)* "eye, face" + the *-tl* / *-atl* absolutive; the medial is open. Soft.
- **Mahuatzi, Artisan of Desire**[57] (boss of *Gifts of the Red Queen*, i.e. Atziri). *māhua* "to spread, thicken; to quarrel, infect" (attested in the *tomāhua* / *chamāhua* family)[10] plus the reverential *-tzin* to *-tzi* (committed, as *Atziri*, *Guatelitzi*). Root and reverential attested, exact sense open. Soft.

With **Mahuxotl** (above) and the already-carried **Xolotl** (the Commander, §6), the *māhua-* and *xolotl* roots recur across the later architect names.

12.8 THE VAAL TEMPLE TRIO (K'AJ Y'ARAAZ, K'AJ Q'URA, K'AJ A'ALAI)

The three bosses of the **Vaal Temple** map[58] are the oldest Vaal-named content examined here (PoE1 1.1.0, 2014), variants of the Atziri trio, tagged "unique demons." They visibly **break the Nahuatl naming pattern**: no *-tl* / *-tzin* absolutes, but ejectives (*K'*, *Q'*), glottal stops, and a tap */r/*, which is Maya-shaped, and specifically the K'iche' / Highland seam (Yucatec lost */r/* and has no uvular *q*). A dedicated form-first pass against Cordemex (Yucatec)[7] and Christenson (K'iche')[47] gives a mixed, honest result:

- Shared prefix **K'aj-**: a candidate in K'iche' / Yucatec *kaj* "sky" (plain *k*, not the ejective the spelling shows), or Yucatec *k'aj* "bitter; to remember"; weak and ambiguous.
- **A'alai**: lands on an attested Maya root, Yucatec *a'al* "to say, command" (also "child"; "heavy") or K'iche' *alaj* "small; to give birth"; plausible.
- **Q'ura** and **Y'ara'az**: no clean attestation; *q'ur* appears only inside *muqur* "owl" and *quruqutik*, and K'iche' *ara* forms are Spanish loans (*arado*, *bara*). These two do not resolve.

The verdict is partial, consistent with pre-conlang phonaesthetic naming that happens to fall in Maya phonotactic space: one plausible root (*A'alai*), a possible *K'aj* "sky," and two elements (*Q'ura*, *Y'ara'az*) with no isolable source. They are logged **open**, not committed, and flagged as the clearest test of whether the 2014 content carries a real Maya layer. If sourced at all, the ejectives and */r/* say it would be Highland Maya, the same seam *jare* opened.

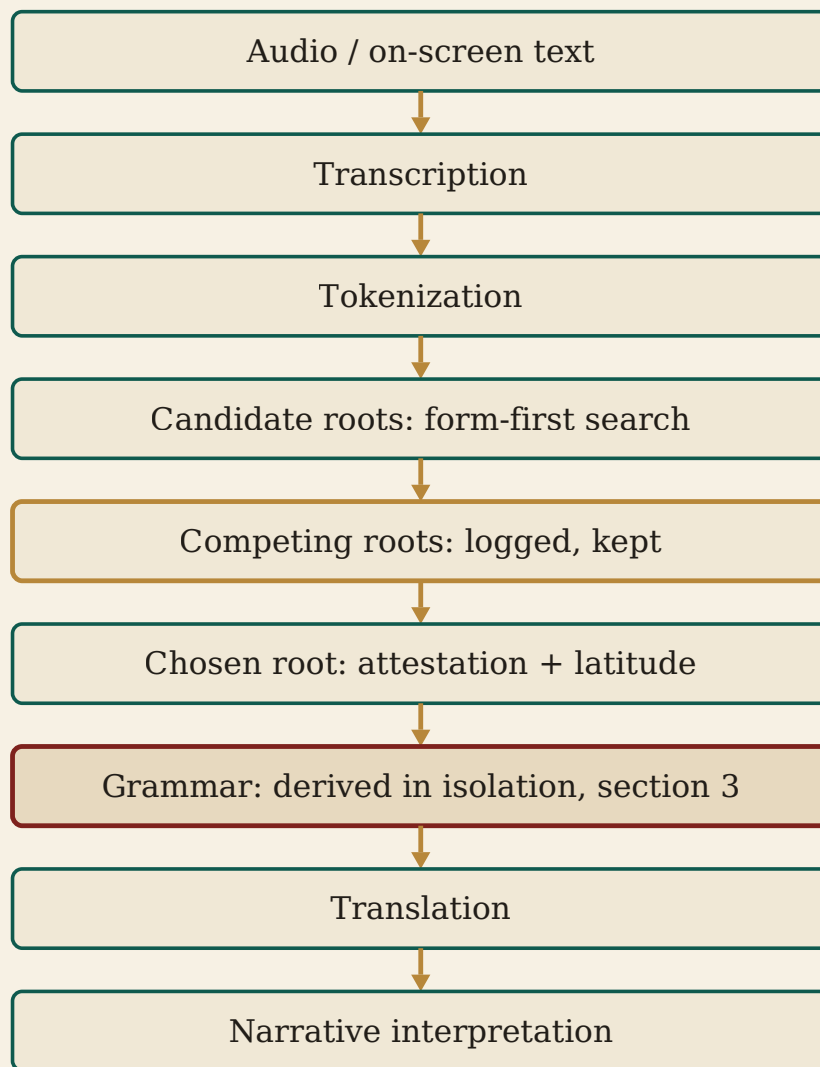


13. METHOD NOTES

- **Geographic-palette principle.** Prefer attested Maya/Nahuatl roots over cross-family guesses (Quechua, Polynesian, Taíno, Tagalog); reject any root that needs invented grammar. Applied late to swap out the four guesses in §10.3 and to reject unverifiable items (*máax* "crush," *óoxol* "heat," *ma'ax* "not even").
- **Candidate source pools beyond the core three.**[19][24] When a token resists the Yucatec / Nahuatl / K'iche' palette, two adjacent pools are worth testing before defaulting to a loan. **Xinkan (Xinca)** is a small, now-extinct non-Mayan family of southeastern Guatemala with no demonstrated affiliation but heavy Mayan-loan contact; it abuts the Highland-Maya seam, so it is a natural next stop for a stubborn root, with one caveat: some Xinkan languages carry voiced stops (*b*, *d*) and ejectives, so a Xinkan source would blur the clean /d/ = Spanish reading of §2.4 and must be weighed against it, not used to wave it away. The wider **Aztecan (Nahuan)** branch of Uto-Aztecan runs, loosely, from the western U.S. down into Central America (the Mesoamerican-languages literature stretches the span from roughly Oregon to Panama); beyond Classical Nahuatl this includes **Pipil / Nawat** of El Salvador, geographically apt for a Vaal language drawn from Central-American sources. Both are recorded as places to look, not committed layers, anything drawn from them must still clear the attested-root test.
- **Result of the Xinkan / Nahuan cross-check.** Run against attested Xinka vocabulary,[26] the committed lexicon turns up no cleaner fit: Xinka core words (*uy* "water," *ura* "fire," *parri* "sun," *tz'uona* "black," *mowa* "white") match no Vaal token, so nothing committed needs revising. Two points are recorded honestly rather than acted on: (a) the zone's other families were run against both anomalies and neither moved, for *jare*, Aztecan has no native /r/ at all and its demonstratives (*inin* / *inon*, *in*) show none,[27] while Xinka has a tap /r/ but only a third-person *nah* demonstrative,[26] so K'iche' *are'* / *ri* stays the fit; for *daka*, Aztecan *maca* / *maka* "to give" matches the imperative slot's sense but not the /d/ onset, and Xinka's native /d/ comes with no attested *daka*-like "give" verb, so Spanish *daca* keeps the form; (b) the living Nahuan branch **Pipil / Nawat** of El Salvador corroborates a Nahuatl stratum this far south but supplies the same roots as Classical Nahuatl, confirming geography without displacing a reading. The one open token the zone's languages *did* resolve is *Guatelitzi* (§10.8), via the Nahuatl tree-root behind *Guatemala*.
- **Living modern reflexes as corroboration.** Several of these roots survive in modern Latin-American Spanish, giving a second, independent attestation channel, especially useful as a fallback or tiebreaker when the classical dictionaries are thin. *poch* "gluttonous, hungry" is carried into Mexican Spanish and listed by the RAE (→ *pochiti*); Nahuatl *chantli* "home" lives on as *chante* "house" (→ *i'chian*); Nahuatl *pōchōtl* "ceiba" survives as *pochote*, and colloquial *pochotón* "husky, thick-built" preserves the tree's "big, sturdy" sense (→ the *pochiti* co-reading). The channel is not limited to Spanish: within living **Maya** itself, *poch* "eager, craving" is still everyday Yucatec, a Belize Yucatec community word-list records *poch* "anxious, eager" (*poch de comer* "craving to eat"), a native-speaker witness that the *poch* "Hungering One" reading rests on a word the core layer never lost.[30] When nothing in the classical sources settles a token, a surviving regional reflex can confirm both the root and its semantic range.[8,22]
- **Uniform directionality.** All three texts are offerings flowing up to Atziri, she receives and consumes; the worshippers give themselves.

- **"Alien" letters as evidence.** The /g/ and /r/ that do not fit the Yucatec-plus-Nahuatl core each revealed an additional source (Nahuatl *yancuic*; K'iche'); the /f/ revealed the Romance seam.
- **Cross-graft morphology.** The conlang freely puts Nahuatl affixes on Maya roots (*qexcan* = Maya *k'ex* + Nah. *-can*; *pochiti* = Maya *poch* + Nah. inchoative *-ti*), a structural signature worth using when parsing new tokens.

The end-to-end path from raw signal to interpretation, form-first at every step, is:



Form-first throughout: the search starts from the token's shape, never from the desired gloss.





14. STATUS

All five texts are documented, and the reconstruction now rests on a measured statistical footing (§17). Texts 1 and 2 are closed and source-validated against the game files; Texts 3, 4, and 5 are documented, with their unresolved tokens tracked in §10 rather than forced into the committed lexicon. The committed lexicon (§9) carries 65 tokens run through the two-gate hardening protocol (the original 62 plus three names later blind-certified by Battery E, §17.7), and the full morpheme index (§15) inventories roughly 147 elements across the whole reconstruction. Text 3, **Quemalani, the Elite Commander** (§6), contributes a five-line corpus with committed content tokens (folded into §9) and several flagged soft tokens (§10.4). Text 4, the **Drill Sergeant and Vaal Regiment** drill (§7), adds a fixed call-and-response built on committed roots (*Otsuks! ... U'te mucane!*), with its soft tokens (*a'tul, cheyel, Axba, quxzeh, Kiibsa'*) tracked in §10.9 and none promoted to §9.

Statistical standing (full detail in §17). The committed readings are distinguishable from chance: on the headline base of 65 committed tokens, 26 reconstruct under strict rules where noise would yield about 3 (binomial $p \approx 3 \times 10^{-18}$), holding when related tokens are collapsed to 56 distinct roots (25 of 56, $p \approx 5 \times 10^{-19}$, §17.7), so the all-coincidence explanation is strongly disfavored under the tested model. The robustly verified (hardened) core is 25 tokens: 38.5% of the committed lexicon by token, 44.6% by distinct root (the original 62-token run gave 22 tokens, 35.5% and 41.5%, retained as the prior epoch, §17.7). Per reading, a hardened gloss is genuine with probability around 75 to 81% at the data-anchored base rate, and correspondingly lower across a whole line. Every committed token now carries a tier from the confidence taxonomy (**H** hardened, **C** committed but latitude-dependent, **C*** committed with a logged competitor, **S** soft, **L** lore-anchored), and every new token is assigned one by the protocol in `HARDENING_PROTOCOL.md`.

Texts 1 and 2 are source-validated against the game files. Text 1 is the datamined `VaalSermon_01` to `08`; Text 2's sixteen lines match both Cuachic Vault bosses' chant pools exactly, with zero drift, and the speaker roles are fixed: **Zelina** (Priestess) sings the call, **Zolin** (Priest) the response. *Ma'oxe* is confirmed verbatim.

Text 5, the stray-capture log (§8), now holds three entries. The Kamasan Smith's *Ti ek tala jare'yantul!* parses fully on committed roots and added *tala* "to come" to the lexicon. **Ahuatotli, the Blind** (Nahuatl combat-cries) and **Mektul** (Yucatec combat-cries) contribute mostly skill-names, each given a candidate root but held soft in §10, none promoted to §9. All three speakers are gathered with the other figures in §12.

The Xinkan and Nahuan cross-check is complete (§13). Run against attested Xinka and Pipil/Nawat vocabulary, the committed lexicon turned up no cleaner fit, so no committed reading was revised; the living Nawat branch corroborates a Nahuatl stratum this far south without displacing a reading.

Open items are narrow, and now include the logged adversarial competitors. Still open: the exact K'iche' lemma behind *jare'* (its apostrophe form now attested in Text 5); the unadopted *Ma'oxe* alternates; *fukuur*'s phonetic fit (*fulgor* adopted but soft); *ta'* (*ti'* vs *taak*); and *ukto* in the Text 4 catechism, whose leading reading is now Nahuatl *ocotl* "pine torch" but stays soft (§10.7). Seven committed tokens carry a logged cross-language competitor (**C***, §10.12), the strongest being *itsok* vs Nahuatl *itzli* "obsidian"; these are live co-readings, not settled glosses. *Guatelitzi* has left the open list (an attested in-game name, *Guatelitzi, Architect of Flesh* of the Temple of Atzoatl, now in §9, §10.8), and two former holdouts have rule-abiding leading candidates in §10.4: *buxa* ← *ba'ax* "what /

what-thing" (keeps $x = /ʃ/$ intact) and *Eche lu* ← Spanish *échelo* "pour/cast it out" (supplies the *ch* /tʃ/ that *he'ela'* could not).

Project status: substantially complete, pending new material. With all five known texts documented, the lexicon hardened and tiered (65 committed tokens, after Battery E blind-certified three later onomastic names, *Quecholli*, *Panquetzaliztli*, *Ixchel*, §17.7), the syntax derived and blind-confirmed, and the null-model batteries closed out (the full latitude-by-access 2x2 as Batteries A-D, extended by Battery E, §17), the reconstruction has reached the limit of what the current corpus can support. The open items listed above are narrow, and most cannot be settled without evidence this corpus does not contain. The natural next input is simply more Vaal text: further dataminced lines, newly voiced barks, or fresh in-game captures. New material would help in three ways, in rough order of value: it would enlarge the sample and tighten the confidence intervals of §17, which are wide chiefly because the corpus is small; it would supply the minimal pairs needed to close specific open tokens (a second environment for *jare'*, *ta'*, or *ukto*, say); and it would test the §3 grammar against clauses it has not yet seen. Until then, the document is best read as stable rather than in progress. The method, the tooling (HARDENING_PROTOCOL.md and the null-model battery), and the confidence taxonomy are all in place, so any new token or line can be folded in and scored the same way without reopening the framework.

Corpus versioning (for future patches). Each revision of this document is tied to the game patches it covers (currently *Path of Exile 2* 0.5.0 and *Path of Exile 1* through 3.27, §1). When a patch adds Vaal text, audio, or lore, the procedure is fixed: snapshot the current version first, so the pre-patch logic is preserved; fold the new material into the corpus and re-run the tests (the null-model battery, the strict-latitude and adversarial gates, the blind syntax pass); and let the results stand wherever they fall, rewriting rules or the lexicon wholesale if the enlarged corpus demands it, on the same verifiable-data-first basis as the current work. A reading a new patch overturns is not deleted but moved to the §10 appendix with the patch that overturned it, so the record shows why each change was made. The hardened core (§17) is the most stable and should be the last to move; the soft and open parses (the starred tokens, §10) are where new data will land first.





15. MORPHEME INDEX

This index inverts the **entire** reconstruction, not the committed lexicon alone: every root and affix we have analysed anywhere in the document, with its source layer, gloss, a status mark, and the tokens it appears in. It is both a lookup and a consistency check, the *Appears in* column shows each element doing the same job across different words and figures.

Status records the confidence tier of the token(s) each element heads, on the §2b hardening taxonomy rather than the former committed/soft/candidate scale. The tiers, written out in full here for readability (§9 uses the same tiers in their short H / C / C* form): **Hardened**; **Committed** (latitude-dependent); **Committed (Competitor)**, committed but with a logged cross-language competitor; **Soft** candidate (§10); and **Lore**, an attested root whose support is a proper name or narrative rather than a hardened derivation (this tier absorbs the former "candidate" name-roots). The orthogonal lore tag that §9 marks as +L on some committed tokens is folded into the confidence tier here to keep the column to a single axis; see §9 for a committed token's full mark. Where an element heads several committed tokens at different tiers, all are listed, most-settled first (for example the relational *te' / ti'* is Hardened / Committed / Committed (Competitor) / Lore). A Hardened or Committed root and a Lore root are not equal evidence, and the column keeps them apart. Entries are grouped by source layer, then alphabetical. Rejected alternates (logged in §10 and §12) and unanalysable proper names (*Vaal*, *Zerphi*, and the opaque architects) carry no isolable morpheme and so are not listed; names sourced from outside the Mesoamerican palette are noted separately at the foot of the section.

Totals: 147 elements. By the best tier each element reaches: 32 Hardened, 38 Committed, 6 Committed (Competitor), 22 Soft, 49 Lore (equivalently, 76 head a committed token, 22 a soft one, and 49 rest on a name or lore anchor). The six added with the Battery-E names (*quechōl-*, *pan(tli)*, *quetza*, *-liztli*, *ix-*, *chel*) all head hardened onomastic names (§12.5–12.6).

Maya (Yucatec and Classic Maya)

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>a-</i>	Maya	you, your (Set A 2nd person)	Soft	a'tul
<i>akal</i>	Maya	pond; still water	Hardened	akal
<i>anaab</i>	Classic Maya	court-servant, attendant	Committed	anab
<i>-ane</i>	Maya	honorific; agentive plural	Committed / Committed (Competitor)	ikba'yucane, mucane
<i>at / yan</i>	Maya	presentative: behold, here is	Committed (Competitor)	A'te / U'Te / Yatle
<i>ba'</i>	Maya	self, reflexive	Committed	elba, ik'bala, ikba'yucane, kifba, nochbe
<i>ba'ax</i>	Maya	what (interrogative)	Soft	buxa, Axba
<i>-bil</i>	Maya	passive, participle	Committed	'Ibil
<i>che'</i>	Maya	tree, wood	Hardened	che', kahk-tche

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>chel</i>	Yucatec	rainbow (the goddess <i>Ix Chel</i>)	Hardened	Ixchel
<i>chiikul</i>	Maya	sign, omen	Lore	chikula'
<i>-e</i> (voc.)	Maya / Nah.	vocative	Committed	Ik'eche, Teoyuxtlane
<i>-ech</i>	Maya	you (2nd person)	Committed	Ik'eche
<i>ek'</i>	Maya	star; black, dark	Hardened	ek
<i>el / elel</i>	Maya	to burn	Committed	Ela, elba
<i>-el</i>	Maya	abstractive, intransitive	Committed	ik'el, cheyel
<i>-en / teen</i>	Maya	me, I (1st person)	Hardened	-en
<i>ha'</i>	Maya	water	Committed	sakilja
<i>haay</i>	Yucatec	to raze, level	Lore	Hayoxi
<i>han-</i>	Maya	to eat	Lore	ko'janti
<i>ich / ichil</i>	Maya	in, within; eye, face	Hardened	ich
<i>iitz</i>	Maya	sap, essence; resin	Committed (Competitor)	itsok / itzil
<i>ik'</i>	Maya	wind, breath, spirit; dark	Committed	ik'bala, ikba'yucane, Ik'eche, ik'el, ka'tse-ik
<i>il</i>	Maya	to see	Committed	'Ibil, inib
<i>in</i>	Maya	my, I	Soft	inib
<i>ix-</i>	Yucatec	feminine / agentive prefix	Hardened	Ixchel
<i>ka(j)</i>	Yucatec	dependent, subjunctive: that, when	Soft	ka
<i>ka'</i>	Maya	two	Lore	ka'tse-ik
<i>ka'a</i>	Maya	again	Committed	cha'tsoke
<i>k'áak'</i>	Maya	fire	Hardened	k'áak', kahk-tche
<i>k'áat</i>	Maya	to ask, want	Lore	ipkaat
<i>k'ex</i>	Maya	substitution, exchange; transform	Committed	Kextal, qexcan
<i>k'i'ik'</i>	Maya	blood	Committed	kifba
<i>ki'</i>	Maya	good, delicious, sweet	Hardened	ki'
<i>kíim / kíimil</i>	Maya	death, to die	Lore	kíimil, Kíibsa'
<i>ko'</i>	Maya	come (hortative)	Lore	ko'janti
<i>k'u</i>	Maya	god, sacred	Lore	Yugul
<i>kutz</i>	Maya	turkey, sacrificial bird	Committed (Competitor)	kutsen
<i>kuxtal / kux</i>	Maya	life, to live	Lore	kux / kuxkal / kuxte'
<i>k'ux</i>	Maya	to bite, gnaw; rancor	Hardened	k'ux (quxzeh)

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>le...o'</i>	Yucatec	the (article)	Lore	le / le'
<i>lu'um</i>	Maya	earth, ground	Lore	líimek
<i>ma'</i>	Maya	not, without (negation)	Hardened / Lore	ma, Ma'oxe
<i>máax</i>	Yucatec	who? (interrogative)	Hardened	máax
<i>mux</i>	Maya	to grind, crush	Lore	te'moxti
<i>muk'</i>	Maya	strength, force	Committed / Committed (Competitor)	mucane, mujuk'
<i>náach</i>	Maya	far, distant	Hardened	náach
<i>nuk / nojoch</i>	Maya	big, great	Committed	ta'nuk
<i>ól</i>	Yucatec	spirit, will	Lore	Ma'oxe (alt)
<i>poch</i>	Maya	hungry, gluttonous, craving	Committed (Competitor)	pochiti
<i>pox</i>	Highland Maya	posh, ceremonial corn/ cane liquor	Soft	puxe
<i>pul / puul</i>	Maya	to throw, cast, hurl	Hardened	upulché
<i>puuch'</i>	Maya	to crush	Hardened	uch' / pu uch'
<i>sak</i>	Maya	white, pure	Committed	sakilja
<i>taak / taak'</i>	Maya	to want	Soft	ta'
<i>taal</i>	Maya	to come	Committed (Competitor)	tala, a'tul
<i>te' / ti'</i>	Maya	relational: to, for, at (also <i>te'</i> tree)	Hardened / Committed / Committed (Competitor) / Lore	te, Ti, ko'janti, xu'te, ta'
<i>til</i>	Maya	to burn, kindle	Committed	til
<i>to'on</i>	Maya	we, us	Soft	Donuks
<i>ts'ook</i>	Maya	end, to finish	Committed / Lore	cha'tsoke, Tzokan'te / tzok
<i>tsuk</i>	Yucatec	group, cluster, company	Soft	Otsuks
<i>tul</i>	Maya	to decline, wane	Hardened	tul
<i>u-</i>	Maya	Set A 3rd person: his, her, its	Hardened	u
<i>uk'</i>	Maya	to drink	Hardened	uch' / pu uch', Donuks
<i>uts</i>	Yucatec	good	Committed	Yutsal / yutsal, Teoyuxtane
<i>waaj</i>	Maya	bread; altar-offering	Hardened	waaj, xecwa
<i>xiib</i>	Yucatec	male, man	Lore	Xibaqua
<i>xok</i>	Maya	to count, read	Lore	Ma'oxe
<i>xul</i>	Maya	end, limit	Committed	xu'te
<i>yax</i>	Maya	first, new, green	Lore	yax / Yaxe
<i>yuk</i>	Yucatec	to shake, set trembling	Lore	Yugul
<i>-s</i>	Maya	causative	Soft	Kíibsa'

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
-a'	Maya	deictic, terminal	Soft	K'úbsa'

K'iche' (Highland Maya)

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>are'</i>	K'iche'	focus: it is, that is	Committed	jare
<i>kame</i>	K'iche'	death (in <i>Camazotz</i>)	Lore	Kamasa
<i>ko-</i>	K'iche'	hard, strong, firm	Soft	koriek
<i>kor-</i>	K'iche'	to loosen	Lore	koriek (alt)
<i>k'or-</i>	K'iche'	to knock, tap	Lore	koriek (alt)
<i>ri</i>	K'iche'	the (article)	Committed	jare
<i>-(i)rik</i>	K'iche'	inchoative: to become X	Soft	koriek
<i>sotz'</i>	K'iche'	bat (in <i>Camazotz</i>)	Lore	Kamasa
<i>xib'</i>	K'iche'	fear (root of <i>Xibalba</i>)	Lore	Xibaqua

Nahuatl

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>ahua-</i> / <i>ahuatl</i>	Nah.	oak; thorn	Lore	Ahuana, Ahuatotli
<i>aocmo</i> / <i>ayoc</i>	Nah.	no more, no longer	Committed	Aiokmo, 'Ayok
<i>atl</i> / <i>ātl</i>	Nah.	water	Committed	atla, Atziri, Atzoatl
<i>āzcatl</i>	Nah.	ant	Lore	Azcapa, azcado
<i>calli</i>	Nah.	house, structure	Lore	Kuetzakala
<i>-can</i>	Nah.	locative: place of	Committed	qexcan
<i>chantli</i>	Nah.	home, dwelling	Committed	i'chian
<i>chiya</i> / <i>chīa</i>	Nah.	to watch, wait, lie in wait	Soft	cheyel
<i>choloa</i>	Nah.	to flee, leap	Lore	Cholotl
<i>cua</i> / <i>qua</i>	Nah.	to eat	Lore	quiquate, Citaqualotl
<i>cualli</i>	Nah.	good, worthy	Hardened	kujkuali
<i>cuauh-</i>	Nah.	tree, eagle	Lore	huatat, Guatelitzi (superseded)
<i>cuitla-</i>	Nah.	rear; behind	Lore	cutlotl
<i>eztli</i>	Nah.	blood	Hardened	Eztli Pilli
<i>i-</i>	Nah.	his, her (3rd-person possessive)	Committed	i'chian
<i>-liztli</i>	Nah.	action/result nominalizer	Hardened	Panquetzaliztli
<i>mā</i>	Nah.	optative: may, let	Hardened / Committed	ma, kilya
<i>māitl</i> / <i>-mac</i>	Nah.	hand; loc. "in the hands of"	Lore	Yaomac

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>mahui</i>	Nah.	to fear	Lore	Mahuxotl
<i>matlatl</i>	Nah.	net, snare	Lore	Matatl
<i>nochi / mochi</i>	Nah.	all, every	Committed	nochira, nochbe, te'moxti
<i>ocotl / ocutl</i>	Nahuatl	pine torch, firebrand	Soft	ukto
<i>-pa(n)</i>	Nah.	locative: at	Lore	Azcapa, Zilquapa
<i>pan(tli)</i>	Nah.	banner, flag	Hardened	Panquetzaliztli
<i>paqui</i>	Nah.	to be glad	Lore	Paquate
<i>pilli</i>	Nah.	noble, prince	Hardened	Eztli Pilli
<i>pōchōtl</i>	Nah.	ceiba, silk-cotton tree; protector	Lore	pochiti (co-reading)
<i>poloa</i>	Nah.	to lose, destroy	Lore	Quipolatl
<i>poyahua</i>	Nah.	to darken, redden	Soft	puyao
<i>quechōl-</i>	Nah.	roseate spoonbill; precious-feather bird	Hardened	Quecholli
<i>quetza</i>	Nah.	to raise, erect (verb root of <i>quetzal(li)</i>)	Hardened	Panquetzaliztli
<i>quetzal(li)</i>	Nah.	quetzal-plume; precious (from <i>quetza</i> "to raise")	Lore	Kuetzakala, Ketzuli
<i>qui-</i>	Nah.	3rd-person object prefix	Soft	quiquate
<i>quil-</i>	Nah.	green, verdant	Committed	kilya
<i>teōtl</i>	Nah.	god, divine	Committed	Teoyuxtlane
<i>tepetl</i>	Nah.	hill, mountain	Lore	xictep
<i>-ti</i>	Nah.	inchoative (to become)	Committed (Competitor)	pochiti
<i>tla-</i>	Nah.	nonspecific object prefix	Committed	tlayeb
<i>tlācah / tlāl-</i>	Nah.	people; land, earth	Lore	Tlaxye'
<i>tlahtoa</i>	Nah.	to speak	Lore	tatlat (alt)
<i>-tlān</i>	Nah.	place of (locative)	Committed	atla, Teoyuxtlane
<i>tlapechtli</i>	Nah.	platform, scaffold, altar-bed	Committed	tlapēc / yotlapēc
<i>tlatla</i>	Nah.	to burn	Lore	tatlat
<i>tlayohua</i>	Nah.	night; it grows dark	Committed	tlayeb
<i>tototl</i>	Nah.	bird	Lore	Ahuatotli
<i>tzacua</i>	Nah.	to close, imprison	Lore	Tzamoto
<i>-tzin</i>	Nah.	reverential	Committed	Atziri, Guatelitzi
<i>tzoatl</i>	Nah.	dirty water, slops	Soft	Atzoatl
<i>-tl / -tli / -atl / -otli</i>	Nah.	absolutive (noun) suffix	Lore	Cholotl, Matatl, Quipolatl, Citaqualotl, Juatalotli
<i>xi-</i>	Nah.	imperative: do!, make!	Hardened	xi, xictep

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>Xīpe</i>	Nah.	the Flayed One (deity)	Lore	Xipocado
<i>xōtla</i>	Nah.	to kindle, blaze, glow	Lore	Xartlene
<i>xoxoc-</i>	Nah.	raw, fresh, green	Soft	soxsal
<i>-ya</i>	Nah.	inchoative: to come to be	Committed	kilya
<i>yancuic</i>	Nah.	new	Lore	Gyan'uks
<i>yāō(tl)</i>	Nah.	war, enemy	Lore	Yaomac
<i>yocoŷa</i>	Nah.	to create, devise	Lore	yuquia

Spanish / Romance

ELEMENT	SOURCE	GLOSS	STATUS	APPEARS IN
<i>-ada</i>	Romance	past-participle suffix	Hardened	ascensionada / ascenada
<i>ascensión</i>	Spanish	ascension, rising	Hardened	ascensionada / ascenada
<i>daca</i>	Spanish	give it here (imperative)	Soft	daka
<i>echar / eche</i>	Spanish	to throw, pour, cast out	Soft	Eche lu
<i>fulgor</i>	Spanish	brilliance, blaze	Soft	fukuur
<i>jefe / xefe</i>	Spanish	chief, commander	Hardened	xefe
<i>lo</i>	Spanish	it (object pronoun)	Soft	Eche lu
<i>otra</i>	Spanish	another, other	Committed	u'tra

Names sourced outside the Mesoamerican palette. A handful of figure-names are not built from palette morphemes and so carry no rows above; they are recorded here for completeness: *Apep* (Egyptian *Apep* / *Apophis*, the chaos-serpent), *Ralakesh* (Sanskrit *Rākṣasa*), *Arakaali* (a blend of Greek *Arachne* and Hindu *Kālī*), and the *Vaal Omnitect* (Latin *omni-* "all" + *-tect*, from *architect*). See §12.3 and §12.4.





16. CONTRIBUTORS AND ACKNOWLEDGMENTS

This reconstruction has been improved by members of the Path of Exile community, credited here and at the relevant points in the text.

- **AdministratorQotra** corrected the phonological framework. He flagged that an earlier draft overgeneralized the Nahuatl consonant system, conflating the plain glottal stop (the saltillo, which Nahuatl has) with a glottalized ejective series (which it does not). That correction is the basis of the saltillo distinction in the §2.4 key and prompted the full phoneme-inventory tables now set out in §2.3.
- **Tenebris-Umbra** identified *Guatelitzi* as an attested in-game proper name, *Guatelitzi, Architect of Flesh* of the Temple of Atzoatl (§10.8), resolving a token that had previously been reconstructed from its phonology.
- **Murky-Definition-625** compiled the community reference on Vaal architect lore[48], drawn on for the architect roles in §12.4.
- **Substantial_Bat_8440** supplied the fuller Drill Sergeant catechism (Text 4, §7), including the second response *Atziri!*, beyond the opening and variant calls of the earlier capture.





17. STATISTICAL ANALYSIS: LEXICON HARDENING AND SYNTAX CONFIRMATION

To test whether the committed readings are distinguishable from chance, three blind experiments were run against a null model. Full protocol, seeds, and per-token data live beside this document: `NULL_MODEL_PROTOCOL.md`, `NULL_MODEL_RESULTS.md`, `ADVERSARIAL_RESULTS.md`, `HARDENING_PROTOCOL.md`, and the run history in **EXPERIMENT_LOG.md**.

17.1 METHODOLOGY

- **Null model.** Each run mixes 100 phonotactically Vaal-like nonsense strings (generated from a model trained only on Vaal surface forms; fixed seed; reproducible) with 30 real committed tokens, shuffles them with labels hidden, and has a fresh decoder blind to the key root every item at the project's standard. Any root found for a nonsense string is a false positive by construction.
- **Five batteries,** 5 seeds each (1729, 9001, 271828, 42, 55555): **A** loose latitude / offline dictionaries; **B** loose / online Classical Nahuatl; **C** strict / offline; **D** strict / online (A-D each 500 pseudo + 150 real pooled, forming the latitude x dictionary-access 2x2, §17.3); **E** strict / online certification of the three post-battery names as blind plants (§17.7). A-D are the original epoch; E extends it to the enlarged headline base.
- **Adversarial pass.** Each committed token, surface form only, is handed to a blind adversary that hunts the best root in a *different* source language. The token falls if a strong different-language, different-meaning competitor exists.
- **Hardening.** A committed token earns the top tier (**H**) only if it reconstructs under the strict latitude AND survives the adversarial pass (`HARDENING_PROTOCOL.md`).

17.2 METRICS

For the pseudo population P and real population R, each item labeled C (committed-quality), `soft`, or `none`; `found` = C or `soft`. P_C is the count of pseudo labeled C, $|P|$ the pseudo total, etc.

- **FPR(C)** = $P_C / |P|$. False-positive rate: the fraction of nonsense strings wrongly certified committed-quality. Lower is better.
- **FPR(C+soft)** = $(P_C + P_{\text{soft}}) / |P|$. Same at the loosest bar (any root found).
- **TPR(C)** = $R_C / |R|$. True-positive rate: the fraction of real tokens certified committed-quality. Higher is better.
- **TPR(found)** = $(R_C + R_{\text{soft}}) / |R|$. Same at the loosest bar.
- **Discrimination(C)** = $TPR(C) - FPR(C)$. Absolute separation between signal and noise; 0 means the method cannot tell them apart.
- **Enrichment(C)** = $TPR(C) / FPR(C)$. How many times likelier a C is on a real token than on noise; 1x is worthless.

17.3 RESULTS (POOLED PER BATTERY)

BATTERY	FPR(C)	FPR(C+SOFT)	TPR(C)	TPR(FOUND)	DISCRIMINATION(C)	ENRICHMENT(C)
A loose / offline	0.062	0.670	0.207	0.767	+0.145	3.3x
B loose / online	0.080	0.576	0.173	0.653	+0.093	2.2x
C strict / offline	0.034	0.064	0.127	0.180	+0.093	3.7x
D strict / online	0.046	0.094	0.140	0.173	+0.094	3.0x

The four batteries form a latitude x dictionary-access 2x2 (loose/strict against offline/online). Battery D (strict + online) was run last, with fresh blind agents grepping the offline dictionaries plus consulting the Wired Humanities Nahuatl Dictionary and the Gran Diccionario Nahuatl online (NULL_MODEL_RESULTS.md, blind_test_results_tight_online_s{seed}.csv).

Findings: (1) the **soft tier is near-noise**, under loose latitude a root is found for about 67% of nonsense versus 77% of real tokens, a ten-point gap, so a soft reading is close to what decoding noise yields and is treated as a candidate, not evidence. (2) The **committed tier clears the noise floor by about 3x**, stably across seeds. (3) Adding online Nahuatl access did **not** widen the gap on either latitude: under loose rules (B vs A) it slightly narrowed it, and under strict rules (D vs C) it lifted FPR and TPR together (0.034 to 0.046 and 0.127 to 0.140), leaving discrimination flat (+0.094), so the completed 2x2 confirms dictionary coverage is not the bottleneck, latitude is. (4) Tightening latitude (C) collapsed noise-matching (FPR at the loose bar 0.670 to 0.064) but also the committed lexicon's own reconstructability (0.767 to 0.180), showing most committed readings depend on the permissive latitude. (5) The hypothesis that Vaal is **entirely random noise is strongly disfavored under the tested model**: 23 of the 62 committed tokens reconstruct under strict rules, against ~3 expected if each were noise passing at the **4.6% strict-and-online false-positive rate (Battery D, the resources the real decode uses; the offline 3.4% would give an even smaller p)** (binomial $p \approx 2 \times 10^{-15}$, a ~12.2-sigma departure). Because some committed tokens share a root and so are not independent trials, the same test taken over the 53 distinct roots (§17.7) gives 22 strict passes against ~2.4 expected ($p \approx 4 \times 10^{-16}$): the rejection is not an artifact of counting related forms more than once. (These are the original A-D figures; folding in the three Battery-E names gives the headline base of 26 of 65 tokens, $p \approx 3 \times 10^{-18}$, and 25 of 56 distinct roots, $p \approx 5 \times 10^{-19}$; see the Battery E note below and §17.7.)

Battery E (target certification, not a floor re-measurement). After the 2x2, three names hardened post-battery (*Quecholli*, *Panquetzaliztli*, *Ixchel*) were run as blind plants under the strict-and-online conditions of Battery D, across the same five seeds, shuffled among pseudo distractors and real controls (BATTERY_E_RESULTS.md). All three passed strict Gate-1 in every pass (15/15) and the real controls calibrated to their known tiers, certifying the three as null-tested and lifting the headline base to 65 tokens / 26 strict passes (§17.7). Battery E did not re-measure the noise floor; the 4.6% Battery D floor still governs the null-rejection test.

17.4 ADVERSARIAL SURVIVAL AND THE HARDENED TIER

The adversarial pass found no genuine cross-language competitor for about 88% of committed tokens (52 of 59 testable, counting *pul*); seven have one (logged in §10.12), the strongest being *itsok* vs Nahuatl *itztli* "obsidian." Combining the strict-latitude decode with adversarial survival, of the 62 committed tokens in the original A-D run **22 are Hardened (H)**, 7 are committed with a logged competitor (C*), and 33 are committed but latitude-dependent (C); the three Battery-E names (§17.7) add three more H, giving **25 H of 65** in the enlarged headline

base. The hardened core, the readings that clear the noise floor by a real margin, is: *akal, ascensionada, che', ek, -en, Eztli (Pilli), ich, k'áak', ki', kujkuali, k'ux, ma, máax, náach, pul, Ti, tul, u, uch', waaj, xefe, xi*. (Several are short function words or clean loans that reconstruct trivially; H marks robustness to testing, not depth of analysis.) Collapsing morphologically related tokens to distinct roots (§17.7) leaves the split at 22 H / 6 C* / 25 C of 53 (25 H of 56 with the three Battery-E names), because the merged duplicates are almost all non-hardened relatives.

Going forward, every token is assigned a tier via the two-gate test in `HARDENING_PROTOCOL.md`, and no token may carry H without both gates recorded. (The hardening battery first covered 61 committed tokens; a 62nd, *pul* "to throw, cast," was restored from the pre-import chat copy and then run through both gates, passing as Hardened, so the committed lexicon is 62 tokens with 22 hardened. Battery E (§17.7) later blind-certified three more Hardened names, *Quecholli, Panquetzaliztli, Ixchel*, giving the enlarged headline base of 65 tokens with 25 hardened; 62 / 22 is the prior epoch.)

17.5 CONFIDENCE INTERVAL FOR THE TRANSLATION

Full working, with Wilson 95% intervals and the base-rate sensitivity table, is in `STATISTICAL_SUMMARY.md`.

Measured coverage (assumption-free). The hardened core of the committed lexicon is **38.5%, 95% CI [27.6%, 50.6%]** (25 of 65 committed tokens, the enlarged base after Battery E, §17.7); counted over the 56 distinct roots it is **44.6%, 95% CI [32.4%, 57.6%]** (25 of 56). The original 62-token run gives **35.5% [24.7, 47.9]** and **41.5% [29.3, 54.9]** (22 of 62 / 53) and is retained as the prior epoch. Composition note: the three Battery-E names are exact attested lexemes and so are easy plants; they lift the coverage figure slightly and are not representative of the harder text tokens, though they do not affect the null rejection, which is set by the noise floor, not by plant difficulty. Either way, between a third and two-fifths of the vocabulary the translations rest on is robust to both the strict-latitude and adversarial tests; the remainder is latitude-dependent, plausible but not distinguishable from chance at the strict bar. One caution on reading these figures: they are the *protocol's certification rate* (how often a reading survives hardening), not a direct measure of translation correctness. The correctness estimate is the per-reading positive predictive value below, which folds in the base rate *b*.

Per-reading confidence by tier. Treating the tests as a diagnostic, the probability that a reading is a genuine root rather than a chance dictionary hit is the positive predictive value, $PPV = (TPRb) / (TPRb + FPR(1-b))$, where *b* is the base rate (fraction of Vaal built from real palette roots). Because *b* is uncertain, each tier is given as a range from pessimistic (low *b*) to optimistic (high *b*) so a reader can take their own outlook; the data-anchored *b* ≈ 58%* (the §12 name roster, §17.6), a little above the 50% neutral prior. The anchored column below spans *b* = 50–58%. The hardened row uses the strict-and-online null rates (Battery D, TPR 0.140 / FPR 0.046, §17.3, the resources the real decode uses); the other rows use their looser bars.

TIER	PESSIMISTIC (B=20%)	DATA-ANCHORED (B≈50-58%)	OPTIMISTIC (B=75%)
Hardened (H)	43%	75–81%	90%
Committed, latitude-dependent (C)	45%	77–82%	91%
Committed + competitor (C*)	~23%	~39–41%	~46%
Soft / candidate (S)	22%	53–61%	77%

C* values are the confidence in the *specific* gloss (the root is genuine at the C level, but a comparable cross-language competitor roughly halves which reading is right). Soft readings straddle a coin-flip across the whole range and are not evidence on their own.

Sentence-level confidence (floor and ceiling). Per-token confidence combines across a line into an *interval*, not a point. The **floor** is the independence case (tokens uncorrelated), the product of the per-token values; the **ceiling** is the Fréchet upper bound, $P(\text{line}) \leq \min_i P(\text{token}_i)$, reached when syntax forces the remaining words once the least-certain one is fixed. Real language is positively correlated, so the true value lies between. Both vantage points below use the data-anchored prior ($b \approx 58\%$, §17.6).

The average line. Across the connected corpus (Texts 1–4 and the Kamasan Smith: 51 lines, 186 tokens) a line runs **3.65 tokens** on average. A line of that length is:

- all Hardened: [46%, 81%]
- all Committed: [49%, 82%]
- all Soft: [17%, 61%]

The texts run mostly on committed and hardened vocabulary, so a typical line sits in the [~49%, ~81%] band, an even chance or better of being wholly correct at the pessimistic end, better than four-in-five at the optimistic end.

Two concrete lines, best and worst. No short line in the corpus is built purely of hardened tokens (a hardened word almost always shares its line with a softer one), so the strongest real cases are committed-grade. The **best case** is *Tlayeb kifba!* "In the dark, the heart!", two committed tokens, at [68%, 82%], the highest-confidence line in the corpus; the most hardened-heavy complete line, *Atziri, Atziri, ascenada akal!* "risen into the eternal waters" (two hardened content words, *ascenada* and *akal*, plus the queen's committed name), reaches [44%, 81%], its hardened tokens lifting the ceiling while the four-token length holds the floor near even. The **worst case** is the Kamasan Smith's *Ti ek tala jare'yantul!* (six tokens, two of them competitor-flagged C*): [7%, 41%], dragged down by length and by the two C* tokens whose ~0.41 confidence caps the ceiling. Most lines fall between these, in the average band above.

The product alone is a lower estimate, not the answer (and not even an absolute floor: negative correlation could dip below it). The pattern holds, longer lines are less certain than short ones and than any single token, so the honest unit of confidence remains the individual token.

17.6 THE BASE RATE B, ESTIMATED FROM DATA

The PPVs above turn on **b**, the fraction of Vaal actually built from real palette roots, which cannot be observed directly. Rather than assume a value, we bound it empirically.

Proper-noun proxy (palette-scoped). The named roster in §12 is a closed, un-cherry-pickable sample of the developers' naming behaviour. Four of its figures are documented borrowings from *outside* the Mesoamerican palette entirely, Apep (Egyptian), Ralakesh (Sanskrit), Arakaali (Greek/Hindu), and the Omnitect (Latin); these were never attempts at a palette root and are out of scope for this estimate. Among the **33 in-scope names, 19 carry an attested palette root (Nahuatl, Maya, K'iche', or Spanish): $\approx 58\%$** . Many rest on structural markers hard to produce by chance, the Nahuatl *-tl* / *-tli* / *-atl* absolutive endings on *Matatl*, *Cholotl*, *Citaqualotl* and the like, which is why the name roster is a comparatively strong signal.

Bounds on that figure. Counting only clean, fully-worked parses (setting soft candidates aside) gives a conservative floor of $\approx 15\text{--}18\%$; the 58% is soft-inclusive and so upper-leaning. Folding the four out-of-scope borrowings back into the denominator drags the ratio to $19/37 \approx 51\%$, but that penalizes *b* for deliberate design choices the palette was never meant to catch, so it understates palette usage and is not the right denominator here.

Mixture model, and why it is not usable here. In principle $b = (O - \text{FPR}) / (\text{TPR} - \text{FPR})$ from the corpus's observed pass rate *O*. With our numbers this overflows ($b > 1$), because the null-model TPR (0.140, Battery D) was measured on a deliberately hard subset of plant tokens and understates the real committed lexicon's strict-pass rate (37%); when *O* exceeds TPR the equation has no valid solution. So the mixture model cannot be applied to the data we have, and any value forced out of it (for instance by mixing a strict-tier *O* with loose-tier rates) is an artifact, not a derivation.

Consequence. The palette-scoped proxy puts ***b* $\approx 58\%$** , above the 50% neutral prior, so the data-anchored confidence sits in the upper-middle of the reported range: a hardened reading is genuine with probability $\approx 75\%$ (***b* = 50%) to $\approx 81\%$ (*b* = 58%).** We still do not stretch to $b = 75\%$; the roster does not reach that far. The full PPV-versus-*b* table is kept in `STATISTICAL_SUMMARY.md` for readers who prefer a different prior.

17.7 NON-INDEPENDENCE OF RELATED TOKENS (DISTINCT-ROOT RECOUNT)

The counts above treat each committed token as an independent trial. That overstates the evidence wherever several tokens descend from one root: *ik'bala*, *ikba'yucane*, *Ik'eché*, and *ik'el* are all built on Yucatec *ik'* "spirit, breath," so they are four surface forms of a single lexical bet, not four independent confirmations. Counting them singly inflates both the sample size and the strict-pass tally. This subsection reruns the tests with morphological relatives collapsed to their shared root, so the result is visible on both bases. (The recount below is stated on the original 62-token A-D run; the enlarged headline base, with the three Battery-E names, is in "A third base" at the end of this subsection.)

Levels of dependence. Related observations inflate the evidence at four nested levels. The recount here corrects the first two (which a raw token count conflates); the third is handled separately as syntax (§17.8), and the fourth is flagged rather than counted:

LEVEL	WHAT REPEATS	EXAMPLE	HOW THIS DOCUMENT TREATS IT
1. Token	the same surface form, reused	repeated <i>Atziri</i>	counted per occurrence for coverage, per lemma for the independence claim
2. Root	different forms of one lemma	<i>ik'bala</i> , <i>ik'el</i> , <i>Ik'eché</i> all from <i>ik'</i>	collapsed to 53 distinct roots (below); removes 9 dependent forms
3. Construction	the same grammatical frame, reused	pronominal article plus noun	tested as syntax, not counted as lexical evidence (§17.8)
4. Semantic field	glosses clustering in one domain	water, dark, and blood imagery	not corrected numerically; flagged as a caveat on thematic over-reading

Grouping rule. Two committed tokens are merged when they share the same attested head root (the same §9 source lemma), regardless of affixes or compounding partners. Shared *affixes* (the reflexive *-ba'*, the honorific *-ane*) do not merge tokens whose head roots differ. Sixteen of the 62 tokens collapse into 7 root families:

ROOT	TOKENS MERGED	STRICT (GATE 1)
<i>ik'</i> "spirit, breath"	ik'bala, ikba'yucane, Ik'eche, ik'el	all fail
<i>aocmo</i> "no more"	Aiokmo, 'Ayok	all fail
<i>ātl</i> "water"	atla, Atziri	all fail
<i>el</i> "burn"	Ela, elba	all fail
<i>muk'</i> "strength"	mucane, mujuk'	all fail
<i>k'ex</i> "transform"	Kextal, qexcan	all fail
<i>ti' / te'</i> relational	te, Ti	both pass

The 62 tokens thus reduce to **53 distinct roots** (the three Battery-E names each add a further distinct root, so the enlarged headline base has **56**; see "A third base" below). Only one family (*ti'/te'*) contained a strict-pass, so the collapse removes exactly one strict-pass (23 to 22) while dropping nine mostly-failing forms from the denominator.

Results on both bases.

METRIC	PER TOKEN (N = 62, A-D)	PER DISTINCT ROOT (N = 53, A-D)
Strict Gate-1 passes	23 (expected 2.9)	22 (expected 2.4)
Null-rejection p	2×10^{-15}	4×10^{-16}
Hardened tier	35.5% [24.7, 47.9]	41.5% [29.3, 54.9]
Gate-1 strict	37.1% [26.2, 49.5]	41.5% [29.3, 54.9]
Gate-2 adversarial	88.7% [78.5, 94.4]	90.6% [79.7, 95.9]
Tier split H / C* / C	22 / 7 / 33	22 / 6 / 25

Reading. The dependency correction does not weaken the case; it slightly strengthens it. The related tokens are concentrated in the strict-failing families (*ik'-*, *el-*, *muk'-*, *aocmo-*, *ātl-*, *k'ex-*), so removing the duplicates prunes the denominator faster than the signal: 22 of the 23 strict passes survive as distinct roots, now over a count of 53 rather than 62. The hardened fraction therefore rises from 35.5% to 41.5% on the original run (38.5% to 44.6% on the enlarged headline base, "A third base" below), and the null model stays rejected at the same overwhelming level (p on the order of 10^{-15} to 10^{-16} , and 10^{-18} to 10^{-19} on the enlarged base). The recount is defensive, not promotional: non-independence of related forms is a standard objection, and here it overturns none of the conclusions.

Both bases are kept on purpose. The per-token figures are not superseded. They are the honest raw account of how the committed lexicon behaves form by form; the per-root figures are the honest account of how it behaves lemma by lemma. Neither is privileged. The per-token base is the more conservative denominator for the *coverage* claims (it counts every attested form the translations actually use), the per-root base is the correct one for the *independence* claim (it counts each lexical bet once). Where a single headline number is needed, the per-token figure is quoted first with the per-root figure beside it.

A third base: folding in the post-battery hardened names. Three onomastic tokens (*Quecholli*, *Panquetzaliztli*, *Ixchel*, §9, §12.5–12.6) were hardened after the original four-battery run. Rather than fold them in post-hoc, they

were then put through the same machinery as blind plants: **Battery E** (`BATTERY_E_RESULTS.md`, §17.3) ran all three, shuffled among pseudo distractors and real controls, past five fresh blind decoders under the strict-and-online conditions of Battery D. They passed 15/15 (each 5/5), the controls calibrated to their known tiers, and no cross-language competitor surfaced. The three are therefore null-tested by the same procedure as the original lexicon, so the enlarged base below now carries the headline, with the original 62-token run (Batteries A-D) retained as the prior epoch.

METRIC	PRIOR EPOCH (62 TOKENS / 53 ROOTS, A-D)	HEADLINE BASE (65 TOKENS / 56 ROOTS, + BATTERY E)
Hardened tier, per token	35.5% [24.7, 47.9]	38.5% [27.6, 50.6]
Hardened tier, per distinct root	41.5% [29.3, 54.9]	44.6% [32.4, 57.6]
Strict Gate-1 passes, per token	23 (expected 2.9)	26 (expected 3.0)
Null-rejection p, per token	2×10^{-15} (~12.2 sigma)	3×10^{-18} (~13.6 sigma)
Null-rejection p, per distinct root	4×10^{-16} (~12.8 sigma)	5×10^{-19} (~14.3 sigma)
Hardened PPV band (data-anchored)	75–81%	75–81%

(The enlarged distinct-root count assumes the three names add three new roots; if *quetza* in *Panquetzaliztli* is treated as a root already present in the corpus, the per-root figure is $25/55 = 45.5\%$.) Each name is counted as one observation by its head root rather than split into its constituent morphemes (*Panquetzaliztli* as a single *quetza*-headed bet, not *pan* + *quetza* + *-liztli*; *Ixchel* as a single *chel*-headed bet, not *ix-* + *chel*); the morphemes themselves are catalogued in §15, but splitting one naming decision into several strict-passes would manufacture the independence this recount exists to remove, and decomposition would if anything shrink the count, since *quetza* is the root already behind *Ketzuli* and *Kuetzakala* (§12.3). The enlarged base moves every figure slightly in the favourable direction and rejects the noise null even more decisively, yet changes no conclusion; the per-reading PPV band does not move at all, since three confirmed-real names cannot shift the base-rate estimate (§17.6) by more than a point. Both columns stay on the page: the enlarged base is the current headline (null-tested via Battery E), and the original 62-token run is retained beside it as the prior epoch, since it is the larger, harder-plant sample the four-battery program was built on. Re-running the full battery when new material arrives remains the versioning trigger (§14).

17.8 CONFIRMATION OF THE SYNTACTIC ANALYSIS (§3)

The grammar in §3 was tested the same way the lexicon was, blind and against chance. Three fresh analysts were each given only the Vaal corpus (surface lines, English translations, and a token glossary) and the isolation rule, with no access to §3's conclusions, and were asked to derive the grammar from distribution alone. Full protocol and per-analyst results are in `SYNTAX_CONFIRMATION_PROTOCOL.md` and `SYNTAX_EXPERIMENT_LOG.md`.

Six testable conclusions were scored: (1) clause type is set by a clause-initial particle; (2) article, possessive, and modifier precede the noun; (3) the possessed precedes the possessor; (4) verb precedes object; (5) equational clauses take no copula; (6) operators are proclitic and derivation is suffixed.

Result. All three analysts confirmed all six conclusions: 18 of 18 confirmations, a 100% agreement rate (Wilson 95% CI [82.4%, 100%]); per conclusion 3 of 3 (Wilson 95% CI [43.8%, 100%]). The confidence gradings the

analysts assigned tracked ours: all three rated verb-before-object as the single medium-confidence claim, since it rests on one clean transitive clause (*Xi daka puxe*), and all three independently reproduced the open-questions list (the particles *ka* and *ti*, the *le le* doubling, the segmentation of *jare'yantul*). One analyst added a refinement now recorded in §3.4 and §3.7: *tlayeb* "dark" may be a fronted locative in some lines rather than a pure attributive. Counting that as a partial rather than a full confirmation lowers the rate only to 17 of 18, 94.4% (Wilson 95% CI [74.2%, 99.0%]). A formal inter-rater kappa is not reported: with unanimous confirmation on a single category it reduces to 0/0 and is undefined, so the negative control below is the substantive specificity check in its place.

Against chance. Under a null in which each analyst picks a word-order value at random, three analysts agreeing on one binary parameter has probability $(1/2)^3 = 0.125$, and all six parameters reaching unanimous agreement has probability about 3.8×10^{-6} (about 2.6×10^{-9} if the parameters are treated as three-way rather than binary). The reproducibility is far beyond coincidence.

Negative control (specificity). Confirmation shows the grammar is reproducible; a control shows it is specific to the real corpus. Two further blind analysts were given a scrambled corpus, the same lines and glossary with the words randomly reordered within each line (seed 1729), which destroys word order while leaving vocabulary and morpheme shapes intact, and were asked the same questions. The four word-order conclusions collapsed: both analysts reported "undetermined / no consistent order" for clause-initial type marking, noun-phrase order, possessed-before-possessor, and verb-object, each of which the real corpus had yielded 3 of 3. Counting whether an analyst recovers a consistent order rule, the real corpus gives 12 of 12 (3 analysts by 4 rules) and the scrambled corpus 0 of 8 (2 analysts by 4 rules), a clean dissociation (Fisher exact p about 7.9×10^{-6}). The two order-independent conclusions, the absence of a copula and the front-operator / back-suffix morphology, held in both conditions, which is the correct result: neither depends on word order, so neither is evidence of linear syntax, and the control properly declines to credit them as such. Vaal's word-order regularities are therefore real signal, not an artifact of analyst expectation, while the copula and morpheme-edge observations are true but weaker, since they would survive even in a bag of words. Full run in `SYNTAX_EXPERIMENT_LOG.md`.

The honest caveat. The analysts shared one corpus and one glossary, so this measures that the §3 grammar is reproducibly derivable from the data, not that the grammar was independently the designers' intent. It is the syntactic analogue of the base-rate limit on the lexicon (§17.7): the test shows the reading is not idiosyncratic to one analyst, which is what a small closed corpus can support, and no more. Two structural points strengthen it within that limit: the conclusions are about surface position and so are largely independent of whether individual glosses are correct, and the corpus, though small (about two dozen clauses), is closed and exhaustively examined.

17.9 ORTHOGRAPHIC HARDENING OF THE CORRESPONDENCE RULES (§2.6)

The stylization rules in §2.6 cannot be tested against sound without audio (see the scope note there), but they can be tested as spelling correspondences, and in fact already were: the strict-versus-loose latitude that the null model varies (§17.1, §17.3) is precisely the permissiveness of these rules. Hardening them is therefore a matter of reading the existing results at the level of the rules rather than the tokens.

Rules split into two classes by latitude. Strict-safe correspondences change nothing the source did not already license: written *u* realized /o/ (systematic, audio-confirmed), silent initial *g*, *Gua-* / *Gue-* for /k^w/, the reverential *-tziin* reducing to *-tzi*, and the underspecified marking of ejectives. Permissive (loose-only) correspondences add or move material the source does not have: the *-tl* to *-to* reduction and reorder, terminal-vowel padding (*el* to *ela*, and

the *-e* / *-a* on closed roots), the non-morphemic codas *-uks* / *-s*, and the soft tails *-che* / *-zeh*. The strict latitude forbids exactly this second class (no metathesis, no vowel epenthesis, no ad hoc coda changes), which is what makes it strict.

Result. Of the 62 committed tokens in the original run, **23 (37.1%, Wilson 95% CI [26.2%, 49.5%]) derive from their source using only strict-safe correspondences**; the other 39 (62.9%) need at least one permissive rule (on the enlarged headline base it is 26 of 65, since the three Battery-E names are un-stylised exact lexemes). These are the same tokens that pass Gate 1 in §17.4, now read as a statement about phonology: a little over a third of the lexicon's spellings map to their roots without any lossy stylization. The specificity of the two classes is the null model's own measurement: applied to nonsense strings, the strict bar manufactures a committed-quality match 3.4% of the time offline (Battery C), or 4.6% with online access (Battery D, the resources the real decode uses), while the loose bar does so 67% of the time (Battery A found-rate). So a derivation that leans on a permissive rule is, as a spelling correspondence, close to a coin-flip, and a derivation that does not is specific. The phonologically hardened set and the lexically hardened set are the same set.

What this does and does not establish. It hardens the *orthographic* correspondences: it shows which spelling-to-source mappings are tight and which are permissive, and it puts the permissive ones on the same near-noise footing the lexicon already reports. It does not recover the actual phonemes; that is the audio work the project has chosen not to pursue (§2.6). The reproducible tooling is the null-model battery (`NULL_MODEL_PROTOCOL.md`, `TIGHTENED_LATITUDE.md`); no new experiment is introduced here, only the reading of the existing one at the rule level.

17.10 SOURCES OF UNCERTAINTY (WHAT DOMINATES)

The confidence intervals above quantify sampling uncertainty, which is not the largest unknown. Ranked by how much each could move the conclusions, the dominant sources are conceptual rather than statistical:

SOURCE	WHAT IT AFFECTS	DIRECTION AND SIZE	WHERE IT IS HANDLED
Base rate b (fraction of Vaal built from real roots)	every PPV in §17.5	dominant; unmeasurable, proxied near 58% from the name roster	reported as a range, not a point (§17.6)
Developer stylization	which spelling-to-root maps count	can inflate loose-latitude matches	strict-versus-loose latitude test (§17.3, §17.9)
Unidentified source languages	false negatives and mis-attributions	widens the admissible pool, raises FPR	pre-committed palette; Xinkan and Nawat cross-checks (§13)
Transcription ambiguity	the token forms themselves	small but unquantified	corpus-only forms; audio phonology out of scope (§2.6)
Semantic drift in glossing	whether a real root yields the right sense	elastic (green to evergreen to undying)	competitor logging (C*); "genuine root" kept separate from "correct gloss"
Game localization	the English barks used as ground truth	mild circularity	flagged; syntax derived independently of the glosses (§3, §17.8)

The single largest lever is the base rate b. The statistics constrain the test's error rates tightly, but the translation's overall correctness still rides on how much of Vaal the designers built from real roots, which no internal test can measure.



18. CITATIONS

Numbered to match the bracketed markers in §§2–12 and the Source column of §9. APA 7th style; every entry carries a working URL (FAMSI and Wired Humanities are word-searchable, so any root can be confirmed by lookup). Print works are marked. Candidate parses raised in collaboration (including external-model suggestions) were adopted only after verification against these sources.

Primary Vaal text & game data

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4. LocalIdentity. (2025). *poe2-data* [Data set: NPCTextAudio.json, English & French]. GitHub. <https://github.com/LocalIdentity/poe2-data>
5. Path of Exile 2 Wiki (Fextralife). (n.d.). *Fate of the Vaal; Recruit the Vaal; The Cuachic Vault; Atziri's Temple*. Retrieved June 16, 2026, from <https://pathofexile2.wiki.fextralife.com/Fate+of+the+Vaal>, <https://pathofexile2.wiki.fextralife.com/Recruit+the+Vaal>, <https://pathofexile2.wiki.fextralife.com/The+Cuachic+Vault>, <https://pathofexile2.wiki.fextralife.com/Atziri's+Temple>. Path of Exile Wiki. (n.d.). *Atziri, Queen of the Vaal; The Temple of Atzoatl; The Vaal*. https://www.poewiki.net/wiki/Atziri,_Queen_of_the_Vaal, https://www.poewiki.net/wiki/The_Temple_of_Atzoatl, https://www.poewiki.net/wiki/The_Vaal
6. Grinding Gear Games. (2024–2026). *Path of Exile 2* [Video game; in-game dialogue and item text], *pochiti* (poe2db: *Atziri, the Red Queen*, https://poe2db.tw/us/Atziri%2C_the_Red_Queen); the *Eztli Pilli* tome; the Blood Priests' English barks; the *Quemalani* chat-feed lines. **Yucatec Maya**
7. Bolles, D. (2001). *Combined dictionary–concordance of the Yucatecan Mayan language*. Foundation for the Advancement of Mesoamerican Studies. <http://www.famsi.org/reports/96072/index.html>, searchable online. Print companion: Barrera Vásquez, A., Bastarrachea Manzano, J. R., & Brito Sansores, W. (1980). *Diccionario maya Cordemex: Maya-español, español-maya*. Ediciones Cordemex. [Yucatec roots: *ik'*, *ba'*, *iitz*, *kuxtal*, *kíimil*, *il*, *uk'*, *kutz*, *sak*, *ha'*, *muk'*, *náach*, *ich*, *lu'um*, *tul*, *el*, *ts'ook*, *k'ex*, *xul*, *yax*, *akal*, *anaab*, *k'i'ik'*, *ki'*, *pox*; the *-x* interrogatives.]
8. Real Academia Española. (n.d.). *Poch*. In *Diccionario de la lengua española*. Retrieved June 16, 2026, from <https://dle.rae.es/poch>
9. Encyclopædia Britannica. (n.d.). *Yucatec language*. Retrieved June 16, 2026, from <https://www.britannica.com/topic/Yucatec-language> **Classical Nahuatl**
10. Wired Humanities Projects. (n.d.). *Nahuatl dictionary*. University of Oregon. <https://nahuatl.wired-humanities.org/>, per-word entries: *eztli* (<https://nahuatl.wired-humanities.org/content/eztli>), *pilli* (<https://nahuatl.wired-humanities.org/content/pilli>), *pochotl* (<https://nahuatl.wired-humanities.org/content/pochotl>), *chantli* (<https://nahuatl.wired-humanities.org/content/chantli>), *yocoya* (<https://nahuatl.wired-humanities.org/content/yocoya>); also *atl*, *teōtl*, *cualli*, *nochi*, *yancuic*, *aocmo*, *tlapechtli*, *tlayohua*, *mochi*; the attested collocation *huey pilli*, *cualli eztli*.
11. Gran Diccionario Náhuatl. (2012). Universidad Nacional Autónoma de México. *Pochotl*. <https://gdn.iib.unam.mx/diccionario/pochotl/175246>, *ceiba*; protector–ruler metaphor.
12. Wiktionary. (n.d.). *pochotl*; *teotl*; *yancuic*; *waaj*. <https://en.wiktionary.org/wiki/pochotl>, <https://en.wiktionary.org/wiki/teotl>, <https://en.wiktionary.org/wiki/yancuic>, <https://en.wiktionary.org/wiki/waaj>
13. Wikipedia. (n.d.). *Teotl*. Retrieved June 16, 2026, from <https://en.wikipedia.org/wiki/Teotl>
14. Wired Humanities Projects. (n.d.). *Ti* (suffix) & *Tlan*. In *Nahuatl dictionary*. <https://nahuatl.wired-humanities.org/content/ti>, <https://nahuatl.wired-humanities.org/content/tlan>, the inchoative *-ti* and locative *-tlan*.

15. SIL México. (n.d.). *Consonantes del náhuatl*. Instituto Lingüístico de Verano. https://mexico.sil.org/es/lengua_cultura/nahuatl/consonantes-del-nahuatl, Nahuatl phonology (*b, d, g, f, ñ, rr* loan-only; *tl, kʷ*; *u*→[*v~f*]).
16. Wikipedia. (n.d.). *Aztec warfare* (the *cuāchicqueh*, "shorn ones"). Retrieved June 16, 2026, from https://en.wikipedia.org/wiki/Aztec_warfare **K'iche' / wider Mayan family**
17. Wikipedia. (n.d.). *K'iche' language*. Retrieved June 16, 2026, from https://en.wikipedia.org/wiki/K%27iche%27_language, phonology (implosive *b*, tap *r*, no voiced stops; dialectal [ð] for [l]); the *are' / ri* demonstratives behind *jare*.
18. Bennett, R. (2016). Mayan phonology. *Language and Linguistics Compass*, 10(10). Open-access PDF: https://bpb-us-w2.wpmucdn.com/campuspress.yale.edu/dist/c/1125/files/2015/09/Bennett2015_Mayan_phonology-2hbkyox.pdf, Mayan consonant inventories; the uvular *q*.
19. Encyclopædia Britannica. (n.d.). *Mesoamerican Indian languages*. Retrieved June 16, 2026, from <https://www.britannica.com/topic/Mesoamerican-Indian-languages>, uvular *q* in the K'ichean-Mamean / Q'anjob'alan branches; the family's geographic span. **Spanish / Romance**
20. Real Academia Española. (n.d.). *Daca*. In *Diccionario de la lengua española*. Retrieved June 16, 2026, from <https://dle.rae.es/daca>, "Da, o dame, acá" (recorded since the *Diccionario de autoridades*, 1732).
21. Wiktionary. (n.d.). *echar*; *fulgor*. <https://es.wiktionary.org/wiki/echar>, <https://es.wiktionary.org/wiki/fulgor>, *échelo* (← Lat. *iacta re*); *fulgor* (← Lat. *fulgur*); also *otra* (→ *u'tra*), *jefe* (archaic *xefe*), *ascensión*.
22. Diccionario del español de México. (n.d.). *Chante*. El Colegio de México. <https://dem.colmex.mx/> (s.v. *chante*, ← Nahuatl *chantli*). For *pochotón* (Costa Rican usage; ← *pochote* ← Nahuatl *pochotl*): Ecija. (n.d.). *Tico expressions (costarriqueñismos)* 4. <https://www.ecija.com/en/news-and-insights/tico-expressions-costarriqueñismos-4/> **Datamining tool**
23. SnosMe. (n.d.). *poe-dat-viewer* [Computer software]. GitHub. <https://github.com/SnosMe/poe-dat-viewer>, browser reader for PoE2 .*datc64* tables; route to current *NPCTextAudio* data. **Candidate source pools (exploratory; §13)**
24. Encyclopædia Britannica. (n.d.). *Xinca language*. <https://britannica.com/topic/Xinca-language> ; Wikipedia. (n.d.). *Xincan languages*. https://en.wikipedia.org/wiki/Xincan_languages, the non-Mayan Xinkan family of southeastern Guatemala (Mayan-loan contact; some varieties carry voiced *b, d*).
25. *Guatemala* (toponymy). New World Encyclopedia. (n.d.). *Guatemala*. <https://www.newworldencyclopedia.org/entry/Guatemala> ; Wikipedia. (n.d.). *K'iche' people*. https://en.wikipedia.org/wiki/K%CA%BCiche%CA%BC_people, Nahuatl *Cuauhtēmallān* "place of many trees" (*cuahui(tl)* "tree" + *tema* "to fill" + *-tlan* "place"), the calque of K'iche' *k'iche'* "many trees"; the *c uauh-* "tree" root behind *Guate-* (cf. *Cuauhtēmoc* → *Guatemoc*).
26. Xinka lexicon. Native Languages of the Americas. (n.d.). *Xinka words* (Chiquimulilla Xinka). http://www.native-languages.org/xinka_words.htm ; Rogers, C., & Sachse, F. (2025). Xinkan and Lencan. In S. Wichmann (Ed.), *The Languages and Linguistics of Mexico and Northern Central America: A Comprehensive Guide* (pp. 775–814). De Gruyter Mouton. <https://doi.org/10.1515/9783110421705-017> (drawing on Sachse 2010, ~1300-item Xinka vocabulary), basic Xinka forms used for the cross-check (§10.8, §13).
27. Aztec lexical & grammatical data (cross-check; §10.2, §10.4, §10.9, §13). Mexica.Net. (n.d.). *Nahuatl–English online dictionary*. <https://www.mexica.net/dictionary/>, *maca* "to give"; the demonstratives *inin* "this," *inon* "that." Wired Humanities Projects. (n.d.). *namaca* (*na-* + *maca*). <https://nahuatl.wired-humanities.org/content/namaca> . The absence of a native /*r*/ in Nahuatl, and so across the Nahuan branch, incl. Nawat, per [15].
28. Yucatec Maya verbal morphology (for the §10.9 root-search). Blair, R. W., & Vermont-Salas, R. (1965–1967). *Spoken (Yucatec) Maya*. University of Chicago [Chicago "Digital Maya"]. <https://lucy.lib.uchicago.edu/>, *kuxtal* as a stative / positional verb. The Yucatec causative *-s* and verbal-adjective *-kun(s)* (e.g. *séebkuns* "make quick"): parryc.com, *Yucatec Maya* grammar (citing Gorostieta). <https://parryc.com/language/yucatec-maya> . That *d, f, g, j, r, v* are non-native to Yucatec: <https://www.mustgo.com/worldlanguages/yucatec/> . Standard reference work: Bricker, V. R., Po'ot Yah, E., & Dzul de Po'ot, O. (1998). *A Dictionary of the Maya Language as Spoken in Hocabá, Yucatán*. University of Utah Press.

29. Mesoamerican military lexicon (for the §10.9 *Otsuks* semantic search). Nahuatl *yāōquīzqui* "soldier": Wired Humanities Projects, citing Molina (1571). <https://nahuatl.wired-humanities.org/content/yaoquizqui>. Nahuatl *tiacāuh* "valiant man / warrior": Wired Humanities, citing Sahagún, *Florentine Codex*, Bk 10. <https://nahuatl.wired-humanities.org/content/tiacauh>. The *Otōntin* (Otomi) and *Cuāchicqueh* ("Shorn Ones") warrior societies: Wikipedia, *Aztec warfare*. https://en.wikipedia.org/wiki/Aztec_warfare. K'iche' *achi* "man," *ajlab'al* "warrior" (*lab'al* "war"): Kaufman, T. (2003), *A Preliminary Mayan Etymological Dictionary* (FAMSI). Yucatec *holkan* "soldier," *ba'ate'el* "warrior/fight": Barrera Vásquez, A. (1980), *Diccionario Maya Cordemex*.
30. Yucatec Maya (Belize) heritage word-list, native-speaker community attestation, used as living corroboration (not a primary dictionary). Institute of Archaeology (NICH), Belize. (n.d.). *A list of 100 Yucatec Maya words and their English translation* [Facebook post; compiled by Andy Chuc of To'one Masehual'o'on, San Pablo, Orange Walk]. <https://www.facebook.com/IABelizeNICH/posts/a-list-of-100-yucatec-maya-words-and-their-english-translation/10159657738759966/>, records *poch* "anxious, eager" (*poch de comer* "craving to eat"), *he'la* "here it is / tengo," and *ta'* "excrement."
31. Yucatec *tsuk* "group, cluster, bunch" (root for *Otsuks*, §10.9). Barrera Vásquez, A. (1980). *Diccionario Maya Cordemex*, s.v. *tzuc* / *tsuk* ("buche o panza de los animales; cresta o mechón; grupo de árboles pequeños, montecillo") and s.v. *much'*, where *tsuk* is listed as a synonym glossed "*montón de granos o cosas menudas*," used as a **numeral classifier** in the frame *hun tsuk* "one cluster/heap," *ka' tsuk* "two," with *tsuk-en-tsuk* "in clusters." **Now verified directly in the Cordemex text** (the *Diccionario Maya Cordemex* citation-preview, consulted in full from the local HTML copy added to the project, preview at <https://dokumen.pub/diccionario-maya-cordemex-maya-espaol-espaol-maya.html>; complete PDF at https://filosofamaya.com/wp-content/uploads/2024/04/03_yuc_diccionario-maya-cordemex.pdf [32]), superseding the earlier dependence on a secondary Cordemex-phonetic transcription (*Apellidos mayas y sus significados*, México Desconocido, <https://www.mexicodesconocido.com.mx/apellidos-mayas-significados-nombres-maya.html>), which is retained only as corroboration. The classifier sense is decisive: a form that literally counts *groups* is the most natural source for a unit-call. **May a-language dictionary repository (source-of-sources; §10.9, §13)**
32. Filosofía Maya. (2021, July 31; updated 2026-01-14). *Diccionarios de las lenguas mayas*. <https://filosofamaya.com/2021/07/31/diccionarios-de-lenguas-mayas/>, direct-download PDFs of thirteen Maya dictionaries, including the **complete** *Diccionario Maya Cordemex* (Barrera Vásquez 1980; https://filosofamaya.com/wp-content/uploads/2024/04/03_yuc_diccionario-maya-cordemex.pdf, the full text behind the partial dokumen.pub preview used for [31]); the colonial *Calepino de Motul* (Acuña 2001) and *Bocabulario de Maya Than* (Acuña 1993); Bolles' *Combined Vocabularies* (2010); the *Diccionario Maya Popular* (2007); Bricker, Po'ot Yah & Dzul de Po'ot's *Hocabá* (1998); INEA (1998); and, for the conlang's non-Yucatec layers, Itzaj (Hofling & Tesucún 2000), Mopan (Hofling 2011), Lacandón (Hofling 2014), and two Ch'orti' dictionaries (Pérez Martínez et al. 1996; Hull 2016). The repository carries **no K'iche' dictionary**, so the K'iche' seam ([17]) still sources separately. A catalogued index with per-source relevance notes is kept beside this document as *Maya_Dictionary_Sources.md*.
33. New roots for the §10.9 Text 4 upgrades. **Nahuatl** *chiya* / *chīa* "**to wait, await, watch for, observe**" (the sentinel verb behind *cheyel*): Wired Humanities Projects, *Nahuatl dictionary*, s.v. *chiya* (citing Molina 1571; Lockhart 2001; Launey & MacKay 2011). <https://nahuatl.wired-humanities.org/content/chiya>. **Yucatec** *k'ux* "**to bite, gnaw; ache; rancor, anger**" (the root behind *quxzeh*, ejective per the corpus's *q* = /k'/, cf. *qexcan* ← *k'ex*): Barrera Vásquez, A. (1980), *Diccionario Maya Cordemex*, s.v. *k'ux* (*k'uxuk* "doler"; *k'uxil* "rencor, coraje"), verified directly in the project copy; full PDF via [32].
34. Nahuatl exclamatory / presentative particle *o* (the onset of *Otsuks*, §10.9). Wired Humanities Projects, *Nahuatl dictionary*, s.v. *o*, Molina (1571): *o* "es señal de optativo, y también de los pretéritos, e interjección del que está afligido y hace exclamación," and *o* "*he aquí*. adverbio." <https://nahuatl.wired-humanities.org/content/o-o>
35. Maya orthographic standard (for the ejective /k'/ behind the corpus's *q*; §10.9, §2.4). Briceño Chel, F., & Can Tec, G. R. (Coords.). (2014). *U nu'ukbesajil u ts'ibta'al maayat'aan* / *Normas de escritura para la lengua maya*. Instituto Nacional de Lenguas Indígenas (INALI). https://site.inali.gob.mx/pdf/norma_maya.pdf, the standard writes the glottalized stops *k'*, *ts'*, *ch'*, *p'*, *t'* (there is no *q* in Maya), confirming /k'/ as a distinct phoneme and so underwriting the reading of the corpus's stylised *q* (*qexcan* ← *k'ex*; *qux* ← *k'ux*) as a real ejective.
36. Modern Spanish→Maya wordlist (peninsular Yucatec, INALI-style orthography), a present-day source independent of Cordemex, corroborating several §10.9 roots. *Diccionario Maya* (Spanish→Maya), distributed via ilide.info / Scribd (<https://www.scribd.com/document/36429175/Diccionario-Maya>); copy added to the project. Confirms **tsúuk** (listed in *miúuch'* *tsúuk* for *montón* "heap") for the *Otsuks* root; **k'uux** "morder; odiar; rencor" for *quxzeh*; **taal** "venir" for *a'tul*; **ba'ax** "qué; cosa" for *buxa* / *Axba*; and **ka** "que" for the §10.9 particle *ka*. The Yucatec watch/sentinel field weighed against *cheyel*, *ch'úuk* "lie in wait, ambush," *cha'an* "behold, watch," *pa'at* "wait," *kanan* "guard", is also drawn from this source.

37. Text 5, Line 1 (the Kamasan Smith). In-game footage capture of the Vaal **Kamasan Smith**, a boss that appears under randomized names (captured as *Xomatl* and *Zantico*), a fire/forge-themed boss in **Atziri's Temple at Lira Vaal**, encountered at the height of Vaal power before the Cataclysm; he works the forge until the player enters, then speaks *Ti ek tala jare'yantul!* and attacks. Line tokens corroborated by Cordemex[7] (*ti', ek', taal, yan, tul*) and the modern wordlist[36] (*estrella* = *eeek'*; *venir* = *taal*); *jare'* per [17].
38. *Guatelitzi, Architect of Flesh*. Path of Exile Wiki. https://www.poewiki.net/wiki/Guatelitzi,_Architect_of_Flesh, a Vaal Architect of the **Temple of Atzoatl** (Incursion, v3.3.0; core v3.5.0); resident of *Pools of Restoration / Sanctum of Vitality / Sanctum of Immortality* (room flavour "*Flesh rending for life unending*"), with maximum-Life, Energy-Shield, and Life/ES-regeneration signature modifiers; the flesh-and-immortality architect named by Zelina's "*Tome of Guatelitzi*" (§10.8). In-game identification contributed by Reddit user *Tenebris-Umbra*.
39. *Xibaqua*. Path of Exile Wiki. <https://www.poewiki.net/wiki/Xibaqua>, the progenitor of all Vaal, "born from the flesh of ancient Vaal gods"; origin myth quoted from the *Demon Stitcher* item (*History of Wraeclast*, https://www.poewiki.net/wiki/History_of_Wraeclast): the gods reclaim their flesh until "a droplet of pure light: The first Vaal" remains.
40. Temple of Atzoatl lore. Path of Exile Wiki, *The Vaal* and *Atziri (lore)*. https://www.poewiki.net/wiki/The_Vaal, [https://www.poewiki.net/wiki/Atziri_\(lore\)](https://www.poewiki.net/wiki/Atziri_(lore)), the temple's pursuits (immortality, corruption, weather, sacrifice), Atziri's later seat, the "Demon of Atzoatl."
41. Vaal gods (*Yaomac*, *Yugul*, *Ralakesh*). Path of Exile Wiki, *The Vaal* and *History of Wraeclast*. https://www.poewiki.net/wiki/The_Vaal, https://www.poewiki.net/wiki/History_of_Wraeclast, *Yaomac* (three serpentine heads, shepherd of Vaal souls; *Yaomac's Accord, Triumvirate Authority*), *Yugul* ("Reflection of Terror," Act 8 pantheon), *Ralakesh* ("Master of a Million Faces").
42. *Kopec* and *Kamasa* (Vaal gods). Path of Exile Wiki, *God*; and the Chaos-Bloodline ascendancies *Priest of Kopec / Priest of Kamasa / Priest of Yaomac*. <https://www.poewiki.net/wiki/God>, https://www.poewiki.net/wiki/Priest_of_Kopec
43. Vaal Architects of the Temple of Atzoatl (the architect roster). Path of Exile Wiki, *Incursion room*. https://www.poewiki.net/wiki/Incursion_room
44. Text 5, Lines 2 and 3 (*Ahuatotli, the Blind*; *Mektul*). In-game footage captures of two Vaal bosses' combat shouts (*Ahuatotli* is the Delve boss "Ahuatotli, the Blind"). Recognisable tokens corroborated by Cordemex[7] and the modern wordlist[36]: *táak'* in "gold," *k'áak'* "fire," *ik'* "wind / spirit."
45. Comparative onomastics for the pantheon names (the cross-check pass behind §12.3). *Camazotz*, the Popol Vuh death-bat (K'iche' *kame* "death" + *sotz'* "bat"): <https://en.wikipedia.org/wiki/Camazotz> . *Apep / Apophis*, Egyptian chaos-serpent: <https://en.wikipedia.org/wiki/Apep> . *Rakshasa*, the shape-shifting demons of Hindu / Buddhist lore: <https://en.wikipedia.org/wiki/Rakshasa> . *Arachne* (Greek): <https://en.wikipedia.org/wiki/Arachne> ; *Kali* (Hindu): <https://en.wikipedia.org/wiki/Kali> . The Nahuatl locative *-mac* "in the hands of" (from *mātl* "hand") and *calli* "house" per [10]. These foreign roots are recorded as onomastic comparison, not as additions to the committed Maya / Nahuatl palette.
46. *Nahuatl 1100* (vocabulary course). My Little Word Land. <https://mylittlewordland.com/course/304181/nahuatl-1100> . A community-built Nahuatl word-list, used here as a cross-reference pool. It glosses *atzoatl* as "dirty water" (§12.2). Treated as an **informal primary source requiring secondary validation**: the gloss is borne out by *tzoatl* "dirty water, slops" (*lauazas o lauaduras*, Molina 1571) in the Wired Humanities Nahuatl Dictionary[10], <https://nahuatl.wired-humanities.org/content/tzoatl> . Other entries from this list are to be confirmed against a primary dictionary before any adoption.
47. Christenson, A. J. (n.d.). *K'iche'-English Dictionary and Guide to Pronunciation of the K'iche'-Maya Alphabet*. Brigham Young University; offered to FMSI as a research tool. https://www.famsi.org/mayawriting/dictionary/christenson/quidic_complete.pdf . Compiled from field work with native K'iche' speakers in Momostenango and Totonicapán, 1978-1985. The project's first dedicated K'iche' dictionary; the lexical authority for the Highland-Maya seam (*jare*, *koriek*), where the Wikipedia phonology note[17] previously stood alone.
48. Vaal Architect lore, and some connections I found while studying it. Community lore compilation by Reddit user Murky-Definition-625, *r/Wraeclast*. https://www.reddit.com/r/Wraeclast/comments/1shprfn/vaal_architect_lore_and_some_connections_i_found/ . Used here only for its consolidation of in-game, wiki-attested architect roles (room functions, boss identities, and flavour text); the author's speculative connections are not adopted. Primary roster authority remains the Incursion room page[43].

49. Roots for the *kilya* re-analysis (§9). Nahuatl *quil-* "green, verdant": Wired Humanities Projects, Nahuatl dictionary, s.v. *quiltic* "the color green" (Molina 1571, f. 89v), <https://nahuatl.wired-humanities.org/content/quiltic>, and *quilitl* "greens, edible verdure" (Molina; Karttunen), <https://nahuatl.wired-humanities.org/content/quilitl>. The inchoative *-ya* "to come to be X" is the suffix of *hueyiya* "to grow, become big" (from *huēi* "big") [46] and of *celiya* "to sprout, blossom, turn green again, revive" (*prender, brotar, retoñecer la planta*, Molina), <https://nahuatl.wired-humanities.org/content/celiya-0>, the sense behind *kil ya* but a /s/-onset near-miss on the form. The fronting *mā* is the Classical Nahuatl optative particle (see the *ma* entry; [10]).
50. Phoneme inventories of the source languages (§2.3). Classical Nahuatl: Andrews (2003), *Introduction to Classical Nahuatl*, as reproduced in *Research in Language* 14(3), 2016 (<https://www.researchgate.net/publication/309517030>); SIL Mexico, *The Consonants of Nahuatl* (https://mexico.sil.org/language_culture/aztec/consonants-of-nahuatl); Wikipedia, *Nahuatl* (<https://en.wikipedia.org/wiki/Nahuatl>). Yucatec Maya: Wikipedia, *Yucatec Maya language* (https://en.wikipedia.org/wiki/Yucatec_Maya_language); Frazier (2011), *Tonal Dialects and Consonant-Pitch Interaction in Yucatec Maya* (<https://www.melfraz.com/ling/Frazier2011-TonalDialects.pdf>); Bennett (2016), *Mayan phonology*, Language and Linguistics Compass ([https://people.ucsc.edu/~rbennett/resources/papers/pdfs/Bennett%20\(2016\)%20-%20Mayan%20phonology.pdf](https://people.ucsc.edu/~rbennett/resources/papers/pdfs/Bennett%20(2016)%20-%20Mayan%20phonology.pdf)). K'iche': Wikipedia, *K'iche' language* (https://en.wikipedia.org/wiki/K%27iche%27CA%BCiche%27CA%BC_language); Bennett (2016), op. cit.; Par Sapón & Can Pixabaj (2000), *Variación Dialectal en K'iche'* (OKMA/Cholsamaj).
51. Yucatec *uts* "good" and the form *yutsal* (§9). Diccionario Maya Cordemex (full text: https://filosofamaya.com/wp-content/uploads/2024/04/03_yuc_diccionario-maya-cordemex.pdf) glosses *uts* as "cosa buena" (a good thing; *ma' uts* "cosa mala," not-good = bad) and carries the form *yutsal* in *yutsal ti* "todo es bueno lo que hay en él" (all that is in it is good), beside *uts ti* "todo es bueno" [31][32]. The modern peninsular wordlist lists *uts* for "bien" and *utsil* for "bondad" (goodness) [36]. The third-person possessive *y-* (Set A, pre-vocalic) is Yucatec; *utz* "good" is also a basic K'iche' word, corroborating the root across the Maya layers.
52. Nahuatl *ocotl* / *ocutl* "pine torch" (§10.7, *ukto*). The Wired Humanities Nahuatl Dictionary glosses *ocotl* as a pine tree and a torch made of pine, citing Frances Karttunen, *An Analytical Dictionary of Nahuatl* (Univ. of Oklahoma Press, 1992), 176, and describing it from the Florentine Codex as a resinous torch that gives light. <https://nahuatl.wired-humanities.org/content/ocotl>. The Codex attests the variant spelling *ocutl* (with /u/) and pairs it with the verb "to burn" (*tlatlathiuh in ocutl*). Molina lists *quauhocotl* "tea de pino" (pine torch). The root *oco-* also appears in *ococuahuitl* "pine wood" and *ocototon* "pine splinters" [46].
53. Campbell, L. (1985). *The Pipil Language of El Salvador* (Mouton Grammar Library 1). Mouton Publishers. PDF via TUSHIK: <http://tushik.org/wp-content/uploads/CAM-pipil.pdf>. The standard Nawat / Pipil reference (a Nawat-Spanish-English lexicon with Classical Nahuatl cognates). Used to confirm, form-first, that /f/ and /r/ are loan-only in Nawat (all *f*-headwords tagged Spanish; /r/ only in loans such as *frijolar, carrizal*), so no native Nawat root can host *fukuur* (§10.4); to record living-Nawat *uku-t* "pine, torch pine" (< Proto-Nahua *oko-, CN *ocotl*), an independent reflex of the *ukto* root (§10.7); and it attests Nawat *chiya* "to look, spy, wait" (corroborating *cheyel*, §10.9) and *maka* "to give" (the semantic echo behind *daka*, §10.4).
54. Quecholli (Vaal unique mace, PoE2). Path of Exile 2 Wiki, *Quecholli*, <https://www.poe2wiki.net/wiki/Quecholli>. Inscribed to Doryani of the Vaal. The name is Classical Nahuatl *quecholli*, the 14th veintena (twenty-day month) dedicated to Mixcoatl and to weapon-making, itself a bird-name (roseate spoonbill / precious-feather bird), attested in Sahagun's Florentine Codex and Molina; Nahuatl morphology per the Wired Humanities dictionary [10].
55. Panquetzaliztli (Vaal mace, PoE1; the fated form of Quecholli). Path of Exile Wiki, *Panquetzaliztli*, <https://www.poewiki.net/wiki/Panquetzaliztli>. Also inscribed to Doryani of the Vaal. Classical Nahuatl *panquetzaliztli*, the 15th veintena, "the raising of banners," festival of Huitzilopochtli: *pan(tli)* "banner" + *quetza* "to raise" + *-liztli* (Sahagun; Molina). The *quetza* root also underlies *Quetzalcoatl* and the corpus names *Ketzuli*, *Kuetzakala*.
56. Ixchel / the Trialmaster (a Vaal citizen, "the Godstealer"). Path of Exile Wiki, *The Trialmaster*, https://www.poewiki.net/wiki/The_Trialmaster: "the Trialmaster was formerly a Vaal citizen known as Ixchel," set by the Order of the Djinn to steal the god Yaomac, whence "Godstealer." *Ixchel* is the attested Maya goddess Ix Chel (Yucatec *Ix-* feminine prefix + *Chel* "rainbow").
57. Ixtlatl and Mahuatzi (Vaal Architect bosses of the strongboxes *Gifts of the Sacrificed* and *Gifts of the Red Queen*). Path of Exile Wiki, *Gifts of the Sacrificed*, https://www.poewiki.net/wiki/Gifts_of_the_Sacrificed (assets Metadata/Monsters/VaalArchitect/HoodedVaalArchitectBoss1 and 2). Nahuatl morphology (*ixtli* "face/eye"; *māhua* "to spread"; reverential *-tzin*) per [10].

58. Vaal Temple bosses *K'aj Y'ara'az*, *K'aj Q'ura*, *K'aj A'alai* (PoE1, introduced 1.1.0, 2014; variants of the Atziri trio). Path of Exile Wiki, *Vaal Temple*, https://www.poewiki.net/wiki/Vaal_Temple#Boss . Checked form-first against Cordemex[7] and Christenson[47]; see §12.8.
59. The *-aqua* tail of *Xibaqua* (§12.1). Classical Nahuatl *aquia* "to insert, submerge, drive in, transplant" (Molina, *Vocabulario* 1571, pt. 2 f. 7v), Wired Humanities Nahuatl Dictionary, <https://nahuatl.wired-humanities.org/content/aquia> . Latin *aqua* "water" and its regular development to Spanish *agua* (Latin *qu* to Spanish *gu*): Wiktionary, *aqua*, <https://en.wiktionary.org/wiki/aqua> , and *agua*, <https://en.wiktionary.org/wiki/agua> . The Vaal tail keeps /k/ (*aqua*), not /g/ (*agua*), so the water reading, if a loan, is the Latin form, not the Spanish.